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Methods of detecting inherited myopathies in horses

Abstract

This disclosure describes detecting four genetically distinct kinds of inherited myopathy in horses, referred to as Polysaccharide Storage Myopathy type 2 (PSSM2), or in some cases as Myofibrillar Myopathy (MFM).

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This application is a United States national phase patent application based upon international patent application number PCT/US19/33717 of international filing date May 23, 2019, which claims the benefit U.S. Provisional Patent Application No. 62/717,072 filed Aug. 10, 2018 and U.S. Provisional Patent

Application No. 62/737,295, filed Sep. 27, 2018, all three of which applications are incorporated herein by reference in its entirety.

SEQUENCE LISTING

(1) This application contains a Sequence Listing electronically submitted via EFS-Web to the United States Patent and Trademark Office as an ASCII text file entitled "0310_000142WO01_SL" having a size of 223 kilobytes and created on Apr. 30, 2019. The information contained in the Sequence Listing is incorporated by reference herein.

SUMMARY

(2) This disclosure describes, in one aspect, methods for detecting the presence or absence of a set of biomarkers in a horse. Generally, the method includes obtaining a biological sample from a horse that includes a nucleic acid that includes the coding region for three genes: dysferlin (DYSF), pyridine nucleotide-disulfide oxidoreductase domain-containing protein 1 (PYROXD1), and collagen type VI alpha 3 chain (COL6A3). There are two important biomarkers in DYSF, one important biomarker in PYROXD1, and one important biomarker in COL6A3, to be assayed as follows: (1) Dysferlin (DYSF), determining whether the nucleic acid has the specific substitution of an adenine (A) for a guanine (G) on the forward strand at chr15:31,306,949 of the current horse genome assembly (EquCab2, GCA_000002305.1) as displayed in the UCSC Genome Browser and as shown in FIG. **1**, or the equivalent substitution in the complement thereof. This base substitution, known as rs1145077095 in dbSNP, corresponding to position 31,306,949 in SEQ ID NO:1, results in a nonconservative amino acid substitution in the dysferlin (DYSF) protein. The amino acid substitution caused by this base substitution is shown in FIG. **4**. FIG. **4** shows the partial sequence of an altered dysferlin with tryptophan (W) substituted for arginine (R) at position 253, with SEQ ID NO:10 showing the partial protein sequence encoded by the wild-type or common allele and SEQ ID NO:11 showing the partial protein sequence encoded by the variant. (2) Dysferlin (DYSF), determining whether the nucleic acid has the specific substitution of a thymine (T) for a guanine (G) on the forward strand at chr15:31,225,630 of the current horse genome assembly (EquCab2, GCA_000002305.1) as displayed in the UCSC Genome Browser and as shown in FIG. **13**, or the equivalent substitution in the complement thereof. This base substitution, known as rs1136366555 in dbSNP, corresponding to position 31,225,630 in SEQ ID NO:56, results in a nonconservative amino acid substitution in the dysferlin (DYSF) protein. The amino acid substitution caused by this base substitution is shown in FIG. **16**. FIG. **16** shows the partial sequence of an altered dysferlin with threonine (T) substituted for proline (P) at position 1290, with SEQ ID NO:60 showing the partial protein sequence encoded by the wild-type or common allele and SEQ ID NO:61 showing the partial protein sequence encoded by the variant. (3) Pyridine nucleotide-disulfide oxidoreductase domain-containing protein 1 (PYROXD1), determining whether the nucleic acid has the specific substitution of a cytosine (C) for a guanine (G) on the forward strand at chr6:47,661,977 of the current horse genome assembly (EquCab2, GCA_000002305.1) as displayed in the UCSC Genome Browser and as shown in FIG. **22**, or the equivalent substitution in the complement thereof. This base substitution, known as rs1136260157 in dbSNP, corresponding to position 47,661,977 in SEQ ID NO:103, results in a nonconservative amino acid substitution in the pyridine nucleotide-disulfide oxidoreductase domain-containing protein 1 (PYROXD1) protein. The amino acid substitution caused by this base substitution is shown in FIG. **25**. FIG. **25** shows an altered pyridine nucleotide-disulfide oxidoreductase domain-containing protein 1 (PYROXD1), with histidine (H) substituted for aspartate (D) at position 492, with SEQ ID NO:107 showing the protein sequence encoded by the wild-type or common allele and SEQ ID NO:108 showing the protein sequence encoded by the variant. (4) Collagen type VI alpha 3 chain (COL6A3), determining whether the nucleic acid has the specific substitution of a guanine (G) for a cytosine (C) on the forward strand at chr6:23,480,621 of the current horse genome assembly (EquCab2, GCA_000002305.1) as displayed in the UCSC Genome Browser and

as shown in FIG. **33**, or the equivalent substitution in the complement thereof. This base substitution, known as rs1139437410 in dbSNP, corresponding to position 23,480,621 in SEQ ID NO:163, results in a nonconservative amino acid substitution in the collagen type VI alpha 3 chain (COL6A3) protein. The amino acid substitution caused by this base substitution is shown in FIG. **36**. FIG. **36** shows an altered collagen type VI alpha 3 chain (COL6A3), with alanine (A) substituted for glycine (G) at position 2182, with SEQ ID NO:167 showing the protein sequence encoded by the wild-type or common allele and SEQ ID NO:168 showing the protein sequence encoded by the variant.

(3) In some embodiments, the method further includes amplifying at least a portion of the DYSF, PYROXD1, or COL6A3 coding regions. In some of these embodiments, all or part of (1) Exon B of the DYSF coding region as identified in FIG. **3** (SEQ ID NO:5) is amplified. This specified exon corresponds to the gene models presented in FIG. **2** in this disclosure; the specific base substitutions detected are presented in FIG. **1**, even if alternative gene models or different isoforms result in this exon being named or numbered differently. In another aspect, this disclosure describes a method for detecting the presence or absence of a biomarker in a physiological sample. Generally, the method includes obtaining a physiological sample from a horse that includes a nucleic acid that includes at least a portion of SEQ ID NO:1 that includes nucleotide 31,306,949 of SEQ ID NO:1, determining whether the nucleic acid has an adenine (A) substituted for a guanine (G) at nucleotide 31,306,949 on the forward strand of SEQ ID NO:1; this single-nucleotide polymorphism (SNP) is also known as rs1145077095 in dbSNP. (2) Exon I of the DYSF coding region as identified in FIG. **15** (SEQ ID NO:59) is amplified. This specified exon corresponds to the gene models presented in FIG. **14** in this disclosure; the specific base substitutions detected are presented in FIG. **13**, even if alternative gene models or different isoforms result in this exon being named or numbered differently. In another aspect, this disclosure describes a method for detecting the presence or absence of a biomarker in a physiological sample. Generally, the method includes obtaining a physiological sample from a horse that includes a nucleic acid that includes at least a portion of SEQ ID NO:56 that includes nucleotide 31,225,630 of SEQ ID NO:56, determining whether the nucleic acid has an thymine (T) substituted for a guanine (G) at nucleotide 31,225,630 of the forward strand of SEQ ID NO:56; this single-nucleotide polymorphism (SNP) is also known as rs1136366555 in dbSNP. (3) Exon 12 of the PYROXD1 coding region, as identified in FIG. **24** (SEQ ID NO:106) is amplified. This specified exon corresponds to the gene models presented in FIG. **23** in this disclosure; the specific base substitutions detected are presented in FIG. **22**, even if alternative gene models or different isoforms result in this exon being named or numbered differently. In another aspect, this disclosure describes a method for detecting the presence or absence of a biomarker in a physiological sample. Generally, the method includes obtaining a physiological sample from a horse that includes a nucleic acid that includes at least a portion of SEQ ID NO:103 that includes nucleotide 47,661,977 of SEQ ID NO:103, determining whether the nucleic acid has a cytosine (C) substituted for a guanine (G) at nucleotide 47,661,977 of the forward strand of SEQ ID NO:103; this single nucleotide polymorphism (SNP) is also known as rs1136260157 in dbSNP. (4) Exon 26 of the COL6A3 coding region, as identified in FIG. **35** (SEQ ID NO:166) is amplified. This specified exon corresponds to the gene models presented in FIG. **34** in this disclosure; the specific base substitutions detected are presented in FIG. **33**, even if alternative gene models or different isoforms result in this exon being named or numbered differently. In another aspect, this disclosure describes a method for detecting the presence or absence of a biomarker in a physiological sample. Generally, the method includes obtaining a physiological sample from a horse that includes a nucleic acid that includes at least a portion of SEQ ID NO:163 that includes nucleotide 23,480,621 of SEQ ID NO:163, determining whether the nucleic acid has a guanine (G) substituted for a cytosine (C) at nucleotide 23,480,621 of the forward strand of SEQ ID NO:163; this single nucleotide polymorphism (SNP) is also known as rs1139437410 in dbSNP.

(4) In all cases, the nucleotide at the specified position of the forward strand may be inferred by the determination of the nucleotide at the specified position on the reverse (complementary) strand. In some embodiments, the method further includes amplifying at least a portion of the nucleic acid.
(5) In another aspect, this disclosure describes a method for detecting the presence or absence of a biomarker in a physiological sample. Generally, the method includes obtaining a physiological sample from a horse that includes a nucleic acid encoding (1) Dysferlin polypeptide, then determining whether the nucleic acid encodes a dysferlin polypeptide altered as described as follows: a dysferlin polypeptide having, in part, the amino acid sequence of SEQ ID NO:10 (FIG. **4**) or a dysferlin polypeptide having a tryptophan (W) substituted for arginine (R) at position 253 as shown in SEQ ID NO:11 (FIG. **4**). (2) Dysferlin polypeptide, then determining whether the nucleic acid encodes a dysferlin polypeptide altered as described as follows: a dysferlin polypeptide having, in part, the amino acid sequence of SEQ ID NO:60 (FIG. **16**) or a dysferlin polypeptide having a threonine (T) substituted for proline (P) at position 1290 as shown in SEQ ID NO:61 (FIG. **16**). (3) Pyridine nucleotide-disulfide oxidoreductase domain-containing protein 1 (PYROXD1) polypeptide, then determining whether the nucleic acid encodes a pyridine nucleotide-disulfide oxidoreductase domain-containing protein 1 polypeptide altered as described as follows: a pyridine nucleotide-disulfide oxidoreductase domain-containing protein 1 polypeptide having, in part, the amino acid sequence of SEQ ID NO:107 (FIG. **25**) or a pyridine nucleotide-disulfide oxidoreductase domain-containing protein 1 polypeptide having a histidine (H) for an aspartate (D) at position 492 as shown in SEQ ID NO:108 (FIG. **25**). (4) Collagen type VI alpha 3 chain (COL6A3) polypeptide, then determining whether the nucleic acid encodes a collagen type VI alpha 3 chain polypeptide altered as described as follows: a collagen type VI alpha 3 chain polypeptide having, in part, the amino acid sequence of SEQ ID NO:167 (FIG. **36**) or a collagen type VI alpha 3 chain polypeptide having an alanine (A) for a glycine (G) at position 2182 as shown in SEQ ID NO:168 (FIG. **36**).
(6) The above summary is not intended to describe each disclosed embodiment or every implementation. The description that follows more particularly exemplifies illustrative embodiments. In several places throughout the application, guidance is provided through lists of examples, which examples can be used in various combinations. In each instance, the recited list serves only as a representative group and should not be interpreted as an exclusive list.

Description

BRIEF DESCRIPTION OF THE FIGURES
(1) FIG. **1**. A portion of the current horse genome assembly (EquCab2, GCA_000002305.1) with coordinates as displayed in the UCSC Genome Browser centered on the chr15:31,306,949 position, the site of a substitution of an adenine (A) for a guanine (G) on the forward strand that results in the substitution of a tryptophan (W) for arginine (R) at amino acid position 253 in dysferlin as shown in FIG. **4** (SEQ ID NO:11). The reverse complement sequence is shown, with the site of a substitution of a thymine (T) for a cytosine (C) as indicated (SEQ ID NO:1). The single nucleotide polymorphism (SNP) defined by this base substitution is identified as rs1145077095 in dbSNP.
(2) FIG. **2**. A portion of the normal equine DYSF Coding DNA Sequence (SEQ ID NO:2) and a portion of the mutant DYSF Coding DNA Sequence (SEQ ID NO:3) bearing the C to T mutation at nucleotide position at position 1027 in this figure, corresponding to chr15:31,306,949 as shown in SEQ ID NO:1 (FIG. **1**). This sequence is a region of perfect consensus among 21 different experimentally predicted mRNA isoforms. The numbering in FIG. **2** is that of isoform X1 (XM_023618907.1), which for this segment perfectly matches the numbering of isoforms X2 (XM_023618908.1), X3 (XM_023618909.1), X4 (XM_023618910.1), X5 (XM_023618911.1), and X6 (XM_023618912.1). The numbering of the start and end positions for other isoforms is

described in detail in the text. In both sequences, the sequence of Exon B as shown in FIG. **3** is indicated in bold. The site of a C to T mutation at nucleotide position 1027, corresponding to 31,306,949 in SEQ ID NO:1 (FIG. **1**), and to rs1145077095 in dbSNP, is underlined. The region of sequence comprising Exon B as shown in FIG. **3** is displayed as codons in the correct reading frame for both SEQ ID NO:2 and SEQ ID NO: 3 in FIG. **5**.

(3) FIG. **3**. A view of the current horse genome assembly (EquCab2, GCA_000002305.1) in the UCSC Genome Browser with exon sequences that match the partial DYSF Coding DNA Sequence (SEQ ID NO:2) and the mutant partial DYSF Coding DNA Sequence (SEQ ID NO:3). The DYSF Coding DNA Sequences correspond to Exons A, B, C, D, E, and F as indicated by the regions of sequence similarity of the translated genomic DNA to DYSF protein sequences from human (Homo), cattle (Bos), rat (Rattus), mouse (Mus), and DYSF protein sequences from the more distantly-related zebrafish (Danio) and African claw-toed frog (Xenopus). Partial matches to the paralogous protein myoferlin (MYOF) from human (Homo), cattle (Bos), mouse (Mus), rat (Rattus), and zebrafish (Danio) are also seen. The sequences of Exons A-F with 10 bp of flanking intron sequence and their coordinates in the current horse assembly are displayed below the image from the UCSC Genome Browser. Sequence IDs are Exon A (SEQ ID NO:4), Exon B (SEQ ID NO:5), Exon C (SEQ ID NO:6.), Exon D (SEQ ID NO:7), Exon E (SEQ ID NO:8), and Exon F (SEQ ID NO:9).

(4) FIG. **4**. Models of part of the normal protein sequence encoded by horse DYSF (XP_023474694.1, presented here as SEQ ID NO:10) corresponding to a translation of SEQ ID NO:2 shown in FIG. **2** and part of the altered protein sequence encoded by horse DYSF with the base substitution at chr15:31,306,949 (based on XP_023474694.1, presented here as SEQ ID NO:11) corresponding to a translation of SEQ ID NO:3 shown in FIG. **2**. The portion of the protein encoded by Exon B as shown in FIG. **3** is indicated in bold, while amino acid position 253 affected by the base substitution of an adenine (A) for a guanine (G) on the forward strand at the chr15:31,306,949 position, corresponding to rs1145077095 in dbSNP as shown in FIG. **1**, is underlined. The amino acid positions in XP_023474694.1 are indicated at the beginning and end of the sequence.

(5) FIG. **5**. Horse DYSF Exon B and flanking genomic DNA sequence from which PCR primers to amplify genomic DNA containing the site of the DYSF-R253W mutation would be most appropriately derived. Genomic coordinates are as in FIG. **1**. Exon B from chr15:31,307,036 to chr15:31,306,908 is shown broken into codons in the correct reading frame for the wild-type allele (SEQ ID NO:12) and the DYSF-R253W allele (SEQ ID NO:13). Only the reference sequence from the assembly is shown for the flanking sequences. The codon affected by the G to A mutation site at nucleotide position chr15:31,306,949 on the forward strand, corresponding to rs1145077095 in dbSNP as shown in FIG. **1** (C to T in the reverse complement as shown), is shown in bold, with the position of the base substitution indicated by underlining. The base substitution changes the bold three base codon from one coding for an arginine (CGG) to one coding for a tryptophan (TGG). Example primers used experimentally to amplify genomic DNA containing the mutation site are shown in lower case [5′-CCCGAGATTTCTGGCTTTCT-3′ (SEQ ID NO:14) and 5′-CTCGACAAGTTCTGGGGTGT-3′(SEQ ID NO:15)].

(6) FIG. **6**. Traces from Sanger DNA sequencing of amplified DYSF genomic DNA using primers shown in FIG. **5** (SEQ ID NO:14 and SEQ ID NO:15). The sequence of the reverse strand is shown. The arrows in the figure indicate nucleotide position chr15:31,306,949, the site of a substitution of a thymine (T) for a cytosine (C) in this position, corresponding to rs1145077095 in dbSNP, that creates the DYSF-R253W variant. The traces show, from left to right, results for a horse homozygous for the wild-type or common allele (N/N), results for a horse heterozygous for the substitution (N/P5), and results for a horse homozygous for the substitution (P5/P5).

(7) FIG. **7**. Sequence of the human DYSF coding sequence derived from NM_003494.3 (SEQ ID NO:16). The 5′ UTR and 3 UTR have been removed; the sequence begins with the ATG start codon

and ends with the TGA stop codon. The numbering of the first and last nucleotides corresponds to that of NM_003494.3. The sequence of the human exon corresponding to Exon B in horse as shown in FIG. **3** is indicated in bold.

(8) FIG. **8**. Sequence of the human DYSF protein sequence, equivalent to NP_003485.1 (SEQ ID NO:17). The numbering of the first and last amino acids corresponds to that of NP_003485.1. The sequence encoded by the human exon corresponding to Exon B in horse as shown in FIG. **3** is indicated in bold.

(9) FIG. **9**. Comparison of that portion of the protein sequence of DYSF encoded by horse Exon B from wild type (XP_023474694.1, shown here as SEQ ID NO:19) and DYSF-R253W (R253W, shown here as SEQ ID NO:20) to the protein sequence of DYSF encoded by human Exon 7 (NP_003485.1, shown here as SEQ ID NO:18). Between the sequences of the horse and human proteins in the alignment, an asterisk (*) indicates an identical amino acid in that position, while a space ( ) indicates the nonconservative substitution of an arginine (R) in horse for a glycine (G) in human at human position 247, and a plus sign (+) indicates the conservative substitution of an arginine (R) in horse for a lysine (K) in human at human position 256. No other amino acid substitutions are seen in comparison of the human sequence (SEQ ID NO:18) and the wild-type horse sequence (SEQ ID NO:19). The sequence from horse bearing the DYSF-R253W mutation (SEQ ID NO:20) has a nonconservative substitution of a tryptophan (W) for an arginine (R) that is found in human (SEQ ID NO:18) and the wild-type horse sequence (SEQ ID NO:19). This position, corresponding to position 253 in horse (SEQ ID NO:19, SEQ ID NO:20) and position 251 in human (SEQ ID NO:18) is indicated by bold and underlining.

(10) FIG. **10**. Features of the dysferlin protein encoded by the human DYSF gene. (A) The canonical human isoform has 2080 amino acids. The protein contains seven C2 calcium-binding domains designated C2A through C2G, shaded in gray. The secondary structure of the protein sequence encoded by each of the C2 domains consists of eight segments assembling into beta sheet, with two different topologies described. Domains C2C, C2D, C2F, and C2G are designated as topology type I, while domains C2A, C2B, and C2E are designated as topology type II (C.Therrien et al. 2006 J. Neurological Sciences 250: 71-78). The positions of four additional conserved domains (DysfN, DysfC, annexin binding, and transmembrane) are also indicated (C.Therrien et al. 2006 J. Neurological Sciences 250: 71-78), also shaded in gray. (B) topology type II C2 calcium-binding domains C2B, C2A, and C2E are shown expanded, with positions of pathogenic missense alleles listed in TABLE 1 (C2B) and TABLE 2 (C2A and C2E) shown. The DYSF-R253W substitution in the horse C2B domain precisely corresponds to the DYSF-R251W substitution in human.

(11) FIG. **11**. Amino acid sequences of proteins encoded by the C2B domain of DYSF, including the position of the equine DYSF-R253W substitution. Species included in the analysis are described in the text. The next to the last line (labeled CLUSTAL) shows the consensus sequence, where positions with fully conserved amino acids are represented by an asterisk (*), positions with strongly conserved amino acids are indicated by a colon (:), positions with weakly conserved amino acids are indicated are indicated by period (.), and nonconserved positions are indicated by a blank space ( ). The last line shows the position of the DYSF-R253W substitution in horse in bold. The position of the DYSF-R253W substitution is indicated in bold in all of the aligned sequences.

(12) FIG. **12**. Sequence comparison of the C2A, C2B, and C2E domains of dysferlin encoded by human DYSF to the C2B domain of dysferlin encoded by horse DYSF. (A) The amino acid sequences of two isoforms of the C2A domain of human dysferlin (SEQ ID NO:51 and SEQ ID NO:52), the C2B domain of human dysferlin (SEQ ID NO:53), the C2E domain of human dysferlin (SEQ ID NO:54), and the C2B domain of horse dysferlin (SEQ ID NO:55), are shown. (B) Clustal Omega was used to align these four sequences. The fifth line (labeled CLUSTAL) shows the consensus sequence, where positions with fully conserved amino acids are represented by an asterisk (*), positions with strongly conserved amino acids are indicated by a colon (:), positions

with weakly conserved amino acids are indicated are indicated by period (.), and nonconserved positions are indicated by a blank space ( ). The remaining lines show the position of the horse DYSF-R253W substitution and various pathogenic human substitutions (presented in TABLE 1 and TABLE 2) aligned to the consensus.

(13) FIG. **13**. A portion of the current horse genome assembly (EquCab2, GCA_000002305.1) with coordinates as displayed in the UCSC Genome Browser centered on the chr15:31,225,630 position, the site of a substitution of a thymine (T) for a guanine (G) on the forward strand that results in the substitution of a threonine (T) for proline (P) at amino acid position 1290 in dysferlin as shown in FIG. **16** (SEQ ID NO:61). The reverse complement sequence is shown, with the site of a substitution of an adenine (A) for a cytosine (C) as indicated (SEQ ID NO:56). The single nucleotide polymorphism (SNP) defined by this base substitution is identified as rs1136366555 in dbSNP.

(14) FIG. **14**. A portion of the normal equine DYSF Coding DNA Sequence (SEQ ID NO:57) and a portion of the mutant DYSF Coding DNA Sequence (SEQ ID NO:58) bearing the C to A mutation at nucleotide position at position 4174 in this figure, corresponding to chr15:31,225,630 as shown in SEQ ID NO:56 (FIG. **13**). [This sequence is a region of perfect consensus among 21 different experimentally predicted mRNA isoforms. The numbering in FIG. **2** is that of isoform X1 (XM_023618907.1), which for this segment perfectly matches the numbering of isoforms X2 (XM_023618908.1), X3 (XM_023618909.1), X4 (XM_023618910.1), X5 (XM_023618911.1), and X6 (XM_023618912.1). The numbering of the start and end positions for other isoforms is described in detail in the text.] In both sequences, the sequence of Exon I as shown in FIG. **15** is indicated in bold. The site of a C to A mutation at nucleotide position 4174, corresponding to 31,225,630 in SEQ ID NO:56 (FIG. **13**), and to rs1136366555 in dbSNP, is underlined. The region of sequence comprising Exon I as shown in FIG. **15** is displayed as codons in the correct reading frame for both SEQ ID NO:57 and SEQ ID NO: 58 in FIG. **17**.

(15) FIG. **15**. A view of the current horse genome assembly (EquCab2, GCA_000002305.1) in the UCSC Genome Browser with exon sequences that match the partial DYSF Coding DNA Sequence (SEQ ID NO:57) and the mutant partial DYSF Coding DNA Sequence (SEQ ID NO:58). The DYSF Coding DNA Sequences correspond to Exons G, H, and I as indicated by the regions of sequence similarity of the translated genomic DNA to DYSF protein sequences from human (Homo), orangutan (Pongo), cattle (Bos), mouse (Mus), rat (Rattus) and DYSF protein sequences from the more distantly-related zebrafish (Danio). The sequence of Exon I with 10 bp of flanking intron sequence and its coordinates in the current horse assembly is displayed below the image from the UCSC Genome Browser (SEQ ID NO: 59).

(16) FIG. **16**. Models of part of the normal protein sequence encoded by horse DYSF (XP_023474694.1, presented here as SEQ ID NO:60) corresponding to a translation of SEQ ID NO:57 shown in FIG. **14** and part of the altered protein sequence encoded by horse DYSF with the base substitution at chr15:31,225,630 (based on XP_023474694.1, presented here as SEQ ID NO:61) corresponding to a translation of SEQ ID NO:58 shown in FIG. **14**. The portion of the protein encoded by Exon I as shown in FIG. **15** is indicated in bold, while amino acid position 1290 affected by the base substitution of a thymine (T) for a guanine (G) at the chr15:31,225,630 position on the forward strand, corresponding to rs1136366555 in dbSNP as shown in FIG. **13** (C to A in the reverse complement as shown), is underlined. The amino acid positions in) XP_023474694.1 are indicated at the beginning and end of the sequence.

(17) FIG. **17**. Horse DYSF Exon I and flanking genomic DNA sequence from which PCR primers to amplify genomic DNA containing the site of the DYSF-P1290T mutation would be most appropriately derived. Genomic coordinates are as in FIG. **13**. Exon I from chr15:31,225,648 to chr15:31,225,619 is shown broken into codons in the correct reading frame for the wild-type allele (SEQ ID NO:62) and the DYSF-P1290T allele (SEQ ID NO:63). Only the reference sequence from the assembly is shown for the flanking sequences. The codon affected by the G to T mutation site at

nucleotide position chr15:31,225,630 on the forward strand, corresponding to rs1136366555 in dbSNP as shown in FIG. **13** (C to A in the reverse complement as shown), is shown in bold, with the position of the base substitution indicated by underlining. The base substitution changes the bold three base codon from one coding for a proline (CCT) to one coding for a threonine (ACT). Example primers used experimentally to amplify genomic DNA containing the mutation site are shown in lower case [5′-GGTTGCAAACTCCCAACTGT-3′ (SEQ ID NO:64) and 5 GATTTTTCAAGCTGCCGAAG-3′ (SEQ ID NO:65)].
(18) FIG. **18**. Traces from Sanger DNA sequencing of amplified DYSF genomic DNA using primers shown in FIG. **17** (SEQ ID NO:64 and SEQ ID NO:65). The sequence of the forward strand is shown. The arrows in the figure indicate nucleotide position chr15:31,225,630, the site of a substitution of an thymine (T) for a guanine (G) in this position, corresponding to rs1136366555 in dbSNP, that creates the DYSF-P1290T variant. The traces show, from left to right, results for a horse homozygous for the wild-type or common allele (N/N), results for a horse heterozygous for the substitution (N/P6), and results for a horse homozygous for the substitution (P6/P6).
(19) FIG. **19**. Comparison of that portion of the protein sequence of DYSF encoded by horse Exon I from wild type (XP_023474694.1, shown here as SEQ ID NO:67) and DYSF-P1290T (P1290T, shown here as SEQ ID NO:68) to the protein sequence of DYSF encoded by human (NP_003485.1, shown here as SEQ ID NO:66). Between the sequences of the horse and human proteins in the alignment, an asterisk (*) indicates an identical amino acid in that position, while a space ( ) indicates the nonconservative substitution of a glycine (G) in horse for an arginine (R) in human at human position 1297, and a plus sign (+) indicates the conservative substitution of an alanine (A) in horse for a serine (S) in human at human position 1267, a tyrosine (Y) in horse for a histidine (H) in human position 1285, an aspartic acid (D) in horse for a glutamic acid (E) in human position 1294, and a glutamic acid (E) in horse for an aspartic acid (D) in human position 1300. No other amino acid substitutions are seen in comparison of the human sequence (SEQ ID NO:66) and the wild-type horse sequence (SEQ ID NO:67). The sequence from horse bearing the DYSF-P1290T mutation (SEQ ID NO:68) has a nonconservative substitution of a threonine (T) for a proline (P) that is found in human (SEQ ID NO:66) and the wild-type horse sequence (SEQ ID NO:67). This position, corresponding to position 1290 in horse (SEQ ID NO:67, SEQ ID NO:68) and position 1288 in human (SEQ ID NO:66) is indicated by bold and underlining.
(20) FIG. **20**. Features of the dysferlin protein encoded by the human DYSF gene. (A) The canonical human isoform has 2080 amino acids. The protein contains seven C2 calcium-binding domains designated C2A through C2G, shaded in gray. The positions of four additional conserved domains (DysfN, DysfC, annexin binding, and transmembrane) are also indicated (C.Therrien et al. 2006 J. Neurological Sciences 250: 71-78), also shaded in gray. The interdomain region between the C2D and C2E domains, affected by the horse DYSF-P1290T mutation, is indicated in light gray. (B) The interdomain region between the C2D and C2E domains is shown expanded, with positions of pathogenic and potentially pathogenic missense alleles listed in TABLE 3 shown. The horse DYSF-P1290T substitution corresponds to a proline at position 1288 in human, with no pathogenic or potentially pathogenic allele identified at that position in human.
(21) FIG. **21**. Amino acid sequences of proteins encoded by the C2D-C2E interdomain region of DYSF, including the position of the equine DYSF-P1290T substitution. Species included in the analysis are described in the text. The next to the last line (labeled CLUSTAL) shows the consensus sequence, where positions with fully conserved amino acids are represented by an asterisk (*), positions with strongly conserved amino acids are indicated by a colon (:), positions with weakly conserved amino acids are indicated are indicated by period (.), and nonconserved positions are indicated by a blank space ( ). The last line shows the position of the DYSF-P1290T substitution in horse in bold. The position of the DYSF-P1290T substitution is indicated in bold in all of the aligned sequences.
(22) FIG. **22**. A portion of the current horse genome assembly (EquCab2, GCA_000002305.1) with

coordinates as displayed in the UCSC Genome Browser centered on the chr6:47,661,977 position, the site of a substitution of a cytosine (C) for a guanine (G) on the forward strand that results in the substitution of a histidine (H) for an aspartate (D) at amino acid position 492 in pyridine nucleotide-disulfide oxidoreductase domain-containing protein 1 (PYROXD1) as shown in FIG. **25** (SEQ ID NO:108). The single nucleotide polymorphism (SNP) defined by this base substitution is identified as rs1136260157 in dbSNP.

(23) FIG. **23**. A portion of the normal equine PYROXD1 Coding DNA Sequence (SEQ ID NO:104) and a portion of the mutant PYROXD1 Coding DNA Sequence (SEQ ID NO:105) bearing the G to C mutation at nucleotide position 1548 in this figure, corresponding to chr6:47,661,977 as shown in SEQ ID NO:103 (FIG. **22**). This sequence is a region of perfect consensus among five different experimentally predicted mRNA isoforms. The numbering in FIG. **23** is that of isoform X1 (XM_001502130.5). The numbering of the start and end positions for other isoforms is described in detail in the text. In both sequences the sequence of Exon 12 as shown in FIG. **24** is indicated in bold. The site of a G to C mutation at nucleotide position 1548, corresponding to 47,661,977 in SEQ ID NO:103 (FIG. **22**), and to rs1136260157 in dbSNP, is underlined. The region of sequence comprising Exon 12 as shown in FIG. **24** is displayed as codons in the correct reading frame for both SEQ ID NO:109 and SEQ ID NO:110 in FIG. **26**.

(24) FIG. **24**. A view of the current horse genome assembly (EquCab2, GCA_000002305.1) in the UCSC Genome Browser with exon sequences that match the PYROXD1 Coding DNA Sequence (SEQ ID NO:104) and the mutant PYROXD1 Coding DNA Sequence (SEQ ID NO:105) in a BLAT search. The PYROXD1 Coding DNA Sequences in horse correspond to PYROXD1 coding sequences in other species as indicated by the regions of sequence similarity of the translated genomic DNA to PYROXD1 protein sequences from human (Homo), orangutan (Pongo), cattle (Bos), mouse (Mus), rat (Rattus), African clawed frog (Xenopus), zebrafish (Danio), chicken (Gallus), and fruit fly (Drosophila). Exon 12, which contains the guanine (G) to cytosine (C) variant at chr6:47,661,977 as shown in SEQ ID NO:103 (FIG. **22**), is indicated below the image of the browser window. The sequence of horse Exon 12 with 10 nucleotides of flanking intron sequence and its coordinates in the current horse assembly is displayed below the image from the UCSC Genome Browser (SEQ ID NO:106).

(25) FIG. **25**. Models of part of the normal protein sequence encoded by horse PYROXD1 (XP_001502180.3, presented here as SEQ ID NO:107) corresponding to a translation of SEQ ID NO:104 shown in FIG. **23** and part of the altered protein sequence encoded by horse PYROXD1 (adapted from XP_001502180.3, presented here as SEQ ID NO:108) with the base substitution at chr6:47,661,977 corresponding to a translation of SEQ ID NO:105 shown in FIG. **23**. The portion of the protein encoded by Exon 12 is indicated in bold, while amino acid position 492 affected by the base substitution of a cytosine (C) for a guanine (G) at the chr6:47,661,977 position on the forward strand as shown in FIG. **22**, is underlined. The amino acid positions in XP_001502180.3 are indicated at the beginning and end of the sequence.

(26) FIG. **26**. Horse PYROXD1 Exon 12 and flanking genomic DNA sequence from which PCR primers to amplify genomic DNA containing the site of the PYROXD1-D492H mutation would be most appropriately derived. Genomic coordinates are as in FIG. **22**. Exon 12 from chr6:47,661,764 to chr6:47,662,012 is shown broken into codons in the correct reading frame for the wild-type allele (SEQ ID NO:109) and the PYROXD1-D492H allele (SEQ ID NO:110). Only the reference sequence from the assembly is shown for the flanking sequences. The codon affected by the G to C mutation site at nucleotide position chr6:47,661,977, as shown in FIG. **22** is shown in bold, with the position of the base substitution indicated by underlining. The base substitution changes the bold three base codon from one coding for an aspartate (GAT) to one coding for a histidine (CAT). Example primers used experimentally to amplify genomic DNA containing the mutation site [5′-CAGATTTTCTGCTGGCCATT-3′ (SEQ ID NO:111) and 5′-TGGTCATCATTAAATCAGTGCAA-3′ (SEQ ID NO:112)] are shown in lower case in the figure.

(27) FIG. **27**. Traces from Sanger DNA sequencing of amplified PYROXD1 genomic DNA using primers shown in FIG. **26** (SEQ ID NO:111 and SEQ ID NO:112). The sequence of the forward strand is shown. The arrows in the figure indicate nucleotide position chr6:47,661,977, the site of a substitution of a cytosine (C) for a guanine (G) in this position that creates the PYROXD1-D492H variant. The traces show, from left to right, results for a horse homozygous for the wild-type or common allele (n/n), results for a horse heterozygous for the substitution (n/P8), and results for a horse homozygous for the substitution (P8/P8).
(28) FIG. **28**. Partial sequence of the human PYROXD1 coding sequence derived from NM_024854.4 (SEQ ID NO:113). This sequence is a region of perfect consensus among three different experimentally predicted mRNA isoforms. The numbering of the first and last nucleotides corresponds to that of NM_024854.4. The numbering of the start and end positions for other isoforms is described in the text. The sequence of Exon 12 is indicated in bold. The sequence begins with beginning of the consensus among the three isoforms and ends with the TAA stop codon.
(29) FIG. **29**. Partial sequence of the human PYROXD1 protein sequence, showing a translation of SEQ ID NO:113, equivalent to NP_079130.2 (SEQ ID NO:66). The numbering of the first and last amino acids corresponds to that of NP_079130.2.
(30) FIG. **30**. Comparison of that portion of the protein sequence of PYROXD1 encoded by horse Exon 12 from wild type (XP_001502180.3, shown here as SEQ ID NO:116) and PYROXD1-D492H (D492H, derived from XP_001502180.3 and shown here as SEQ ID NO:117) to the protein sequence of PYROXD1 encoded by human Exon 12 (NP_079130.2, shown here as SEQ ID NO:115). Between the sequences of the horse and human proteins in the alignment, an asterisk (*) indicates an identical amino acid in that position, while a plus sign (+) indicates the following conservative substitutions: a glutamine (Q) for an arginine (R) at human position 444, a valine (V) for an isoleucine (I) at human position 447, an alanine (A) for a serine (S) at human position 483, and an aspartate (D) for an asparagine (N) at human position 492. The sequence from horse bearing the PYROXD1-D492H mutation (SEQ ID NO:117) has a nonconservative substitution of a histidine (H) for an aspartate (D) at horse position 492, corresponding to human position 490. This position is indicated in bold for all three sequences. The wild-type horse sequence (SEQ ID NO:116) matches the human sequence (SEQ ID NO:115) at this position.
(31) FIG. **31**. Features of the pyridine nucleotide-disulfide oxidoreductase domain-containing protein 1 protein encoded by the human PYROXD1 gene. The human protein has 500 amino acids and two domains: the pyridine nucleotide-disulfide oxidoreductase domain (amino acids 39-361) and the NADH-dependent nitrite reductase domain (447-494) as described in O'Grady et al. 2016 Am J Hum Genet. 99:1086-1105. Positions of two human pathogenic missense alleles (N1555 and Q372H, O'Grady et al. 2016 Am J Hum Genet. 99:1086-1105 DOI: 10.1016/j.ajhg.2016.09.005) are shown, as is the position of the horse D492H mutation described herein.
(32) FIG. **32**. Partial amino acid sequences of proteins encoded by PYROXD1, including the position of the equine PYROXD1-D492H substitution. Species included in the analysis are described in the text. The next to the last line (labeled CLUSTAL) shows the consensus sequence, where positions with fully conserved amino acids are represented by an asterisk (*), positions with strongly conserved amino acids are indicated by a colon (:), positions with weakly conserved amino acids are indicated are indicated by period (.), and nonconserved positions are indicated by a blank space ( ). The last line shows the position of the PYROXD1-D492H substitution in horse in bold. The position of the PYROXD1-D492H substitution is indicated in bold in all of the aligned sequences.
(33) FIG. **33**. A portion of the current horse genome assembly (EquCab2, GCA_000002305.1) with coordinates as displayed in the UCSC Genome Browser centered on the chr6:23,480,621 position, the site of a substitution of a guanine (G) for a cytosine (C) on the forward strand that results in the substitution of an alanine (A) for a glycine (G) at amino acid position 2182 in collagen type VI

alpha 3 chain (COL6A3) as shown in FIG. **36** (SEQ ID NO:168). The reverse complement sequence is shown, with the site of a substitution of a cytosine (C) for a guanine (G) on the reverse strand as indicated (SEQ ID NO:163). The single nucleotide polymorphism (SNP) defined by this base substitution is identified as rs1139437410 in dbSNP.

(34) FIG. **34**. A portion of the normal equine COL6A3 Coding DNA Sequence (SEQ ID NO:164) and a portion of the mutant COL6A3 Coding DNA Sequence (SEQ ID NO:165) bearing the G to C mutation at nucleotide position 6792 in this figure, corresponding to chr6:23,480,621 as shown in SEQ ID NO:163 (FIG. **33**). The numbering in FIG. **34** is that of the COL6A3 coding sequence (CDS) model XM_023642645.1; the sequence presented comprises the coding sequence for the five collagen-like domains in the middle of the protein. See the text for a discussion of the CDS model in NCBI. In both sequences the sequence of the third collagen-like domain, partially encoded by Exon 26, as shown in FIG. **35** is indicated in bold. The site of a G to C mutation at nucleotide position 6792, corresponding to 23,480,621 in SEQ ID NO:163 (FIG. **33**), and to rs1139437410 in dbSNP, is underlined. The region of sequence comprising Exon 26 is displayed as codons in the correct reading frame for both SEQ ID NO:164 and SEQ ID NO:165 in FIG. **37**.

(35) FIG. **35**. A view of the current horse genome assembly (EquCab2, GCA_000002305.1) in the UCSC Genome Browser with exon sequences that match the COL6A3 Coding DNA Sequence (SEQ ID NO:164) and the mutant COL6A3 Coding DNA Sequence (SEQ ID NO:165) in a BLAT search. The COL6A3 Coding DNA Sequences in horse correspond to COL6A3 coding sequences in other species as indicated by the regions of sequence similarity of the translated genomic DNA to COL6A3 protein sequences from human (Homo), mouse (Mus), and dog (Canus). Exon 26, which contains the guanine (G) to cytosine (C) variant at chr6:23,480,621 as shown in SEQ ID NO:163 (FIG. **33**), is indicated below the image of the browser window. See the text for a discussion of the failure to detect sequence similarity of the translation of Exon 26 to protein sequences from other species. The sequence of horse Exon 26 with 10 nucleotides of flanking intron sequence and its coordinates in the current horse assembly is displayed below the image from the UCSC Genome Browser (SEQ ID NO:166).

(36) FIG. **36**. Models of part of the normal protein sequence encoded by horse COL6A3 (XP_023498413.1, presented here as SEQ ID NO:167) corresponding to a translation of SEQ ID NO:164 shown in FIG. **34** and part of the altered protein sequence encoded by horse COL6A3 (adapted from XP_023498413.1, presented here as SEQ ID NO:168) with the base substitution at chr6:23,480,621 corresponding to a translation of SEQ ID NO:165 shown in FIG. **34**. The portion of the protein corresponding to all five collagen-like domains is shown; the portion corresponding to the third collagen-like domain, partially encoded by Exon 26, is indicated in bold, while the amino acid at position 2182 affected by the base substitution of a cytosine (C) for a guanine (G) at the chr6:23,480,621 position as shown in FIG. **33**, is underlined. The amino acid positions in XP_023498413.1 are indicated at the beginning and end of the sequence.

(37) FIG. **37**. Horse COL6A3 Exon 26 and flanking genomic DNA sequence from which PCR primers to amplify genomic DNA containing the site of the COL6A3-G2182A mutation would be most appropriately derived. Genomic coordinates are as in FIG. **33**. Exon 26 from chr6:23,480,631 to chr6:23,480,578 is shown broken into codons in the correct reading frame for the wild-type allele (SEQ ID NO:169) and the COL6A3-G2182A allele (SEQ ID NO:170). Only the reference sequence from the assembly is shown for the flanking sequences. The codon affected by the C to G mutation site at nucleotide position chr6:23,480,621 on the forward strand, corresponding to rs1139437410 in dbSNP as shown in FIG. **33** (G to C in the reverse complement as shown), is shown in bold, with the position of the base substitution indicated by underlining. The base substitution changes the bold three base codon from one coding for a glycine (GGG) to one coding for an alanine (GCG). Example primers used experimentally to amplify genomic DNA containing the mutation site [5′-AGATGGGGCACAGATCAAAC-3′ (SEQ ID NO:172) and 5′-TTCCCAGACTCTCCTGTGCT-3′ (SEQ ID NO:171)] are shown in lower case in the figure.

(38) FIG. **38**. Traces from Sanger DNA sequencing of amplified COL6A3 genomic DNA using primers shown in FIG. **37** (SEQ ID NO:171 and SEQ ID NO:172). The sequence of the forward strand is shown. The arrows in the figure indicate nucleotide position chr6:23,480,621, the site of a substitution of a cytosine (C) for a guanine (G) in this position on the reverse strand that creates the COL6A3-G2182A variant. The traces show, from left to right, the sequence of the forward strand for a horse homozygous for the wild-type or common allele (n/n), and results for a horse heterozygous for the substitution (n/K1).
(39) FIG. **39**. Partial sequence of the human COL6A3 coding sequence derived from NM_004369.3 (SEQ ID NO:173). This sequence is a region of perfect consensus among multiple different experimentally predicted mRNA isoforms. The numbering of the first and last nucleotides corresponds to that of NM_004369.3. The sequence encoding the third collagen-like domain is indicated in bold.
(40) FIG. **40**. Partial sequence of the human COL6A3 protein sequence, showing a translation of SEQ ID NO:173, equivalent to NP_004360.2 (SEQ ID NO:174). The numbering of the first and last amino acids corresponds to that of NP_004360.2. The sequence of the third collagen-like domain is indicated in bold.
(41) FIG. **41**. Comparison of the portion of the protein sequence of COL6A3 comprising the five collagen-like repeats from human (NP_004360.2, shown here as SEQ ID NO:175) and horse (XP_023498413.1, shown here as SEQ ID NO:176). The position of the COL6A3-G2182A substitution is shown below the horse sequence. Between the sequences of the horse and human proteins in the alignment, an asterisk (*) indicates an identical amino acid in that position, while a plus sign (+) indicates a conservative substitution, and a space ( ) indicates a nonconservative substitution. The positions of glycine residues that are part of the Gly-X-Y structure of the collagen triple helix are indicated by an asterisk (*) in reverse text (white text on a black background) between the horse and human sequences. All of these glycine residues are conserved between human and horse. The horse G2182A sequence has a nonconservative substitution of an alanine (A) for a glycine (G) at position 2182; the wild-type horse sequence matches the human sequence at this position.
(42) FIG. **42**. Features of the collagen type VI alpha 3 chain protein encoded by the human COL6A3 gene. (A) The human protein has 3,177 amino acids. It contains five collagen-like domains consisting of Gly-X-Y repeats, labeled as Collagen 1-5 in the figure. Toward the N-terminus of the collagen-like repeats are 10 Von Willebrand Factor-like repeats, labeled as VWFA 1-10 in the figure. Toward the C-terminus of the collagen-like repeats are two additional Von Willebrand Factor-like repeats, labeled as VWFA 11 and 12 in the figure. Near the C-terminus is a fibronectin-like repeat, labeled as Fibronectin in the figure, and a Protease Inhibitor domain (Pancreatic trypsin inhibitor Kunitz domain). (B) The portion of the protein containing the five collagen-like domains extends from position 2038 to 2373 of the human protein. The five collagen-like domains are numbered. Positions of 19 pathogenic human alleles listed in TABLE 4 are indicated above the segment shown, as is the position of the horse G2182A variant. The position of the cysteine involved in the interchain disulfide bond (C2087S-S) is indicated below the segment shown.
(43) FIG. **43**. Partial amino acid sequences of proteins encoded by COL6A3, including the position of the equine COL6A3-G2182A substitution. Species included in the analysis are described in the text. The next to the last line (labeled CLUSTAL) shows the consensus sequence, where positions with fully conserved amino acids are represented by an asterisk (*), positions with strongly conserved amino acids are indicated by a colon (:), positions with weakly conserved amino acids are indicated are indicated by period (.), and nonconserved positions are indicated by a blank space ( ). The last line shows the position of the COL6A3-G2182A substitution in horse in bold. The position of the COL6A3-G2182A substitution is indicated in bold in all of the aligned sequences.
(44) FIG. **44**. An Arabian horse (E682) homozygous for the PYROXD1-D492H substitution

(P8/P8), heterozygous for MYOT-S232P (n/P2), and homozygous for the wild-type alleles of FLNC-E753K (P3a), FLNC-A1207T (P3b), MYOZ3-S42L (P4), reported in a prior disclosure. Muscle wasting in the pelvic girdle (hindquarters), shoulder girdle (topline), and proximal limbs is evident. The horse is pregnant in the photo. The horse was reported to be symptomatic with gait abnormalities ("rope walking"), and died of a choking incident. Difficulty in swallowing is observed in human patients with Myofibrillar Myopathy 8 (MFM8) due to mutations in PYROXD1 (O'Grady et al. 2016 Am J Hum Genet. 99:1086-1105 DOI: 10.1016/j.ajhg.2016.09.005).

DETAILED DESCRIPTION OF ILLUSTRATIVE EMBODIMENTS

(45) This disclosure describes methods for detecting the presence or absence of a biomarker in horses associated with a defect of muscle integrity that causes a form of inherited exercise intolerance. This disease condition has been previously described as Polysaccharide Storage Myopathy, type 2 (PSSM2). Another previously described form of exercise intolerance in horses is Polysaccharide Storage Myopathy, type 1 (PSSM1), caused by a semidominant allele of glycogen synthase (GYS1-R309H). The term PSSM2 is commonly used to describe horses that show exercise intolerance, a negative test result for the GYS1-R309H variant of Glycogen Synthase 1 that is associated with Polysaccharide Storage Myopathy, type 1 (PSSM1), and abnormal findings on muscle biopsy, including abnormally shaped muscle fibers, nuclei displaced to the center of muscle fibers rather than the normal position at the edge of fibers, and pools of glycogen granules of normal size in regions of disorganization that give the false appearance of a glycogen storage disease. In one embodiment, the method involves obtaining a physiological sample from a horse and determining whether the biomarker is present in the sample. As used herein, the phrase "physiological sample" refers to a biological sample obtained from a horse that contains nucleic acid. For example, a physiological sample can be a sample collected from an individual horse such as, for example, a cell sample, such as a blood cell, e.g., a lymphocyte, a peripheral blood cell; a sample collected from the spinal cord; a tissue sample such as cardiac tissue or muscle tissue, e.g., cardiac or skeletal muscle; an organ sample, e.g., liver or skin; a hair sample, e.g., a hair sample with roots; and/or a fluid sample, such as blood.

(46) Examples of breeds of affected horse include, but are not limited to, Shires, Clydesdales, Percheron Horses, Belgian Horses, Draft Horses, Quarter Horses, Paint Horses, Warmblood Horses, or related or unrelated breeds. The phrase "related breed" is used herein to refer to breeds that are related to a breed, such as Quarter Horse, Draft Horse, or Warmblood Horse. Such breeds include, but are not limited to stock breeds such as the American Paint horse, the Appaloosa, and the Palomino. The term "Draft Horse" includes many breeds including but not limited to Clydesdale, Belgian, Percheron, and Shire horses. The term "Warmblood" is also a generic term that includes a number of different breeds. "Warmblood" simply distinguishes this type of horse from the "cold bloods" (draft horses) and the "hot bloods" (Thoroughbreds and Arabians). The method described herein also may be performed using a sample obtained from a crossed or mixed breed horse.

(47) The term "biomarker" generally refers herein to a biological indicator, such as a particular molecular feature, that may affect, may be an indicator, and/or be related to diagnosing or predicting an individual's health. For example, in certain embodiments, the biomarker can refer to (1) a mutation in the equine dyferlin (DYSF) coding region (SEQ ID NO:1), such as a polymorphic allele of DYSF that has a substitution of an adenine (A) for a guanine (G) on the forward strand at chr15:31,306,949 of the current horse genome assembly (EquCab2, GCA_000002305.1) as displayed in the UCSC Genome Browser and as shown for the reverse strand in FIG. **1**, (2) a mutation in the equine dyferlin (DYSF) coding region (SEQ ID NO:56), a polymorphic allele of DYSF that has a substitution of a thymine (T) for a guanine (G) on the forward strand at chr15:31,225,630 of the current horse genome assembly (EquCab2, GCA_000002305.1) as displayed in the UCSC Genome Browser and as shown for the reverse strand in FIG. **13**, (3) a mutation in the equine pyridine nucleotide-disulfide oxidoreductase domain-containing protein 1

(PYROXD1) coding region (SEQ ID NO:103), such as a polymorphic allele of PYROXD1 that has a substitution of a cytosine (C) for a guanine (G) at chr6:47,661,977 on the forward strand of the current horse genome assembly (EquCab2, GCA_000002305.1) as displayed in the UCSC Genome Browser and as shown in FIG. **22**, or (4) a mutation in the equine collagen type VI alpha 3 chain (COL6A3) coding region (SEQ ID NO:163), such as a polymorphic allele of COL6A3 that has a substitution of a guanine (G) for a cytosine (C) on the forward strand at chr6:23,480,621 of the current horse genome assembly (EquCab2, GCA_000002305.1) as displayed in the UCSC Genome Browser and as shown for the reverse strand in FIG. **33**. The specified nucleotide substitution may be inferred by the detection of the complementary base on the reverse strand.

(48) “Oligonucleotide probe” can refer to a nucleic acid segment, such as a primer, that is useful to amplify a sequence in the DYSF, PYROXD1, or COL6A3 coding regions that is complementary to, and hybridizes specifically to, a particular nucleotide sequence in DYSF, PYROXD1, or COL6A3, or to a nucleic acid region that flanks DYSF, PYROXD1, or COL6A3.

(49) As used herein, the term “nucleic acid” and “polynucleotide” refers to deoxyribonucleotides or ribonucleotides and polymers thereof in either single-stranded or double-stranded form. Unless specifically limited, the term encompasses nucleic acids containing known analogs of natural nucleotides that have similar binding properties as the reference nucleic acid and are metabolized in a manner similar to naturally occurring nucleotides. Unless otherwise indicated, a particular nucleic acid sequence also implicitly encompasses conservatively modified variants thereof (e.g., degenerate codon substitutions) and complementary sequences as well as the sequence explicitly indicated. Specifically, degenerate codon substitutions may be achieved by generating sequences in which the third position of one or more selected (or all) codons is substituted with mixed-base and/or deoxyinosine residues.

(50) A “nucleic acid fragment” is a portion of a given nucleic acid molecule. Deoxyribonucleic acid (DNA) in the majority of organisms is the genetic material while ribonucleic acid (RNA) is involved in the transfer of information contained within DNA into proteins. The term “nucleotide sequence” refers to DNA or RNA that can be single-stranded or double-stranded, optionally containing synthetic, non-natural, or altered nucleotide bases capable of incorporation into DNA or RNA.

(51) In some embodiments, the method can involve contacting the sample with at least one oligonucleotide probe to form a hybridized nucleic acid and then amplifying the hybridized nucleic acid. “Amplifying” utilizes methods such as the polymerase chain reaction (PCR), ligation amplification (or ligase chain reaction, LCR), strand displacement amplification, nucleic acid sequence-based amplification, and amplification methods based on the use of Qβ-replicase. These methods are well known and widely practiced in the art. Reagents and hardware for conducting PCR are commercially available. For example, in certain embodiments, (1) Exon B of the equine dysferlin coding region (also referred to as DYSF) as shown in FIG. **3** or portions thereof, (2) Exon I of the equine dysferlin coding region (also referred to as DYSF) as shown in FIG. **15** or portions thereof, (3) Exon 12 of the equine pyridine nucleotide-disulfide oxidoreductase domain-containing protein 1 coding region (also referred to as PYROXD1) as shown in FIG. **24** or portions thereof, or (4) Exon 26 of the equine collagen type VI alpha 3 chain coding region (also referred to as COL6A3) as shown in FIG. **35** or portions thereof, may be amplified by PCR. In another embodiment, at least one oligonucleotide probe is immobilized on a solid surface or a semisolid surface.

(52) The methods described herein can be used to detect the presence or absence of a biomarker associated with equine Polysaccharide Storage Myopathy type 2 (PSSM2) in a horse (live or dead) regardless of age (e.g., an embryo, a foal, a neonatal foal, aborted foal, a breeding-age adult, or any horse at any stage of life) or sex (e.g., a mare (dam) or stallion (sire)).

(53) As used herein, the term “presence or absence” refers to affirmatively detecting the presence of a biomarker or detecting the absence of the biomarker within the experimental limits of the

detection methods used to detect the biomarker.

(54) This disclosure further provides a method for detecting and/or diagnosing Polysaccharide Storage Myopathy type 2 (PSSM2), also referred to as Myofibrillar Myopathy (MFM), in a horse, the method involving obtaining a physiological sample from the horse and detecting the presence or absence of biomarkers in the sample, wherein the presence of the biomarkers is indicative of the disease. One embodiment of the method further involves contacting the sample with at least one oligonucleotide probe to form a hybridized nucleic acid and amplifying the hybridized nucleic acid. For example, in four embodiments, (1) Exon B of equine DYSF as shown in FIG. **3**, (2) Exon I of equine DYSF as shown in FIG. **15**, (3) Exon 12 of equine PYROXD1 as shown in FIG. **24**, or (4) Exon 26 of equine COL6A3 as shown in FIG. **35**, are amplified using, for example, polymerase chain reaction, strand displacement amplification, ligase chain reaction, amplification methods based on the use of QP-replicase and/or nucleic acid sequence-based amplification. In these embodiments of the method, the biomarkers can include (1) an equine dysferlin (DYSF) coding region having the specific substitution of an adenine (A) for a guanine (G) on the forward strand at chr15:31,306,949 of the current horse genome assembly (EquCab2, GCA_000002305.1) as displayed in the UCSC Genome Browser and as shown for the reverse strand in FIG. **1**, (2) an equine dysferlin (DYSF) coding region having the specific substitution of a thymine (T) for a guanine (G) on the forward strand at chr15:31,225,630 of the current horse genome assembly (EquCab2, GCA_000002305.1) as displayed in the UCSC Genome Browser and as shown for the reverse strand in FIG. **13**, (3) an equine pyridine nucleotide-disulfide oxidoreductase domain-containing protein 1 (PYROXD1) coding region having the specific substitution of a cytosine (C) for a guanine (G) on the forward strand at chr6:47,661,977 of the current horse genome assembly (EquCab2, GCA_000002305.1) as displayed in the UCSC Genome Browser and as shown in FIG. **22**, or (4) an equine collagen type VI alpha 3 chain (COL6A3) coding region having the specific substitution of a guanine (G) for a cytosine (C) on the forward strand at chr6:23,480,621 of the current horse genome assembly (EquCab2, GCA_000002305.1) as displayed in the UCSC Genome Browser and as shown for the reverse strand in FIG. **33**. Biomarkers can also include (1) a coding region that encodes a dysferlin (DYSF) polypeptide (SEQ ID NO:10) having an arginine-to-tryptophan (R to W) substitution at amino acid residue 253 of SEQ ID NO:10, as shown in SEQ ID NO:11, (2) a coding region that encodes a dysferlin (DYSF) polypeptide (SEQ ID NO:60) having an proline-to-threonine (P to T) substitution at amino acid residue 1290 of SEQ ID NO:60, as shown in SEQ ID NO:61, (3) a pyridine nucleotide-disulfide oxidoreductase domain-containing protein 1 (PYROXD1) polypeptide (SEQ ID NO:107) having an aspartate-to-histidine (D to H) substitution at amino acid residue 492 of SEQ ID NO:107, as shown in SEQ ID NO:108, or (4) a collagen type VI alpha 3 chain (COL6A3) polypeptide (SEQ ID NO:167) having an glycine-to-alanine (G to A) substitution at amino acid residue 2182 of SEQ ID NO:167, as shown in SEQ ID NO:168. The method can be used to detect Polysaccharide Storage Myopathy type 2 (PSSM2), also known as Myofibrillar Myopathy (MFM), in a horse.

(55) This disclosure further provides a kit that includes a test for diagnosing and/or detecting the presence of equine Polysaccharide Storage Myopathy type 2 (PSSM2), also known as Myofibrillar Myopathy (MFM), in a horse. The kit generally includes packing material containing, separately packaged, at least one oligonucleotide probe capable of forming hybridized nucleic acids with DYSF, PYROXD1, or COL6A3 and instructions directing the use of the probe in accord with the methods described herein.

(56) Horses affected with Polysaccharide Storage Myopathy type 2 (PSSM2) are either heterozygous or homozygous for the affected DYSF, PYROXD1, or COL6A3 alleles. An “allele” is a variant form of a particular genomic nucleic acid sequence. In the context of the methods described herein, some alleles of the DYSF, PYROXD1, or COL6A3 coding regions cause Polysaccharide Storage Myopathy type 2 (PSSM2) in horses. A “DYSF, PYROXD1, or COL6A3 allele,” refers to a normal allele of the DYSF, PYROXD1, or COL6A3 locus as well as an allele

carrying one or more variations that predispose a horse to develop Polysaccharide Storage Myopathy type 2 (PSSM2), also known as Myofibrillar Myopathy (MFM). The coexistence of multiple alleles at a locus is known as "genetic polymorphism." Any site at which multiple alleles exist as stable components of the population is by definition "polymorphic." An allele is defined as polymorphic if it is present at a frequency of at least 1% in the population. A "single nucleotide polymorphism (SNP)" is a DNA sequence variation that involves a change in a single nucleotide.
(57) The methods described herein involve the use of isolated or substantially purified nucleic acid molecules. An "isolated" or "purified" nucleic acid molecule is one that, by human intervention, exists apart from its native environment and is therefore not a product of nature. An isolated nucleic acid molecule may exist in a purified form or may exist in a non-native environment. For example, an "isolated" or "purified" nucleic acid molecule, or portion thereof, is substantially free of other cellular material, or culture medium when produced by recombinant techniques, or substantially free of chemical precursors or other chemicals when chemically synthesized. In one embodiment, an "isolated" nucleic acid is free of sequences that naturally flank the nucleic acid (i.e., sequences located at the 5′ and 3′ ends of the nucleic acid) in the genomic DNA of the organism from which the nucleic acid is derived. For example, in various embodiments, the isolated nucleic acid molecule can contain less than about 5 kb, 4 kb, 3 kb, 2 kb, 1 kb, 0.5 kb, or 0.1 kb of nucleotide sequences that naturally flank the nucleic acid molecule in genomic DNA of the cell from which the nucleic acid is derived. An isolated or purified nucleic acid molecule can be a fragment and/or variant of a reference nucleotide sequence expressly disclosed herein.
(58) A "fragment" or "portion" of a sequence refers to anything less than full-length of the nucleotide sequence encoding—or the amino acid sequence of—a polypeptide. As it relates to a nucleic acid molecule, sequence, or segment when linked to other sequences for expression, a "portion" or a "fragment" refers to a sequence having, for example, at least 80 nucleotides, at least 150 nucleotides, or at least 400 nucleotides. Alternatively, when not employed for expressing—e.g., in the context of a probe or a primer—a "portion" or a "fragment" means, for example, at least 9, at least 12, at least 15, or at least 20 consecutive nucleotides. Alternatively, a fragment or a portion of a nucleotide sequence that is useful as a hybridization probe generally does not encode fragment proteins retaining biological activity. Thus, fragments or portions of a nucleotide sequence may range from at least about 6 nucleotides, about 9, about 12 nucleotides, about 20 nucleotides, about 50 nucleotides, about 100 nucleotides, or more.
(59) A "variant" of a molecule is a sequence that is substantially similar to the sequence of the reference—e.g., native, naturally-occurring, and/or wild-type—molecule. For nucleotide sequences, a variant includes any nucleotide sequence that, because of the degeneracy of the genetic code, encodes the native amino acid sequence of a protein. Naturally occurring allelic variants such as these can be identified with the use of well-known molecular biology techniques, as, for example, with polymerase chain reaction (PCR) and/or hybridization techniques. A variant nucleotide sequence also can include a synthetically-derived nucleotide sequence such as one generated, for example, by using site-directed mutagenesis that encodes the native protein, as well as variant nucleotide sequences that encode a polypeptide having amino acid substitutions. Generally, a nucleotide sequence variant will have at least 40%, at least 50%, at least 60%, at least 70% (e.g., 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%), at least 80% (e.g., 81% 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%), or at least 90% (e.g., 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%) sequence identity to the native (endogenous) nucleotide sequence.
(60) "Synthetic" polynucleotides are those prepared by chemical synthesis.
(61) "Recombinant DNA molecule" is a combination of DNA sequences that are joined together using recombinant DNA technology and procedures that are used to join together DNA sequences as described, for example, in Sambrook and Russell (2001).
(62) "Naturally-occurring," "native," or "wild-type" refers to an amino acid sequence or polynucleotide sequence that can be found in nature, without any known mutation, as distinct from

being produced artificially or producing a mutated, non-wild-type phenotype. For example, a nucleotide sequence present in an organism (including a virus) that can be isolated from a source in nature and that has not been intentionally modified in the laboratory is naturally occurring. Furthermore, “wild-type” refers to a coding region or organism as found in nature without any known mutation.

(63) A “mutant” dysferlin (DYSF) is a polypeptide or a fragment thereof that is encoded by a DYSF coding region having a mutation, e.g., such as might occur at the DYSF locus. A mutation in one DYSF allele may lead to an alteration in the ability of the encoded polypeptide to interact with calcium ions, with alpha-actinin-2 (encoded by ACTN2), annexin A1 (encoded by ANXA1), annexin A2 (encoded by ANXA2), voltage-dependent L-type calcium channel subunit alpha-1S (encoded by CACNA1S), caveolin 3 (encoded by CAV3), desmin (encoded by DES), diacylglycerol kinase delta (encoded by DGKD), filamin C (encoded by FLNC), myoferlin (encoded by MYOF), myomesin-1 (encoded by MYOM1), gamma-sarcoglycan (encoded by SGCG), myomesin-2 (encoded by MYOM2), myosin-binding protein C, slow-type (encoded by MYBPC1), nebulin (encoded by NEBU), optineurin (encoded by OPTN), beta-parvin (encoded by PARVB), Rho family-interacting cell polarization regular 2 (encoded by RIPOR2), SNARE-associated protein Snapin (encoded by SNAPN), titin (encoded by TTN), tripartite motif-containing protein 72 (encoded by TRIM72), or other proteins that are involved in repair of the sarcolemma of myofibrils, or other proteins that are expressed in skeletal or cardiac muscle that are required for the integrity of myofibrils, leading to alterations in the integrity of myofibrils in a horse heterozygous or homozygous for the allele. Alterations in the interactions of specific proteins can be determined by methods known to the art. Mutations in DYSF may be disease-causing in a horse heterozygous or homozygous for the mutant DYSF allele, e.g., a horse heterozygous or homozygous for a mutation leading to a mutant DYSF polypeptide such as substitution mutations in Exon B of DYSF as shown in FIG. **3** and FIG. **4**, such as that designated herein as DYSF-R253W, or in Exon I of DYSF as shown in FIG. **15** and FIG. **16**, such as that designated herein as DYSF-P1290T.

(64) A “mutant” pyridine nucleotide-disulfide oxidoreductase domain-containing protein 1 (PYROXD1) is a polypeptide or a fragment thereof that is encoded by a PYROXD1 coding region having a mutation, e.g., such as might occur at the PYROXD1 locus. A mutation in one PYROXD1 allele may lead to an alteration in the ability of the encoded polypeptide to interact with the cofactors flavin adenine dinucleotide (FAD) or nicotinamide adenine dinucleotide (NAD), with other proteins containing either canonical or non-canonical disulfide (S-S) bonds between cysteine (CVS) residues, such as titin (encoded by TIN), obscurin (encoded by OBSCN), triadin (encoded by TRDN), myosin regulatory light chains (encoded by MYL1, MYL2, MYL3, MYL4, MYL5, MYL6, and others), myosin heavy chains (encoded by MYH1, MYH2, MYH3, MYH4, MYH5, MYH6, and others), the ryanodine receptor (encoded by RYR1), sarcoplasmic reticulum CA.sup.2+-ATPase (encoded by SERCA), nebulin (encoded by NEB), troponin (encoded by TNNT1, TNNT2, and TNNT3), myosin-binding protein C (encoded by MYBPC1), or other proteins that are involved in repair of the sarcolemma of myofibrils, or other proteins that are expressed in skeletal or cardiac muscle that are required for the integrity of myofibrils, leading to alterations in the integrity of myofibrils in a horse heterozygous or homozygous for the allele. Alterations in the interactions of specific proteins can be determined by methods known to the art. It is also possible that a mutation in one or both PYROXD1 alleles may lead to an alteration in the redox state of the cell, which might interfere with signaling processes that are sensitive to the redox state, or to an increased sensitivity to reactive oxygen species such as superoxide, hydrogen peroxide, or other reactive oxygen or reactive nitrogen species. Alterations in the redox state of the cell, as indicated by the ratio of reduced glutathione (GSH) to oxidized glutathione (GSSG), may lead to the irreversible oxidation of sulfhydryl groups in cysteine residues, causing the affected proteins to be ubiquitinated and destroyed. Turnover of particular muscle proteins may alter the

stoichiometry of muscle proteins, causing a temporary decrease in muscle function. Oxidative stress is also known to contribute to muscle fatigue by reducing mitochondrial function through the oxidation of key sulfhydryl groups in sarcoplasmic reticulum CA2+-ATPase (encoded by SERCA) and the ryanodine receptor (encoded by RYR1), which mediate calcium levels. Mutations in PYROXID1 may be disease-causing in a horse heterozygous or homozygous for the mutant PYROXD1 allele, e.g., a horse heterozygous or homozygous for a mutation leading to a mutant PYROXD1 polypeptide such as substitution mutations in Exon 12 of PYROXD1 as shown in FIG. **24** and FIG. **25**, such as that designated herein as PYROXD1-D492H.
(65) A "mutant" collagen type VI alpha 3 chain (COL6A3) is a polypeptide or a fragment thereof that is encoded by a COL6A3 coding region having a mutation, e.g., such as might occur at the COL6A3 locus. A mutation in one COL6A3 allele may lead to an alteration in the ability of the encoded polypeptide to interact with the protein encoded by the common or wild-type allele of COL6A3 in a heterozygote, to interact with proteins encoded by COL6A1, COL6A2, COL6A5, or COL6A6, or to assemble into a correctly-configured collagen triple helix. A mutation in one COL6A3 allele may lead to an alteration in the ability of the encoded polypeptide to interact with the protein products of the COL5A1 gene, which encodes one of the collagen type 5 proteins, with dysferlin (encoded by DYSF), or with other proteins known to interact with collagen. Defects caused by mutations in one or both COL6A3 alleles may interfere with the proper posttranslational modification of collagen, for example by interfering with glycosylation, phosphorylation, the modification of proline residues to hydroxyproline residues, the modification of lysine residues to hydroxylysine residues, or the proper formation of interchain disulfide bonds. Such defects may alter the ability of collagen to interact with other components of the extracellular matrix, or alter the mechanical properties of the extracellular matrix. Alterations in the interactions of specific proteins can be determined by methods known to the art. Mutations in COL6A3 may be disease-causing in a horse heterozygous or homozygous for the mutant COL6A3 allele, e.g., a horse heterozygous or homozygous for a mutation leading to a mutant COL6A3 polypeptide such as substitution mutations in Exon 26 of COL6A3 as shown in FIG. **35** and FIG. **36**, such as that designated herein as COL6A3-G2182A.
(66) A "somatic mutation" is a mutation that occurs only in certain tissues, e.g., in liver tissue, and are not inherited in the germline. A "germline" mutation can be found in any of a body's tissues and is inherited. The present COL6A3 mutation is a germline mutation.
(67) "Homology" refers to the percent identity between two polynucleotide sequences or two amino acid sequences. Two sequences are "homologous" to each other when the sequences exhibit at least 70% (e.g., 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%), at least 80% (e.g., 81% 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%), or at least 90% (e.g., 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%) contiguous sequence identity over a defined length of the sequences.
(68) The following terms are used to describe the sequence relationships between two or more nucleic acids or polynucleotides: "reference sequence," "comparison window," "sequence identity," "percentage of sequence identity," and "substantial identity."
(69) As used herein, "reference sequence" refers to a sequence used as a basis for sequence comparison. A reference sequence may be a subset or the entirety of a specified sequence. For example, a reference sequence may be a segment of a full length cDNA or coding region sequence, or the complete cDNA or coding region sequence.
(70) As used herein, "comparison window" refers to a contiguous and specified segment of a polynucleotide sequence, wherein the polynucleotide sequence in the comparison window may reflect one or more additions and/or deletions (i.e., gaps) compared to the reference sequence (which does not exhibit the additions and/or deletions) for optimal alignment of the two sequences. Generally, the comparison window is at least 20 contiguous nucleotides in length, and optionally can be 30, 40, 50, 100, or longer. To avoid a high similarity to a reference sequence due to inclusion of gaps in the polynucleotide sequence, a gap penalty is typically introduced and is

subtracted from the number of matches. Methods of alignment of sequences for comparison are well known in the art. Thus, the determination of percent identity between any two sequences can be accomplished using a mathematical algorithm.
(71) Computer implementations of these mathematical algorithms can be used for comparing sequences to determine sequence identity. Such implementations include, but are not limited to: Clustal Omega (online at EMBL-EBI), COBALT (online at ncbi.nlm.hih.gov), the ALIGN program (Version 2.0), and GAP, BESTFIT, BLAST, FASTA, and TFASTA in the Wisconsin Genetics Software Package, Version 8 (available from the Genetics Computer Group (GCG) Madison, WI, USA). Alignments using these programs can be performed using the default parameters.
(72) Software for performing BLAST analyses is publicly available through the National Center for Biotechnology Information (see the World Wide Web at ncbi.nlm.nih.gov). This algorithm involves first identifying high scoring pairs (HSPs) by identifying short words of length W in the query sequence, which either match or satisfy some positive-valued threshold score T when aligned with a word of the same length in a database sequence. T is referred to as the neighborhood word score threshold. These initial neighborhood word hits act as seeds for initiating searches to find longer HSPs containing them. The word hits are then extended in both directions along each sequence for as far as the cumulative alignment score can be increased. Cumulative scores are calculated using, for nucleotide sequences, the parameters M (reward score for a pair of matching residues; always >0) and N (penalty score for mismatching residues; always <0). For amino acid sequences, a scoring matrix is used to calculate the cumulative score. Extension of the word hits in each direction are halted when the cumulative alignment score falls off by the quantity X from its maximum achieved value, the cumulative score goes to zero or below due to the accumulation of one or more negative-scoring residue alignments, or the end of either sequence is reached.
(73) In addition to calculating percent sequence identity, the BLAST algorithm also performs a statistical analysis of the similarity between two sequences. One measure of similarity provided by the BLAST algorithm is the smallest sum probability (P(N)), which provides an indication of the probability by which a match between two nucleotide or amino acid sequences would occur by chance. For example, a test nucleic acid sequence is considered similar to a reference sequence if the smallest sum probability in a comparison of the test nucleic acid sequence to the reference nucleic acid sequence is less than about 0.1, less than about 0.01, or even less than about 0.001.
(74) To obtain gapped alignments for comparison purposes, Gapped BLAST (in BLAST 2.0) can be utilized. Alternatively, PSI-BLAST (in BLAST 2.0) can be used to perform an iterated search that detects distant relationships between molecules. When using BLAST, Gapped BLAST, or PSI-BLAST, the default parameters of the respective programs (e.g., BLASTN for nucleotide sequences, BLASTP for proteins) can be used. The BLASTN program (for nucleotide sequences) uses as defaults a wordlength (W) of 11, an expectation (E) of 10, a cutoff of 100, M=5, N=−4, and a comparison of both strands. For amino acid sequences, the BLASTP program uses as defaults a wordlength (W) of 3, an expectation (E) of 10, and the BLOSUM62 scoring matrix. See the World Wide Web at ncbi.nlm.nih.gov. Alignment may also be performed manually by visual inspection. For purposes of the methods described herein, comparison of nucleotide sequences for determination of percent sequence identity to the promoter sequences disclosed herein is preferably made using the BlastN program (version 2.3.0 or later) with its default parameters or any equivalent program. By “equivalent program” is intended any sequence comparison program that, for any two sequences in question, generates an alignment having identical nucleotide or amino acid residue matches and an identical percent sequence identity when compared to the corresponding alignment generated by a BLAST program.
(75) As used herein, “sequence identity” or “identity” in the context of two nucleic acid or polypeptide sequences refers to a specified percentage of residues in the two sequences that are the same when aligned for maximum correspondence over a specified comparison window, as measured by sequence comparison algorithms or by visual inspection. When percentage of

sequence identity is used in reference to a protein, it is recognized that residue positions that are not identical often differ by conservative amino acid substitutions, where amino acid residues are substituted for other amino acid residues with similar chemical properties (e.g., charge or hydrophobicity) and therefore do not change the functional properties of the molecule. When sequences differ in conservative substitutions, the percent sequence identity may be adjusted upwards to correct for the conservative nature of the substitution. Sequences that differ by such conservative substitutions are said to have “sequence similarity” or “similarity.” Methods for making this adjustment are well known to those of skill in the art. Typically, this involves scoring a conservative substitution as a partial rather than a full mismatch, thereby increasing the percentage sequence identity. Thus, for example, where an identical amino acid is given a score of 1 and a non-conservative substitution is given a score of zero, a conservative substitution is given a score between zero and 1. The scoring of conservative substitutions is calculated, e.g., as implemented in the program PC/GENE (Intelligenetics, Mountain View, CA).

(76) A used herein, “percentage of sequence identity” refers to the value determined by comparing two optimally aligned sequences over a comparison window, wherein the portion of the polynucleotide sequence in the comparison window may comprise additions or deletions (i.e. gaps) as compared to the reference sequence (which does not comprise additions or deletions) for optimal alignment of the two sequences. The percentage is calculated by determining the number of positions at which the identical nucleic acid base or amino acid residue occurs in both sequences to yield the number of matched positions, dividing the number of matched positions by the total number of positions in the window of comparison, and multiplying the result by 100 to yield the percentage of sequence identity.

(77) The term “substantial identity,” in the context of polynucleotide sequences, means that a polynucleotide sequence possesses at least 70% (e.g., 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%), at least 80% (e.g., 81% 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%), or at least 90% (e.g., 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%) sequence identity compared to a reference sequence using one of the alignment programs described using standard parameters. These values can be appropriately adjusted to determine corresponding identity of proteins encoded by two nucleotide sequences by taking into account codon degeneracy, amino acid similarity, reading frame positioning, and the like. Substantial identity of amino acid sequences for these purposes normally means sequence identity of at least 70%, or at least 80%, 90%, or even at least 95%.

(78) Another indication that nucleotide sequences are substantially identical is if two molecules hybridize to each other under stringent conditions (see below). Generally, stringent conditions are selected to be about 5° C. lower than the thermal melting point (T.sub.m) for the specific sequence at a defined ionic strength and pH. However, stringent conditions encompass temperatures in the range of about 1° C. to about 20° C., depending upon the desired degree of stringency as otherwise qualified herein. Nucleic acids that do not hybridize to each other under stringent conditions are still substantially identical if the polypeptides they encode are substantially identical. This may occur, e.g., when a copy of a nucleic acid is created using the maximum codon degeneracy permitted by the genetic code. One indication that the two nucleic acid sequences are substantially identical is when the polypeptide encoded by the first nucleic acid is immunologically cross reactive with the polypeptide encoded by the second nucleic acid.

(79) The term “substantial identity,” in the context of a polypeptide, indicates that a polypeptide possesses a sequence with at least 70% (e.g., 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%), at least 80% (e.g., 81% 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%), or at least 90% (e.g., 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%) amino acid sequence identity to the reference sequence over a specified comparison window. An indication that two polypeptide sequences are substantially identical is that one polypeptide is immunologically reactive with antibodies raised against the second polypeptide.

(80) Thus, a polypeptide is substantially identical to a second polypeptide when, for example, the two polypeptides differ only by a conservative substitution. For sequence comparison, typically one amino acid sequence acts as a reference sequence to which test amino acid sequences are compared. When using a sequence comparison algorithm, test and reference amino acid sequences are input into a computer, subsequence coordinates are designated if necessary, and sequence algorithm program parameters are designated. The sequence comparison algorithm then calculates the percent sequence identity for the test sequence(s) relative to the reference sequence, based on the designated program parameters.
(81) As noted above, another indication that two nucleic acid sequences are substantially identical is that two molecules hybridize to each other under stringent conditions. The phrase “hybridizing specifically to” refers to the binding, duplexing, or hybridizing of a molecule only to a particular nucleotide sequence under stringent conditions when that sequence is present in a complex mixture (e.g., total cellular) DNA or RNA. “Bind(s) substantially” refers to complementary hybridization between a probe nucleic acid and a target nucleic acid and embraces minor mismatches that can be accommodated by reducing the stringency of the hybridization media to achieve the desired detection of the target nucleic acid sequence.
(82) “Stringent hybridization conditions” and “stringent hybridization wash conditions” in the context of nucleic acid hybridization experiments such as Southern and Northern hybridizations are sequence dependent, and are different under different environmental parameters. Longer sequences hybridize specifically at higher temperatures. The T.sub.m is the temperature (under defined ionic strength and pH) at which 50% of the target sequence hybridizes to a perfectly matched probe. Specificity is typically the function of post-hybridization washes, the critical factors being the ionic strength and temperature of the final wash solution. For DNA-DNA hybrids, the T.sub.m can be approximated from the equation of Meinkoth and Wahl:
T.sub.m=81.5° C.+16.6 (log M)+0.41(% GC)-0.61 (% form)-500/L
where M is the molarity of monovalent cations, % GC is the percentage of guanosine and cytosine nucleotides in the DNA, % form is the percentage of formamide in the hybridization solution, and L is the length of the hybrid in base pairs. T.sub.m is reduced by about 1° C. for each 1% of mismatching; thus, T.sub.m, hybridization, and/or wash conditions can be adjusted to hybridize to sequences of the desired identity. For example, if sequences with >90% identity are sought, the T.sub.m can be decreased 10° C. Generally, stringent conditions are selected to be about 5° C. lower than the thermal melting point (T.sub.m) for the specific sequence and its complement at a defined ionic strength and pH. However, severely stringent conditions can utilize a hybridization and/or wash at 1° C., 2° C., 3° C., or 4° C. lower than the thermal melting point (T.sub.m); moderately stringent conditions can utilize a hybridization and/or wash at 6° C., 7° C., 8° C., 9° C., or 10° C. lower than the thermal melting point (T.sub.m); low stringency conditions can utilize a hybridization and/or wash at 11° C., 12° C., 13° C., 14° C., 15° C., or 20° C. lower than the thermal melting point (T.sub.m). Using the equation, hybridization and wash compositions, and desired T, those of ordinary skill will understand that variations in the stringency of hybridization and/or wash solutions are inherently described. If the desired degree of mismatching results in a T of less than 45° C. (aqueous solution) or 32° C. (formamide solution), it is preferred to increase the SSC concentration (20×SSC=3.0 M NaCl, 0.3 M trisodium citrate) so that a higher temperature can be used. Generally, highly stringent hybridization and wash conditions are selected to be about 5° C. lower than the thermal melting point (T.sub.m) for the specific sequence at a defined ionic strength and pH.
(83) An example of highly stringent wash conditions is 0.15 M NaCl at 72° C. for about 15 minutes. An example of stringent wash conditions is a 0.2×SSC wash at 65° C. for about 15 minutes. Often, a high stringency wash is preceded by a low stringency wash to remove background probe signal. An example medium stringency wash for a duplex of, e.g., more than 100 nucleotides is 4-6×SSC at 40° C. for 15 minutes. For short probes (e.g., about 10 to 50

nucleotides), stringent conditions typically involve salt concentrations of less than about 1.5 M, more preferably about 0.01 M to 1.0 M, Na.sup.+ ion concentration (or other salts) at pH 7.0 to 8.3, and the temperature is typically at least about 30° C. and at least about 60° C. for long probes (e.g., >50 nucleotides). Stringent conditions may also be achieved with the addition of destabilizing agents such as formamide. In general, a signal to noise ratio of 2× (or higher) than that observed for an unrelated probe in the particular hybridization assay indicates detection of a specific hybridization. Nucleic acids that do not hybridize to each other under stringent conditions are still substantially identical if the proteins that they encode are substantially identical. This occurs, e.g., when a copy of a nucleic acid is created using the maximum codon degeneracy permitted by the genetic code. Very stringent conditions are selected to be equal to the T.sub.m for a particular probe. An example of stringent conditions for hybridization of complementary nucleic acids which have more than 100 complementary residues on a filter in a Southern or Northern blot is 50% formamide, e.g., hybridization in 50% formamide, 1 M NaCl, 1% SDS at 37° C.; and a wash in 0.1×SSC at 60° C. to 65° C. Exemplary low stringency conditions include hybridization with a buffer solution of 30 to 35% formamide, 1 M NaCl, 1% SDS (sodium dodecyl sulfate) at 37° C., and a wash in 1× to 2×SSC at 50° C. to 55° C. Exemplary moderate stringency conditions include hybridization in 40% to 45% formamide, 1.0 M NaCl, 1% SDS at 37° C., and a wash in 0.5× to 1×SSC at 55° C. to 60° C.
(84) The term “variant” polypeptide refers to a polypeptide derived from the native protein by deletion (so-called truncation) and/or addition of one or more amino acids to the N-terminal and/or C-terminal end of the native protein, deletion and/or addition of one or more amino acids at one or more sites in the native protein, and/or substitution of one or more amino acids at one or more sites in the native protein. Such variants may result from, for example, genetic polymorphism or human manipulation. Methods for such manipulations are generally known in the art. Variant DYSF, PYROXD1, or COL6A3 polypeptides may be altered in various ways including, for example, being altered to exhibit one or more amino acid substitutions, one or more deletions, one or more truncations, and/or one or more insertions. For example, an amino acid sequence can be prepared by one or more mutations in the DNA encoding the DYSF, PYROXD1, or COL6A3 polypeptides. Guidance regarding appropriate amino acid substitutions that do not affect biological activity of the protein of interest is well known in the art. Conservative substitutions, such as exchanging one amino acid with another having similar properties, are preferred.
(85) Thus, the nucleotide sequences used to practice the methods described herein can include both naturally-occurring sequences or mutant forms. Likewise, the polypeptides referred to herein can include naturally-occurring polypeptides as well as variations and modified forms thereof. Such variants may continue to possess the desired activity. The deletions, insertions, or substitutions of the polypeptide sequence encompassed herein are not expected to produce radical changes in the characteristics of the polypeptide. However, when it is difficult to predict the exact effect of the substitution, deletion, or insertion in advance of doing so, the effect can be evaluated by routine screening assays.
(86) An individual substitution, deletion, or addition that alters, adds, or deletes a single amino acid or a small percentage of amino acids (typically less than 5%, more typically less than 1%) in an encoded sequence are “conservatively modified variations.”
(87) “Conservatively modified variations” of a particular nucleic acid sequence refers to those nucleic acid sequences that encode identical or essentially identical amino acid sequences, or where the nucleic acid sequence does not encode an amino acid sequence, to essentially identical sequences. Because of the degeneracy of the genetic code, a large number of functionally identical nucleic acids encode any given polypeptide. For instance, the codons CGT, CGC, CGA, CGG, AGA, and AGG all encode the amino acid arginine. Thus, at every position where an arginine is specified by a codon, the codon can be altered to any of the corresponding codons described without altering the encoded protein. Such nucleic acid variations are “silent variations,” which are

one species of “conservatively modified variations.” Every nucleic acid sequence described herein that encodes a polypeptide also describes every possible silent variation, except where otherwise noted. One of skill will recognize that each codon in a nucleic acid (except ATG, which is ordinarily the only codon for methionine, and TGG, which is ordinarily the only codon for tryptophan) can be modified to yield a functionally identical molecule by standard techniques. Accordingly, each “silent variation” of a nucleic acid that encodes a polypeptide is implicit in each described sequence.
(88) Known methods of PCR include, but are not limited to, methods using paired primers, nested primers, single specific primers, degenerate primers, gene-specific primers, vector-specific primers, partially mismatched primers, and the like.
(89) The terms “heterologous DNA sequence,” “exogenous DNA segment,” or “heterologous nucleic acid” refer to a sequence that originates from a source foreign to the particular host cell or, if from the same source, is modified from its original form. Thus, a heterologous coding region in a host cell includes a coding region that is endogenous to the particular host cell but has been modified through, for example, the use of single-stranded mutagenesis. The terms also include non-naturally-occurring multiple copies of a naturally occurring DNA sequence. Thus, the terms refer to a DNA segment that is foreign or heterologous to the cell, or homologous to the cell but in a position within the host cell nucleic acid in which the element is not ordinarily found. Exogenous DNA segments, when expressed, yield exogenous polypeptides.
(90) A “homologous” DNA sequence is a DNA sequence that is naturally associated with a host cell into which it is introduced.
(91) “Genome” refers to the complete genetic material of an organism.
(92) “Coding sequence” refers to a DNA or RNA sequence that codes for a specific amino acid sequence and excludes non-coding (e.g., regulatory) nucleotide sequences. For example, a
(93) DNA “coding sequence” or a “sequence encoding” a particular polypeptide is a DNA sequence that is transcribed and translated into a polypeptide in vitro or in vivo when placed under the control of appropriate regulatory elements. The boundaries of the coding sequence are determined by a start codon at the 5′-terminus and a translation stop codon at the 3′-terminus. A coding sequence can include, but is not limited to, prokaryotic sequences, cDNA from eukaryotic mRNA, genomic DNA sequences from eukaryotic (e.g., mammalian) DNA, and/or synthetic DNA sequences. A transcription termination sequence will usually be located 3′ to the coding sequence. It may constitute an “uninterrupted coding sequence,”—i.e., lacking an intron, such as in cDNA or it may include one or more introns bounded by appropriate splice junctions. An “intron” is a sequence of RNA that is contained in the primary transcript but that is removed through cleavage and re-ligation of the RNA within the cell to create the mature mRNA that can be translated into a protein.
(94) The terms “open reading frame” and “ORF” refer to the nucleotide sequence between translation initiation and termination codons of a coding sequence. The terms “initiation codon” and “termination codon” refer to a unit of three adjacent nucleotides (“codon”) in a coding sequence that specifies initiation and chain termination, respectively, of protein synthesis (mRNA translation).
(95) The term “RNA transcript” refers to the product resulting from RNA polymerase catalyzed transcription of a DNA sequence. When the RNA transcript is a perfect complementary copy of the DNA sequence, it is referred to as a primary transcript or it may be an RNA sequence derived from post transcriptional processing of the primary transcript and is referred to as the mature RNA. “Messenger RNA” (mRNA) refers to the RNA that is without introns and can be translated into protein by the cell. “cDNA” refers to a single- or double-stranded DNA that is complementary to and derived from mRNA.
(96) The term “regulatory sequence” refers to a nucleotide sequence that includes, for example, a promoter, an enhancer, and/or other expression control elements (e.g., polyadenylation signals).

Such regulatory sequences are known to those skilled in the art. The design of an expression vector may depend on such factors as the choice of the host cell to be transfected and/or the amount of fusion protein to be expressed.
(97) The term “DNA control elements” refers collectively to promoters, ribosome binding sites, polyadenylation signals, transcription termination sequences, upstream regulatory domains, enhancers, and the like, that collectively provide for the transcription and translation of a coding sequence in a host cell. Not all of these control sequences need always be present in a recombinant vector so long as the desired coding region is capable of being transcribed and translated.
(98) A control element, such as a promoter, “directs the transcription” of a coding sequence in a cell when RNA polymerase binds to the promoter and transcribes the coding sequence into mRNA, which is then translated into the polypeptide encoded by the coding sequence.
(99) A cell has been “transformed” by exogenous DNA when the exogenous DNA has been introduced inside the cell membrane. Exogenous DNA may or may not be integrated (covalently linked) into chromosomal DNA making up the genome of the cell. In prokaryotes and yeasts, for example, the exogenous DNA may be maintained on an episomal element, such as a plasmid. With respect to other eukaryotic cells, a stably transformed cell is one in which the exogenous DNA has become integrated into the chromosome so that it is inherited by daughter cells through chromosome replication. This stability is demonstrated by the ability of the eukaryotic cell to establish cell lines or clones having a population of daughter cells containing the exogenous DNA.
(100) “Operably linked” refers to the association of nucleic acid sequences on single nucleic acid fragments so that the function of one is affected by the other, e.g., an arrangement of elements wherein the components so described are configured so as to perform their usual function. For example, a regulatory DNA sequence is said to be “operably linked to” a DNA sequence that codes for an RNA or a polypeptide if the two sequences are situated such that the regulatory DNA sequence affects expression of the coding DNA sequence (i.e., that the coding sequence or functional RNA is under the transcriptional control of the promoter). Coding sequences can be operably linked to regulatory sequences in sense or antisense orientation. Control elements operably linked to a coding sequence are capable of effecting the expression of the coding sequence. The control elements need not be contiguous with the coding sequence, so long as they function to direct the expression thereof. Thus, for example, intervening untranslated yet transcribed sequences can be present between a promoter and the coding sequence and the promoter can still be considered “operably linked” to the coding sequence.
(101) “Transcription stop fragment” refers to nucleotide sequences that contain one or more regulatory signals, such as polyadenylation signal sequences, capable of terminating transcription. Examples include the 3′ non-regulatory regions of the genes dysferlin (DYSF), pyridine nucleotide-disulfide oxidoreductase domain-containing protein 1 (PYROXD1), and collagen type VI alpha 3 chain (COL6A3).
(102) “Translation stop fragment” or “translation stop code” or “stop codon” refers to nucleotide sequences that contain one or more regulatory signals, such as one or more termination codons in all three frames, capable of terminating translation. Insertion of a translation stop fragment adjacent to or near the initiation codon at the 5′ end of the coding sequence will result in no translation or improper translation. The change of at least one nucleotide in a nucleic acid sequence can result in an interruption of the coding sequence of the gene, e.g., a premature stop codon. Such sequence changes can cause a mutation in the polypeptide encoded by the DYSF, PYROXD1, or COL6A3 gene. For example, if the mutation is a nonsense mutation, the mutation results in the generation of a premature stop codon, causing the generation of a truncated DYSF, PYROXD1, or COL6A3 polypeptide.
(103) Nucleic Acids
(104) Nucleotide sequences that are subjected to the methods described herein can be obtained from any prokaryotic or eukaryotic source. For example, they can be obtained from a mammalian,

such as equine, cellular source. Alternatively, nucleic acid molecules can be obtained from a library, such as the CHORI-241 Equine BAC library or a similar resource available elsewhere.
(105) As discussed above, the terms "isolated and/or purified" refer to a nucleic acid—e.g. a DNA or RNA molecule—that has been isolated from its natural cellular environment and from association with other components of the cell, such as nucleic acid or polypeptide, so that it can be sequenced, replicated, and/or expressed. For example, an "isolated nucleic acid" may be a DNA molecule that is complementary or hybridizes to a sequence in a coding region of interest—e.g., a nucleic acid sequence encoding an equine collagen type VI alpha 3 chain protein, and remains stably bound under stringent conditions (as defined by methods well known in the art). Thus, an RNA or a DNA is "isolated" in that it is free from at least one contaminating nucleic acid with which it is normally associated in the natural source of the RNA or DNA and in one embodiment of the invention is substantially free of any other mammalian RNA or DNA. The phrase "free from at least one contaminating source nucleic acid with which it is normally associated" includes the case where nucleic acid is reintroduced into the source or natural cell but is in a different chromosomal location or is otherwise flanked by nucleic acid sequences not normally found in the source cell, e.g., in a vector or plasmid.
(106) As used herein, the term "recombinant nucleic acid," e.g., "recombinant DNA sequence or segment" refers to a nucleic acid, e.g., to DNA that has been derived or isolated from any appropriate cellular source, that may be substantially chemically altered in vitro, so that its sequence is not naturally occurring, or corresponds to naturally occurring sequences that are not positioned as they would be positioned in a genome that has not been transformed with exogenous DNA. An example of preselected DNA "derived" from a source would be a DNA sequence that is identified as a useful fragment within a given organism, and which is then chemically synthesized in essentially pure form. An example of such DNA "isolated" from a source would be a useful DNA sequence that is excised or removed from the source by chemical means, e.g., by the use of restriction endonucleases, so that it can be further manipulated, e.g. amplified, for use in the methods described herein. Thus, recovery or isolation of a given fragment of DNA from a restriction digest can employ separation of the digest on polyacrylamide or agarose gel by electrophoresis, identification of the fragment of interest by comparison of its mobility versus that of marker DNA fragments of known molecular weight, removal of the gel section containing the desired fragment, and separation of the gel from DNA. Therefore, "recombinant DNA" includes completely synthetic DNA sequences, semi-synthetic DNA sequences, DNA sequences isolated from biological sources, and DNA sequences derived from RNA, as well as mixtures thereof.
(107) Nucleic acid molecules having base substitutions (i.e., variants) are prepared by a variety of methods known in the art. These methods include, but are not limited to, isolation from a natural source (in the case of naturally occurring sequence variants) or preparation by oligonucleotide-mediated (or site-directed) mutagenesis, PCR mutagenesis, and cassette mutagenesis of an earlier prepared variant or non-variant version of the nucleic acid molecule.
(108) Nucleic Acid Amplification Methods
(109) DNA present in a physiological sample may be amplified by any means known to the art. Examples of suitable amplification techniques include, but are not limited to, polymerase chain reaction (including, for RNA amplification, reverse-transcriptase polymerase chain reaction), ligase chain reaction, strand displacement amplification, transcription-based amplification, self-sustained sequence replication (or "3SR"), the Qβ-replicase system, nucleic acid sequence-based amplification (or "NASBA"), the repair chain reaction (or "RCR"), and boomerang DNA amplification (or "BDA").
(110) The bases incorporated into the amplification product may be natural or modified bases (modified before or after amplification), and the bases may be selected to optimize subsequent electrochemical detection steps.
(111) Polymerase chain reaction (PCR) may be performed according to known techniques. In

general, PCR involves, first, treating a nucleic acid sample (e.g., in the presence of a heat stable DNA polymerase) with one oligonucleotide primer for each strand of the specific sequence to be detected under hybridizing conditions so that an extension product of each primer is synthesized that is complementary to each nucleic acid strand, with the primers sufficiently complementary to each strand of the specific sequence to hybridize therewith so that the extension product synthesized from each primer, when it is separated from its complement, can serve as a template for synthesis of the extension product of the other primer, and then treating the sample under denaturing conditions to separate the primer extension products from their templates if the sequence or sequences to be detected are present. These steps are cyclically repeated until the desired degree of amplification is obtained. Detection of the amplified sequence may be carried out by adding to the reaction product an oligonucleotide probe capable of hybridizing to the reaction product (e.g., an oligonucleotide probe), the probe carrying a detectable label, and then detecting the label in accordance with known techniques. Where the nucleic acid to be amplified is RNA, amplification may be carried out by initial conversion to DNA by reverse transcriptase in accordance with known techniques.
(112) Strand displacement amplification (SDA) may be performed according to known techniques. For example, SDA may be carried out with a single amplification primer or a pair of amplification primers, with exponential amplification being achieved with the latter. In general, SDA amplification primers comprise, in the 5 to 3 direction, a flanking sequence (the DNA sequence of which is noncritical), a restriction site for the restriction enzyme employed in the reaction, and an oligonucleotide sequence (e.g., an oligonucleotide probe) that hybridizes to the target sequence to be amplified and/or detected. The flanking sequence, which serves to facilitate binding of the restriction enzyme to the recognition site and provides a DNA polymerase priming site after the restriction site has been nicked, is about 15 to 20 nucleotides in length in one embodiment. The restriction site is functional in the SDA reaction: the oligonucleotide probe portion is about 13 to 15 nucleotides in length in one embodiment of the invention.
(113) Ligase chain reaction (LCR) also may be performed according to known techniques. In general, the reaction is carried out with two pairs of oligonucleotide probes: one pair binds to one strand of the sequence to be detected; the other pair binds to the other strand of the sequence to be detected; each pair together completely overlaps the strand to which it corresponds. The reaction is carried out by, first, denaturing (e.g., separating) the strands of the sequence to be detected, then reacting the strands with the two pairs of oligonucleotide probes in the presence of a heat stable ligase so that each pair of oligonucleotide probes is ligated together, then separating the reaction product, and then cyclically repeating the process until the sequence has been amplified to the desired degree. Detection may then be carried out in like manner as described above with respect to PCR.
(114) In some embodiments, each exon of the DYSF, PYROXD1, or COL6A3 coding regions is amplified by PCR using primers based on the known sequence. The amplified exons are then sequenced using, for example, an automated sequencer. In this manner, the exons of the DYSF, PYROXD1, or COL6A3 coding regions from horses suspected of having Polysaccharide Storage Myopathy type 2 (PSSM2), also known as Myofibrillar Myopathy (MFM), in their pedigree are then sequenced until a mutation is found. Examples of such mutations include those in Exon B of the DYSF DNA as shown in FIG. **3** and FIG. **4**, in Exon I of the DYSF DNA as shown in FIG. **15** and FIG. **16**, in Exon 12 of the PYROXD1 DNA as shown in FIG. **24** and FIG. **25**, or in Exon 26 of the COL6A3 DNA as shown in FIG. **35** and FIG. **36**. For example, mutations in the DYSF, PYROXD1, or COL6A3 genes include (1) the specific substitution of an adenine (A) for a guanine (G) on the forward strand at chr15:31,306,949 of the current horse genome assembly (EquCab2, GCA_000002305.1) as displayed in the UCSC Genome Browser and as shown for the reverse strand in FIG. **1**, (2) the specific substitution of an thymine (T) for a guanine (G) on the forward strand at chr15:31,225,630 of the current horse genome assembly (EquCab2, GCA_000002305.1)

as displayed in the UCSC Genome Browser and as shown for the reverse strand in FIG. **13**, (3) the specific substitution of a cytosine (C) for a guanine (G) on the forward strand at chr6:47,661,977 of the current horse genome assembly (EquCab2, GCA_000002305.1) as displayed in the UCSC Genome Browser and as shown for the reverse strand in FIG. **22**, and (4) the specific substitution of a guanine (G) for a cytosine (C) on the forward strand at chr6:23,480,621 of the current horse genome assembly (EquCab2, GCA_000002305.1) as displayed in the UCSC Genome Browser and as shown in FIG. **33**. Using this technique, additional mutations causing equine Polysaccharide Storage Myopathy type 2 (PSSM2), also known as Myofibrillar Myopathy, can be identified. Thus, the methods described herein may be used to detect and/or identify an alteration within the wild-type DYSF, PYROXD1, or COL6A3 locus. “Alteration of” a specified locus encompasses all forms of mutations including, for example, a deletion, an insertion, and/or a point mutation in the coding and noncoding regions. A deletion can involve the deletion of all or any portion of the coding region. A point mutation may result in an aberrant stop codon, a frameshift mutation, an amino acid substitution, and/or an alteration in pre-mRNA processing (splicing) that produces a protein with an altered amino acid sequence. Point mutational events may occur in regulatory regions, such as in the promoter of the gene, leading to decreased expression of the mRNA. A point mutation also may interfere with proper RNA processing, leading to decreased expression of the DYSF, PYROXD1, or COL6A3 translation products, decreased mRNA stability, and/or decreased translation efficiency. Polysaccharide Storage Myopathy type 2 (PSSM2), also known as Myofibrillar Myopathy, is a disease caused by point mutations (1) at chr15:31,306,949 of the current horse genome assembly (EquCab2, GCA_000002305.1) as displayed in the UCSC Genome Browser and as shown in FIG. **1**, (2) at chr15: 31,225,630 of the current horse genome assembly (EquCab2, GCA_000002305.1) as displayed in the UCSC Genome Browser and as shown in FIG. **13**, (3) at chr6:47,661,977 of the current horse genome assembly (EquCab2, GCA_000002305.1) as displayed in the UCSC Genome Browser and as shown in FIG. **22**, or (4) chr6:23,480,621 of the current horse genome assembly (EquCab2, GCA_000002305.1) as displayed in the UCSC Genome Browser and as shown in FIG. **33**. Horses predisposed to or having Polysaccharide Storage Myopathy type 2 (PSSM2) may need only one mutated DYSF, PYROXD1, or COL6A3 allele.

(115) Techniques that are useful in performing the methods described herein include, but are not limited to direct DNA sequencing, PFGE analysis, allele-specific oligonucleotide (ASO), dot blot analysis, and/or denaturing gradient gel electrophoresis.

(116) There are several methods that can be used to detect DNA sequence variation. Direct DNA sequencing, either manual or automated (e.g., fluorescent or semiconductor-based sequencing), can detect sequence variation. Another approach is the single-stranded conformation polymorphism assay (SSCA). This method does not detect all sequence changes, especially if the DNA fragment size is greater than 200 bp, but can be used to detect most DNA sequence variation. SSCA allows for increased throughput compared to direct sequencing for mutation detection on a research basis. The fragments that have shifted mobility on SSCA gels are then sequenced to determine the exact nature of the DNA sequence variation. Other approaches based on the detection of mismatches between the two complementary DNA strands include clamped denaturing gel electrophoresis (CDGE), heteroduplex analysis (HA), and chemical mismatch cleavage (CMC). Once a mutation is known, an allele specific detection approach such as allele specific oligonucleotide (ASO) hybridization can be utilized to rapidly screen large numbers of other samples for that same mutation. Such a technique can utilize probes that are labeled with gold nanoparticles to yield a visual color result.

(117) Detecting point mutations may be accomplished by molecular cloning and then sequencing one or more DYSF, PYROXD1, or COL6A3 alleles. Alternatively, the coding region sequences can be amplified directly from a genomic DNA preparation from equine tissue, using known techniques. The DNA sequence of the amplified sequences can then be determined.

(118) Exemplary methods for a more complete, yet still indirect, test for confirming the presence of

a mutant allele include, for example, single stranded conformation analysis (SSCA), denaturing gradient gel electrophoresis (DDGE), an RNase protection assay, allele-specific oligonucleotides (ASOs), the use of a protein that recognizes nucleotide mismatches (e.g., the *E. coli* mutS protein), and allele-specific PCR. For allele-specific PCR, primers are used that hybridize at their 3′ ends to a particular DYSF, PYROXD1, or COL6A3 mutation. If the particular mutation is not present, an amplification product is not observed. Allele-specific PCR may also be carried out using quantitative PCR or real-time PCR using a specialized instrument that is capable of detecting and quantifying the appearance of amplification products during each amplification cycle. An Amplification Refractory Mutation System (ARMS) can also be used. Insertions and deletions of genes can also be detected by cloning, sequencing, and amplification. In addition, restriction fragment length polymorphism (RFLP) probes for the target locus or a surrounding marker locus can be used to score alteration of an allele or an insertion in a polymorphic fragment. Other techniques for detecting insertions or deletions as known in the art can also be used.

(119) In the first three methods (i.e., SSCA, DGGE, and RNase protection assay), a new electrophoretic band appears. SSCA detects a band that migrates differently because the sequence change causes a difference in single-strand, intramolecular base pairing. RNase protection involves cleaving the mutant polynucleotide into two or more smaller fragments. DGGE detects differences in migration rates of mutant sequences compared to wild-type sequences using a denaturing gradient gel. In an allele-specific oligonucleotide assay, an oligonucleotide is designed that detects a specific sequence, and the assay is performed by detecting the presence or absence of a hybridization signal. In the mutS assay, the protein binds only to sequences that contain a nucleotide mismatch in a heteroduplex between mutant and wild-type sequence.

(120) As used herein, a “nucleotide mismatch” refers to a hybridized nucleic acid duplex in which the two strands are not 100% complementary. Lack of total homology may be due to a deletion, an insertion, an inversion, and/or a substitution. Mismatch detection can be used to detect point mutation in the coding region or its mRNA product. While these techniques are less sensitive than sequencing, they are simpler to perform on a large number of samples. An example of a mismatch cleavage technique is the RNase protection method. In the context of detecting a DYSF-associated, PYROXD1-associated, or COL6A3-associated mismatch, the method involves the use of a labeled riboprobe that is complementary to the horse wild-type DYSF, PYROXD1, or COL6A3 coding region sequence. The riboprobe and either mRNA or DNA isolated from tissue are annealed (i.e., hybridized) and subsequently digested with the enzyme RNase A, which is able to detect some mismatches in a duplex RNA structure. If a mismatch is detected by RNase A, it cleaves at the site of the mismatch. Thus, when the annealed RNA preparation is separated on an electrophoretic gel matrix, if a mismatch has been detected and cleaved by RNase A, an RNA product will be seen that is smaller than the full length duplex RNA for the riboprobe and the mRNA or DNA. The riboprobe need not be the full length of the DYSF, PYROXD1, or COL6A3 mRNA or coding region but can be a segment of either. If the riboprobe includes only a segment of the DYSF, PYROXD1, or COL6A3 mRNA or DNA, it may be desirable to use a number of probes to screen the whole mRNA sequence for mismatches.

(121) In a similar fashion, DNA probes can be used to detect a mismatch through enzymatic and/or chemical cleavage. Alternatively, a mismatch can be detected by shifts in the electrophoretic mobility of mismatched duplexes relative to matched duplexes. With either riboprobes or DNA probes, the cellular mRNA or DNA that might contain a mutation can be amplified using PCR before hybridization.

(122) Nucleic Acid Analysis via Microchip Technology

(123) A DNA sequence of the DYSF, PYROXD1, or COL6A3 coding regions that has been amplified by PCR may be screened using an allele-specific probe. Allele-specific probes are nucleic acid oligomers, each of which contains a region of the DYSF, PYROXD1, or COL6A3 coding region harboring a known mutation. For example, one oligomer may be about 30

nucleotides in length, corresponding to a portion of the DYSF, PYROXD1, or COL6A3 coding region sequence. Using a battery of such allele-specific probes, a PCR amplification product can be screened to identify the presence of a previously identified mutation in the DYSF, PYROXD1, or COL6A3 coding region. Hybridizing an allele-specific probe with an amplified DYSF, PYROXD1, or COL6A3 sequence can be performed, for example, on a nylon filter. Hybridizing to a particular probe under stringent hybridization conditions indicates the presence of the same mutation in the tissue as in the allele-specific probe.

(124) An alteration of DYSF, PYROXD1, or COL6A3 mRNA expression can be detected by any technique known in the art. Exemplary techniques include, for example, Northern blot analysis, PCR amplification, and/or RNase protection. Decreased mRNA expression indicates an alteration of the wild-type DYSF, PYROXD1, or COL6A3 locus.

(125) Alteration of wild-type DYSF, PYROXD1, or COL6A3 coding region also can be detected by screening for alteration of a wild-type DYSF, PYROXD1, or COL6A3 polypeptide such as, for example, the wild-type DYSF, PYROXD1, or COL6A3 protein or a portion of the wild-type DYSF, PYROXD1, or COL6A3 protein. For example, a monoclonal antibody immunoreactive with wild-type DYSF, PYROXD1, or COL6A3 (or to a specific portion of the DYSF, PYROXD1, or COL6A3 protein) can be used to screen a tissue. Lack of cognate antigen would indicate a mutation. An antibody specific for a product of a mutant allele also can be used to detect a mutation in the DYSF, PYROXD1, or COL6A3 coding region. Such an immunological assay can be performed using conventional methods. Exemplary methods include, for example, Western blot analysis, an immunohistochemical assay, an ELISA assay, and/or any method for detecting an altered DYSF, PYROXD1, or COL6A3 polypeptide. In some embodiments, a functional assay can be used such as, for example, protein binding determination. In addition, an assay can be used that detects DYSF, PYROXD1, or COL6A3 biochemical function. Finding a mutant DYSF, PYROXD1, or COL6A3 polypeptide indicates a mutation at the DYSF, PYROXD1, or COL6A3 locus.

(126) A mutant DYSF, PYROXD1, or COL6A3 coding region or translation product can be detected in a variety of physiological samples collected from a horse. Examples of appropriate samples include a cell sample, such as a blood cell (e.g., a lymphocyte, a peripheral blood cell), a sample collected from the spinal cord, a tissue sample such as cardiac tissue or muscle tissue (e.g. cardiac or skeletal muscle) an organ sample (e.g., liver or skin), a hair sample, especially a hair sample with the hair bulb (roots) attached, and/or a fluid sample (e.g., blood).

(127) The methods described herein are applicable to any equine disease in which DYSF, PYROXD1, or COL6A3 has a role. The method may be particularly useful for, for example, a veterinarian, a Breed Association, and/or individual breeders, so they can decide upon an appropriate course of treatment, and/or to determine if an animal is a suitable candidate as a brood mare or sire.

(128) Oligonucleotide Probes

(129) As described above, the method may be used to detect the presence and/or absence of a polymorphism in equine DNA. In particular, mutations in the DYSF, PYROXD1, or COL6A3 gene include the specific substitution (1) of an adenine (A) for a guanine (G) on the forward strand at chr15:31,306,949 of the current horse genome assembly (EquCab2, GCA_000002305.1) as displayed in the UCSC Genome Browser and as shown for the reverse strand in FIG. **1**, (2) of a thymine (T) for a guanine (G) on the forward strand at chr15:31,225,630 of the current horse genome assembly (EquCab2, GCA_000002305.1) as displayed in the UCSC Genome Browser and as shown for the reverse strand in FIG. **13**, (3) the specific substitution of a cytosine (C) for a guanine (G) on the forward strand at chr6:47,661,977 of the current horse genome assembly (EquCab2, GCA_000002305.1) as displayed in the UCSC Genome Browser and as shown in FIG. **22**, and the specific substitution of a guanine (G) for a cytosine (C) on the forward strand at chr6:23,480,621 of the current horse genome assembly (EquCab2, GCA_000002305.1) as

displayed in the UCSC Genome Browser and as shown for the reverse strand in FIG. **33**. These substitutions result in (1) an arginine (R) at codon 253 in the dysferlin (DYSF) protein (SEQ ID NO:10) being replaced by a tryptophan (W), as shown in SEQ ID NO: 11, (2) a proline (P) at codon 1290 in the dysferlin (DYSF) protein (SEQ ID NO:57) being replaced by a threonine (T), as shown in SEQ ID NO:58, (3) an aspartate (D) at codon 492 in the pyridine nucleotide-disulfide oxidoreductase domain-containing protein 1 (PYROXD1) protein (SEQ ID NO:104) being replaced by a histidine (H), as shown in SEQ ID NO: 105, and (4) a glycine (G) at codon 2182 in the collagen type VI alpha 3 chain (COL6A3) protein (SEQ ID NO:164) being replaced by an alanine (A), as shown in SEQ ID NO: 165.

(130) A primer pair may be used to determine the nucleotide sequence of a particular DYSF, PYROXD1, or COL6A3 allele using PCR. A pair of single-stranded DNA primers can be annealed to sequences within or surrounding the DYSF, PYROXD1, or COL6A3 coding region in order to prime amplifying DNA synthesis of the DYSF, PYROXD1, or COL6A3 coding region itself. A complete set of primers allows one to synthesize all of the nucleotides of the DYSF, PYROXD1, or COL6A3 coding sequence. In some embodiments, a set of primers can allow synthesis of both intron and exon sequences. In some embodiments, allele-specific primers can be used. Such primers anneal only to particular DYSF, PYROXD1, or COL6A3 mutant alleles, and thus will only amplify product efficiently in the presence of the mutant allele as a template.

(131) The first step of the process involves contacting a physiological sample obtained from a horse, which sample contains nucleic acid, with an oligonucleotide probe to form a hybridized DNA. The oligonucleotide probe can be any probe having from about 4 or 6 bases up to about 80 or 100 bases or more. In one embodiment, the oligonucleotide probe can have between about 10 and about 20 bases.

(132) The primers themselves can be synthesized using conventional techniques and, in some cases, can be made using an automated oligonucleotide synthesizing machine. Given the DYSF, PYROXD1, or COL6A3 genomic sequences as partially set forth in SEQ ID NO:1, SEQ ID NO:56, SEQ ID NO:103, and SEQ ID NO:163, one can design a set of oligonucleotide primers to probe any portion of the DYSF, PYROXD1, or COL6A3 coding sequences. The primers may be designed to hybridize entirely to coding sequence (exons), to noncoding sequence (introns or other noncoding sequences), or to regions spanning the junction of coding and noncoding sequences in genomic DNA.

(133) An oligonucleotide probe may be prepared according to conventional techniques to have any suitable base sequence. Suitable bases for preparing the oligonucleotide probe may be selected from naturally-occurring bases such as adenine, cytosine, guanine, uracil, and thymine. An oligonucleotide probe also can incorporate one or more non-naturally-occurring or "synthetic" nucleotide bases. Exemplary synthetic bases include, for example, 7-deaza-guanine, 8-oxo-guanine, 6-mercaptoguanine, N4-acetylcytidine, 5-(carboxyhydroxyethyl)uridine, 2′-O-methylcytidine, 5-(carboxymethylaminomethyl)-2-thiouridine, 5-carboxymethylaminomethyluridine, dihydrouridine, 2′-O-methylpseudouridine, β,D-galactosylqueuosine, 2″-O-methylguanosine, inosine, N6-isopentenyladenosine, 1-methyladenosine, 1-methylpseudouridine, 1-methylguanosine, 1-methylinosine, 2,2-dimethylguanosine, 2-methyladenosine, N2-methylguanosine, 3-methylcytidine, 5-methylcytidine, N6-methyl adenosine, 7-methylguanosine, 5-methylaminomethyluridine, 5-methoxyaminomethyl-2-thiouridine, β,D-mannosylqueuosine, 5-methloxycarbonylmethyluridine, 5-methoxyuridine, 2-methylthio-N6-isopentenyladenosine, N-((9-β-D-ribofuranosyl-2-methylthiopurine-6-yl)carbamoyl)threonine, N-((9-β-D-ribofuranosylpurine-6-yl)N-methyl-carbamoyl)threonine, uridine-5-oxyacetic acid methylester, uridine-5-oxyacetic acid, wybutoxosine, pseudouridine, queuosine, 2-thiocytidine, 5-methyl-2-thiouridine, 2-thiouridine, 5-methyluridine, N-((9-beta-D-ribofuranosylpurine-6-yl)carbamoyl)threonine, 2′-O-methyl-5-methyluridine, 2′-O-methyluridine, wybutosine, and/or 3-(3-amino-3-carboxypropyl)uridine. Any oligonucleotide backbone may be employed, including DNA, RNA

(although RNA may be less preferred than DNA in certain circumstances), modified sugars such as carbocycles, and sugars containing 2 substitutions (e.g., fluoro or methoxy). The oligonucleotides may be oligonucleotides wherein at least one, or all, of the internucleotide bridging phosphate residues is a modified phosphate such as, for example, a methyl phosphate, a methyl phosphonotlioate, a phosphoroinorpholidate, a phosphoropiperazidate, and/or a phospholioramidate—for example, every other one of the internucleotide bridging phosphate residues may be modified. The oligonucleotide may be a "peptide nucleic acid" such as described in Nielsen et al., Science, 254, 1497-1500 (1991).

(134) The oligonucleotide probe should possess a sequence at least a portion of which is capable of binding to a known portion of the sequence of the nucleic acid in the physiological sample.

(135) In some embodiments, the nucleic acid in the sample may be contacted with a plurality of oligonucleotide probes having different base sequences (e.g., where there are two or more target nucleic acids in the sample, or where a single target nucleic acid is hybridized to two or more probes in a "sandwich" assay).

(136) The oligonucleotide probes provided herein may be useful for a number of purposes. For example, the oligonucleotide probes can be used to detect PCR amplification products and/or to detect mismatches with the DYSF, PYROXD1, or COL6A3 coding region or mRNA.

(137) Hybridization Methodology

(138) The nucleic acid from the physiological sample may be contacted with the oligonucleotide probe in any conventional manner. For example, the sample nucleic acid may be solubilized in solution and contacted with the oligonucleotide probe by solubilizing the oligonucleotide probe in solution with the sample nucleic acid under conditions that permit hybridization. Suitable hybridization conditions are well known to those skilled in the art. Alternatively, the sample nucleic acid may be solubilized in solution with the oligonucleotide probe immobilized on a solid or semisolid support, whereby the sample nucleic acid may be contacted with the oligonucleotide probe by immersing the solid or semisolid support having the oligonucleotide probe immobilized thereon in the solution containing the sample nucleic acid.

(139) Certain embodiments of the methods described herein relate to mutations in the DYSF, PYROXD1, or COL6A3 coding region or the diagnosis of Polysaccharide Storage Myopathy type 2 (PSSM2), also known as Myofibrillar Myopathy (MFM), or the detection of a predisposition for Polysaccharide Storage Myopathy type 2 (PSSM2), also known as Myofibrillar Myopathy (MFM), or to the detection of a mutant DYSF, PYROXD1, or COL6A3 allele in a horse.

(140) Mutations in the equine DYSF, PYROXD1, or COL6A3 coding regions (encoding the dysferlin, pyridine nucleotide-disulfide oxidoreductase domain-containing protein 1, and skeletal muscle protein collagen type VI alpha 3 chain) are present in many populations of horses affected by Polysaccharide Storage Myopathy type 2 (PSSM2), also known as Myofibrillar Myopathy (MFM). The differences in the genomic DNA between horses affected by Polysaccharide Storage Myopathy type 2 (PSSM2), also known as Myofibrillar Myopathy (MFM) include (1) point mutations at nucleic acid chr15:31,306,949 of the current horse genome assembly (EquCab2, GCA_000002305.1) as displayed in the UCSC Genome Browser and as shown in FIG. **1**, (2) point mutations at nucleic acid chr15:31,225,630 of the current horse genome assembly (EquCab2, GCA_000002305.1) as displayed in the UCSC Genome Browser and as shown in FIG. **13**, (3) chr6:47,661,977 of the current horse genome assembly (EquCab2, GCA_000002305.1) as displayed in the UCSC Genome Browser and as shown in FIG. **22**, and (4) chr6:23,480,621 of the current horse genome assembly (EquCab2, GCA_000002305.1) as displayed in the UCSC Genome Browser and as shown in FIG. **33**.

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(141) Dysferlin (DYSF)

(142) A mutation in the equine DYSF coding region (encoding the membrane protein dysferlin) is present in many populations of horses affected by Polysaccharide Storage Myopathy type 2

(PSSM2), also known as Myofibrillar Myopathy (MFM). The difference in the genomic DNA between horses with PSSM2 and control horses include a G-to-A substitution in DYSF Exon B (as defined in FIG. **3**) at nucleotide position chr15:31,306,949.

(143) FIG. **1** shows a portion of the current horse genome assembly (EquCab2, GCA_000002305.1) with coordinates as displayed in the UCSC Genome Browser centered on the chr15:31,306,949 position, the site of a substitution of an adenine (A) for a guanine (G) that results in the substitution of a tryptophan (W) for arginine (R) at amino acid position 253 in dysferlin as shown in FIG. **4** (SEQ ID NO:11). The reverse complement sequence is shown, with the site of a substitution of a thymine (T) for a cytosine (C) as indicated (SEQ ID NO:1). The single nucleotide polymorphism (SNP) defined by this base substitution is identified as rs1145077095 in dbSNP.

(144) There are 21 predicted transcripts of the DYSF coding region in public databases. These models differ somewhat in ways not relevant to the DYSF-R253W mutation. All 21 isoforms share a common segment that includes the chr15:31,306,949 position. This common segment is shown in FIG. **2**.

(145) FIG. **2** shows a portion of the normal equine DYSF Coding DNA Sequence (SEQ ID NO:2) and the mutant DYSF Coding DNA Sequence (SEQ ID NO:3) bearing the C to T mutation at nucleotide position 1027 in this figure, corresponding to chr15:31,306,949 as shown in SEQ ID NO:1 (FIG. **1**). This sequence is a region of perfect consensus among 21 different experimentally predicted mRNA isoforms. The numbering in FIG. **2** is that of isoform X1 (XM_023618907.1), which for this segment perfectly matches the numbering of isoforms X2 (XM_023618908.1), X3 (XM_023618909.1), X4 (XM_023618910.1), X5 (XM_023618911.1), and X6 (XM_023618912.1). The numbering of the start and end positions for X7 (XM_023618913.1) is 671-1699. The numbering for the start and end positions for X8 (XM_023618914.1) is 763-1791. The numbering for the start and end positions of X9 (XM_023618915.1), X11 (XM_023618917.1), X12 (XM_023618918.1), and X13 (XM_023618919.1) is 670-1698. The numbering for the start and end positions of X10 (XM_023618916.1) is 763-1791. The numbering for the start and end positions of X14 (XM_023618920.1) and X15 (XM_023618921.1) is 1075-2103. The numbering for the start and end positions of X16 (XM_023618922.1) and X18 (XM_023618924.1) is 1074-2102. The numbering for the start and end positions of X17 (XM_023618923.1) and X19 (XM_023618925.1) is 981-2009. The numbering for the start and end positions of X20 (XM_023618926.1) and X21 (XM_023618927.1) is 980-2008. In both sequences, the sequence of Exon B as shown in FIG. **3** is indicated in bold. The site of a C to T mutation site at nucleotide position 1027, corresponding to 31,306,949 in SEQ ID NO:1 (FIG. **1**), and to rs1145077095 in dbSNP, is underlined. The region of sequence comprising Exon B as shown in FIG. **3** is displayed as codons in the correct reading frame for both SEQ ID NO:2 and SEQ ID NO: 3 in FIG. **5**.

(146) For the sake of simplicity, the amino acid substitution caused by the substitution of an adenine (A) for a guanine (G) at the chr15:31,306,949 position, identified as rs1145077095 in dbSNP, will be referred to as DYSF-R253W, based on SEQ ID NO:11. The amino acid substitution caused by this mutation will remain the same regardless of the numerical position of the affected codon in alternative gene models. The wild-type (e.g., also referred to herein as “normal” or “unaffected”) allele of this coding region may be referred to as R253 and the mutant allele R253W.

(147) FIG. **3** shows a view of the current horse genome assembly (EquCab2, GCA_000002305.1) in the UCSC Genome Browser with exon sequences that match the partial DYSF Coding DNA Sequence (SEQ ID NO:2) and the mutant partial DYSF Coding DNA Sequence (SEQ ID NO:3). The DYSF Coding DNA Sequences correspond to Exons A, B, C, D, E, and F as indicated by the regions of sequence similarity of the translated genomic DNA to DYSF protein sequences from human (Homo), cattle (Bos), rat (Rattus), mouse (Mus), and DYSF protein sequences from the more distantly-related zebrafish (Danio) and African claw-toed frog (Xenopus). Partial matches to the paralogous protein myoferlin (MYOF) from human (Homo), cattle (Bos), mouse (Mus), rat (Rattus), and zebrafish (Danio) are also seen. The sequences of Exons A-F with 10 bp of flanking

intron sequence and their coordinates in the current horse assembly are displayed below the image from the UCSC Genome Browser. Sequence IDs are Exon A (SEQ ID NO:4), Exon B (SEQ ID NO:5), Exon C (SEQ ID NO:6.), Exon D (SEQ ID NO:7), Exon E (SEQ ID NO:8), and Exon F (SEQ ID NO:9).
(148) FIG. **4** shows a model of part of the normal protein sequence encoded by horse DYSF (XP_023474694.1, presented here as SEQ ID NO:10) corresponding to a translation of SEQ ID NO:2 shown in FIG. **2** and part of the altered protein sequence encoded by horse DYSF with the base substitution at chr15:31,306,949 (based on XP_023474694.1, presented here as SEQ ID NO:11) corresponding to a translation of SEQ ID NO:3 shown in FIG. **2**. The portion of the protein encoded by Exon B as shown in FIG. **3** is indicated in bold, while the amino acid position affected by the base substitution of an adenine (A) for a guanine (G) on the forward strand at the chr15:31,306,949 position, corresponding to rs1145077095 in dbSNP as shown in FIG. **1**, is underlined. The amino acid positions in XP_023474694.1 are indicated at the beginning and end of the sequence.
(149) FIG. **5** shows horse DYSF Exon B and flanking genomic DNA sequence from which PCR primers to amplify genomic DNA containing the site of the DYSF-R253W mutation would be most appropriately derived. Genomic coordinates are as in FIG. **1**. Exon B from chr15:31,307,036 to chr15:31,306,908 is shown broken into codons in the correct reading frame for the wild-type allele (SEQ ID NO:12) and the DYSF-R253W allele (SEQ ID NO:13). Only the reference sequence from the assembly is shown for the flanking sequences. The codon affected by the G to A mutation site at nucleotide position chr15:31,306,949, corresponding to rs1145077095 in dbSNP as shown in FIG. **1** (C to T in the reverse complement as shown), is shown in bold, with the position of the base substitution indicated by underlining. The base substitution changes the bold three base codon from one coding for an arginine (CGG) to one coding for a tryptophan (TGG). Example primers used experimentally to amplify genomic DNA containing the mutation site are shown in lower case
(150) TABLE-US-00001 (SEQ ID NO: 14) [5′-CCCGAGATTTCTGGCTTTCT-3′ and (SEQ ID NO: 15) 5′-CTCGACAAGTTCTGGGGTGT-3′].
(151) Genomic DNA obtained from horses can be genotyped by amplifying a region containing a variant in the DYSF gene using Polymerase Chain Reaction (PCR), then sequencing the amplified DNA using Sanger sequencing. The variant allele DYSF-R253W is abbreviated as P5, while the common or wild-type allele is abbreviated as N. The results can be scored as homozygous for the common or wild-type allele (N/N), heterozygous for the nucleotide substitution (N/P5), or homozygous for the nucleotide substitution (P5/P5).
(152) FIG. **6** shows traces from Sanger DNA sequencing of amplified DYSF genomic DNA using primers shown in FIG. **5** (SEQ ID NO:14 and SEQ ID NO:15). The sequence of the reverse strand is shown. The arrows in the figure indicate nucleotide position chr15:31,306,949, the site of a substitution of a thymine (T) for a cytosine (C) in this position, corresponding to rs1145077095 in dbSNP, that creates the DYSF-R253W variant. The traces show, from left to right, results for a horse homozygous for the wild-type or common allele (N/N), results for a horse heterozygous for the substitution (N/P5), and results for a horse homozygous for the substitution (P5/P5).
(153) Dysferlin is a member of a family of genes with a transmembrane domain. The majority of the protein faces the cytoplasm. Dysferlin has seven C2 domains that are implicated in calcium-dependent membrane fusion events. Dysferlin plays an important role in the repair of muscle fibers.
(154) Mice homozygous for targeted mutations in DYSF that are expected to result in a total loss of function display defects in skeletal muscle. These defects include “dystrophic” muscle that exhibits a progressive muscular weakness. Specific defects in limb grasping are observed. The morphology of muscle fibers in mice homozygous for loss-of-function alleles include centrally located nuclei, increased variability of skeletal muscle fiber size, and skeletal muscle fiber degeneration. Increased serum creatine kinase (CK), a sign of muscle damage, is also seen.
(155) The human ortholog of the equine DYSF gene and the human protein that this gene encodes

are richly annotated with experimental data derived from genetic and biochemical studies. It is informative to compare the amino acid substitutions in DYSF found in horses to the information on protein domains and clinically significant variation in the human dysferlin (DYSF) protein. In order to do this, the equine protein models used in this disclosure must be compared to the canonical or reference sequence of the human protein in a public database that captures data from the published literature, such as UniProt.

(156) A large number of pathogenic mutations in human DYSF are known. Many pathogenic alleles are nonsense mutations, that is, the mutation of a codon encoding an amino acid to a termination codon, causing the truncation of the encoded dysferlin at the site of the mutation. Other pathogenic alleles are frameshift mutations, that is, the addition or deletion of bases within coding regions where the number of bases is not an integral multiple of three, therefore altering the reading frame downstream of the mutation. The altered reading frame typically reaches a termination codon, causing truncation of the encoded dysferlin downstream of the mutation following a segment of altered amino acid sequence. There are also pathogenic missense alleles of DYSF, in which a point mutation causes the alteration of a single amino acid in the dysferlin protein sequence. A set of such alleles is presented in TABLE 1. Cited sources in this table are Aoki 2001 (Aoki et al. 2001 Neurology 57:271-278), Nguyen 2005 (Nguyen 2005 Hum Mut DOI: 10.1002/humu.9355), Krahn 2008 (Krahn et al. 2009 Hum Mut 30:E345-375 DOI: 10.1002/humu.20910), LMDD (http://www.dmd.n1), and ClinVar (https://www.ncbi.nlm.nih.gov/clinvar/).

(157) Human patients homozygous for missense alleles of DYSF as listed in TABLE 1 may present as cases of Limb-Girdle Muscular Dystrophy 2B or other conditions that overlap this condition clinically, such as Miyoshi Myopathy, Proximodistal Myopathy, Pseudometabolic Myopathy, or Isolated HyperCKemia (a disorder in which serum levels of creatine kinase or CK, a muscle enzyme, are elevated, indicating damage to the sarcolemma).

(158) TABLE-US-00002 TABLE 1 Pathogenic missense alleles of the C2B domain of human DYSF. Substitution Domain Reference S209F C2B ClinVar Q221H C2B ClinVar I227T C2B Jin 2016 P233L C2B ClinVar G234E C2B Krahn 2009, ClinVar R251W C2B ClinVar T252M C2B Jin 2016, ClinVar R253W C2B Nguyen 2005, ClinVar H255P C2B ClinVar E264D C2B UMD, ClinVar L266R C2B UMD, ClinVar I282T C2B UMD I284T C2B Krahn 2009, ClinVar T285M C2B ClinVar V286E C2B UMD, ClinVar D288V C2B Jin 2016, ClinVar S289P C2B ClinVar S289F C2B ClinVar G299R C2B Wenzel 2006, Spuler 2008, Krahn 2009, ClinVar G299E C2B UMD, ClinVar G299W C2B Spuler 2008, ClinVar, Hofhuis 2017

(159) Cited sources in this table are: ClinVar (https://www.ncbi.nlm.nih.gov/clinvar/), UMD (http://www.umd.be/DYSF/) Davis 2002 (Davis D B et al. 2002 J Biol Chem 277:22883-22888 DOI: 10.1074/jbc.M201858200) Hofhuis 2017 (Hofhuis J et al. 2017 J Cell Sci 130:841-852 DOI: 10.1242/jcs.198861) Huang 2007 (Huang Y et al. 2007 FASEB J. 21:732-742. DOI: 10.1096/fj.06-6628com) Illarioshkin 2000 (Illarioshkin S N et al. 2000 Neurology 55:1931-1933) Jin 2016 (Jib S-Q et al. 2016 Chin Med J 129:2287-2293 DOI: 10.4103/0366-6999.190671) Krahn 2009 (Krahn M et al. 2009 Hum Mutat. 30:E345-375 DOI: 10.1002/humu.20910) Nguyen 2005 (Nguyen K et al. 2005 Hum Mutat. 26:165 DOI: 10.1002/humu.9355), Spuler 2008 (Spuler S et al. 2008 Ann Neurol. 63:323-328 DOI: 10.1002/ana.21309) Wenzil 2006 (Wenzil K et al. 2006 Hum Mutat. 27:599-600 DOI: 10.1002/humu.9424)

(160) The C2B domain of dysferlin is highly conserved overall. Comparison to the other two C2 domains with type II topology (C2A and C2E) is helpful in assessing the pathogenicity of missense alleles. TABLE 2 shows pathogenic missense alleles in the other two C2 domains with type II topology.

(161) TABLE-US-00003 TABLE 2 Pathogenic missense alleles of the C2A and C2E domains of human DYSF. Substitution Domain Reference M1V C2A ClinVar M1T C2A ClinVar I6N C2A ClinVar W46R C2A UMD W52R C2A Krahn 2009, ClinVar V67D C2A Illarioshkin 2000, Davis

2002, Huang 2007, ClinVar, Hofhuis 2017 V69G C2A ClinVar T74M C2A ClinVar R1331L C2E ClinVar R1331C C2E ClinVar L1341P C2E Wenzel 2006, ClinVar, Hofhuis 2017 R1342Q C2E ClinVar R1342G C2E Jin 2016 R1342W C2E UMD, ClinVar C1361R C2E UMD, Campanaro 2002 P1400R C2E UMD, ClinVar G1418D C2E ClinVar

(162) Cited sources in this table are: ClinVar (https://www.ncbi.nlm.nih.gov/clinvar/), UMD (http://www.umd.be/DYSF/) Davis 2002 (Davis D B et al. 2002 J Biol Chem 277:22883-22888 DOI: 10.1074/jbc.M201858200) Hofhuis 2017 (Hofhuis J et al. 2017 J Cell Sci 130:841-852 DOI: 10.1242/jcs.198861) Huang 2007 (Huang Y et al. 2007 FASEB J. 21:732-742. DOI: 10.1096/fj.06-6628com) Illarioshkin 2000 (Illarioshkin S N et al. 2000 Neurology 55:1931-1933) Jin 2016 (Jib S-Q et al. 2016 Chin Med J 129:2287-2293 DOI: 10.4103/0366-6999.190671) Krahn 2009 (Krahn M et al. 2009 Hum Mutat. 30:E345-375 DOI: 10.1002/humu.20910) Wenzil 2006 (Wenzil K et al. 2006 Hum Mutat. 27:599-600 DOI: 10.1002/humu.9424)

(163) FIG. **7** shows the sequence of the human DYSF coding sequence derived from NM_003494.3 (SEQ ID NO:16). The 5′ UTR and 3′ UTR have been removed; the sequence begins with the ATG start codon and ends with the TGA stop codon. The numbering of the first and last nucleotides corresponds to that of NM_003494.3. The sequence of the human exon corresponding to Exon B in horse as shown in FIG. **3** is indicated in bold.

(164) FIG. **8** shows the sequence of the human DYSF protein sequence, equivalent to NP_003485.1 (SEQ ID NO:17). The numbering of the first and last amino acids corresponds to that of NP_003485.1. The sequence encoded by the human exon corresponding to Exon B in horse as shown in FIG. **3** is indicated in bold.

(165) Comparison of the equine DYSF-R253W missense allele to the pathogenic human alleles presented in TABLE 1 is informative. FIG. **9** shows a comparison of that portion of the protein sequence of DYSF encoded by horse Exon B as defined in FIG. **3** from wild type (XP_023474694.1, shown here as SEQ ID NO:19) and DYSF-R253W (R253W, shown here as SEQ ID NO:20) to the protein sequence of DYSF encoded by human Exon 7 (NP_003485.1, shown here as SEQ ID NO:18). Between the sequences of the horse and human proteins in the alignment, an asterisk (*) indicates an identical amino acid in that position, while a space ( ) indicates the nonconservative substitution of an arginine (R) in horse for a glycine (G) at human position 247, and a plus sign (+) indicates the conservative substitution of an arginine (R) in horse for a lysine (K) at human position 256. No other amino acid substitutions are seen in comparison of the human sequence (SEQ ID NO:18) and the wild-type horse sequence (SEQ ID NO:19). The sequence from horse bearing the DYSF-R253W mutation (SEQ ID NO:20) has a nonconservative substitution of a tryptophan (W) for an arginine (R) that is found in human (SEQ ID NO:18) and the wild-type horse sequence (SEQ ID NO:19). This position, corresponding to position 253 in horse (SEQ ID NO:19, SEQ ID NO:20) and position 251 in human (SEQ ID NO:18) is indicated by bold and underlining.

(166) FIG. **10** shows features of the dysferlin protein encoded by the human DYSF gene. The protein contains seven C2 calcium-binding domains designated C2A through C2G. The secondary structure of the protein sequence encoded by each of the C2 domains consists of eight segments assembling into beta sheet, with two different topologies described. Domains C2C, C2D, C2F, and C2G are designated as topology type I, while domains C2A, C2B, and C2E are designated as topology type II (C.Therrien et al. 2006 J. Neurological Sciences 250: 71-78). The positions of four additional conserved domains (DysfN, DysfC, annexin binding, and transmembrane) are also indicated (C.Therrien et al. 2006 J. Neurological Sciences 250: 71-78). Positions of pathogenic missense alleles listed in TABLE 1 are shown.

(167) The sequence comparison between human and equine dysferlin proteins shows the high degree of sequence conservation between these two species. It also shows that the human equivalent to the equine DYSF-R253W allele would be DYSF-R251W. This allele has been observed in human patients and has been scored as pathogenic (TABLE 1). The extent of sequence

conservation between horse and human is high, and it is of value to examine the extent of sequence conservation among a larger number of species.

(168) FIG. **11** shows amino acid sequences of proteins encoded by the C2B domain of DYSF, including the position of the equine DYSF-R253W substitution. The portion of the human DYSF corresponding to the C2B domain, as identified by UniProt annotation (positions 207-302 of NP_003485.1, identical to O75923) was used as a probe in BLASTP searches of the protein sequence database to identify the corresponding region in the orthologous protein in a set of mammals. Extending this approach to more distantly related organisms (birds, reptile, amphibians, and fish) was not successful; either no match was found, or the top match was to a nonorthologus protein (e.g. myoferlin). In the alignment shown in FIG. **11**, sequences that are identical over this range of positions in different species are clustered as a single SEQ ID NO.

(169) The 44 species in the alignment shown in FIG. **11** are presented below with their common names, their scientific names, and the protein database ID from which the amino acid sequence is derived. Amino acid sequences identical over this range in different species have been clustered into single SEQ ID NOs for the alignment. The species associated with the SEQ ID NOs shown in the alignment are: SEQ ID NO:21 [Human (*Homo sapiens*, XP_005264642.1), Chimpanzee (*Pan troglodytes*, XP_016804238.1), Bonobo (*Pan paniscus*, XP_003808226.1), Western lowland gorilla (Gorilla gorilla gorilla, XP_004029449.2), Sumatran orangutan (*Pongo abelii*, XP_024097762.1)], SEQ ID NO:22 [Olive baboon (*Papio anubis*, XP_021780640.1), Rhesus macaque (*Macaca mulatta*, XP_014968111.1), Black snub-nosed monkey (*Rhinopithecus bieti*, XP_017733098.1)], SEQ ID NO:23 [Mouse (*Mus musculus*, XP_006506246.1), Brown rat (*Rattus norvegicus*, XP_006236829.1)], SEQ ID NO:24 [Praire vole (*Microtus ochrogaster*, XP_005369881.1), Mongolian gerbil (*Meriones unguiculatus*, XP_021497362.1)], SEQ ID NO:25 [Long-tailed chinchilla (*Chinchilla lanigera*, XP_013370151.1), Damora mole-rat (*Fukomys damarensis*, KF021095.1)], SEQ ID NO:26 [David's myotis (*Myotis davidii*, XP_006775358.1), Brandt's bat (*Myotis brandtii*, XP_005867829.1)], SEQ ID NO:27 [Large flying fox (*Pteropus vampyrus*, XP_023389621.1), Alpaca (*Vicugna pacos*, XP_015092494.1), Dromedary (*Camelus dromedaries*, XP_010992957.1), Bactrian camel (*Camelus bactrianus*, XP_010962007.1)], SEQ ID NO:28 [Horse (*Equus caballus*, XP_023474692.1), Donkey (*Equus asinus*, XP_014692144.1), Przewalski's horse (*Equus przewalskii*, XP_008534159.1)], SEQ ID NO:29 [Cat (*Felis catus*, XP_003984159.2), Leopard (*Panthera pardus*, XP_019297096.1)], SEQ ID NO:30 [Goat (*Capra hircus*, XP_017910544.1), Mouflon (*Ovis aries musimon*, XP_011978930.1)], SEQ ID NO:31 [Panamanian white-throated capuchin (*Cebus capucinus imitator*, XP_017377464.1)], SEQ ID NO:32 [Gray mouse lemur (*Microcebus murinus*, XP_012605526.1)], SEQ ID NO:33 [Philipine tarsier (*Carlito syrichta*, XP_008058745.1)], SEQ ID NO:34 [Northern greater galago (*Otolemur garnettii*, XP_012662868.1)], SEQ ID NO:35 [Coquerel's sifika (*Propithecus coquereli*, XP_012501927.1)], SEQ ID NO:36 [Chinese hamster (*Cricetulus griseus*, ERE66849.1)], SEQ ID NO:37 [Thirteen-lined ground squirrel (*Ictidomys tridecemlineatus*, XP_005322205.1)], SEQ ID NO:38 [Alpine marmot (*Marmota marmota marmota*, XP_015335007.1)], SEQ ID NO:39 [American beaver (*Castor canadensis*, XP_020009964.1)], SEQ ID NO:40 [Ferret (*Mustela putorius furo*, XP_004742226.1)], SEQ ID NO:41 [Great roundleaf bat (*Hipposideros armiger*, XP_019503212.1)], SEQ ID NO:42 [African bush elephant (*Loxodonta africana*, XP_023408298.1)], SEQ ID NO:43 [Wild boar (*Sus scrofa*, XP_013851496.2)], SEQ ID NO:44 [Cattle (*Bos taurus*, NP_001095960.1)], SEQ ID NO:45 [Minke whale (*Balaenoptera acutorostrata scammoni*, XP_007175590.1)], SEQ ID NO:46 [Giant panda (*Ailuropoda melanoleuca*, XP_019650380.1)], SEQ ID NO:47 [Killer whale (*Orcinus orca*, XP_004277106.1)], SEQ ID NO:48 [Koala (*Phascolarctos cinereus*, XP_020856837.1)], SEQ ID NO:49 [Tasmanian devil (*Sarcophilus harrisii*, XP_023351344.1)], SEQ ID NO:50 [Platypus (*Ornithorhynchus anatinus*, XP_007665017.1)].

(170) The next to the last line (labeled CLUSTAL) shows the consensus sequence, where positions

with fully conserved amino acids are represented by an asterisk (*), positions with strongly conserved amino acids are indicated by a colon (:), positions with weakly conserved amino acids are indicated are indicated by period (.), and nonconserved positions are indicated by a blank space ( ). The last line shows the position of the DYSF-R253W substitution in horse in bold. The position of the DYSF-R253W substitution is indicated in bold in all of the sequences.
(171) The overall sequence conservation of the C2B domain of dysferlin is quite high across mammals. The amino acid alteration seen in the equine DYSF-R253W substitution is not seen in any of the 44 species; all have an arginine (R) at this position. The comparison of dysferlin C2B sequences presented here refutes the hypothesis that the DYSF-R253W substitution is selectively neutral, and supports the claim that the DYSF-R253W mutation found in horses with PSSM2 is pathogenic.
(172) FIG. **12** shows comparison of the C2A, C2B, and C2E domains of dysferlin encoded by human DYSF to the C2B domain of dysferlin encoded by horse DYSF. (A) The amino acid sequences of two isoforms of the C2A domain of human dysferlin (SEQ ID NO:51 and SEQ ID NO:52), the C2B domain of human dysferlin (SEQ ID NO:53), the C2E domain of human dysferlin (SEQ ID NO:54), and the C2B domain of horse dysferlin (SEQ ID NO:55), are shown. (B) Clustal Omega was used to align these four sequences. The fifth line (labeled CLUSTAL) shows the consensus sequence, where positions with fully conserved amino acids are represented by an asterisk (*), positions with strongly conserved amino acids are indicated by a colon (:), positions with weakly conserved amino acids are indicated are indicated by period (.), and nonconserved positions are indicated by a blank space ( ). The remaining lines show the position of the horse DYSF-R253W substitution and various pathogenic human substitutions (presented in TABLE 1 and TABLE 2) aligned to the consensus.
(173) The comparison of C2A, C2B, and C2E domains reveals a much more limited number of highly conserved positions. The position affected by the equine DYSF-R253W mutation is highly conserved, with only arginine (R) and glutamine (Q) seen in this position. In general, there is a good correspondence between positions that are conserved and the positions of substitutions identified in pathogenic human DYSF alleles. This analysis also refutes the hypothesis that the DYSF-R253W substitution is selectively neutral, and supports the claim that the DYSF-R253W mutation found in horses with Polysaccharide Storage Myopathy type 2 (PSSM2), also known as Myofibrillar Myopathy (MFM) is pathogenic.
(174) FIG. **13** shows portion of the current horse genome assembly (EquCab2, GCA_000002305.1) with coordinates as displayed in the UCSC Genome Browser centered on the chr15:31,225,630 position, the site of a substitution of an thymine (T) for a guanine (G) that results in the substitution of a threonine (T) for proline (P) at amino acid position 1290 in dysferlin as shown in FIG. **16** (SEQ ID NO:61). The reverse complement sequence is shown, with the site of a substitution of an adenine (A) for a cytosine (C) as indicated (SEQ ID NO:56). The single nucleotide polymorphism (SNP) defined by this base substitution is identified as rs1136366555 in dbSNP
(175) FIG. **14** shows a portion of the normal equine DYSF Coding DNA Sequence (SEQ ID NO:57) and the mutant DYSF Coding DNA Sequence (SEQ ID NO:58) bearing the C to A mutation at nucleotide position 4174 in this figure, corresponding to chr15:31,225,630 as shown in SEQ ID NO:56 (FIG. **13**). This sequence is a region of perfect consensus among 21 different experimentally predicted mRNA isoforms. The numbering in FIG. **14** is that of isoform X1 (XM_023618907.1), which for this segment perfectly matches the numbering of isoforms X2 (XM_023618908.1), X4 (XM_023618910.1), and X5 (XM_023618911.1). The numbering of the start and end positions for X3 (XM_023618909.1) and X6 (XM_023618912.1) is 3685-4143. The numbering of the start and end positions for X7 (XM_023618913.1) is 3634-4092. The numbering of the start and end positions for X8 (XM_023618914.1) is 3726-4184. The numbering of the start and end positions for X9 (XM_023618915.1) and X12 (XM_023618918.1) is 3633-4091. The numbering of the start and end positions for X10 (XM_023618916.1) is 3684-4142. The numbering

of the start and end positions for X11 (XM_023618917.1) and X13 (XM_023618919.1) is 3591-4049. The numbering of the start and end positions for X14 (XM_023618920.1) is 4038-4496. The numbering of the start and end positions for X15 (XM_023618921.1) is 3996-4454. The numbering of the start and end positions for X16 (XM_023618922.1) is 4037-4495. The numbering of the start and end positions for X17 (XM_023618923.1) is 3944-4402. The numbering of the start and end positions for X18 (XM_023618924.1) is 3995-4453. The numbering of the start and end positions for X19 (XM_023618925.1) is 3902-4360. The numbering of the start and end positions for X20 (XM_023618926.1) is 3943-4401. The numbering of the start and end positions for X21 (XM_023618927.1) is 3901-4359 In both sequences, the sequence of Exon I as shown in FIG. **15** is indicated in bold. The site of a C to A mutation site at nucleotide position 4174, corresponding to chr15:31,225,630 in SEQ ID NO:56 (FIG. **13**), and to to rs1136366555 in dbSNP, is underlined. The region of sequence comprising Exon I as shown in FIG. **15** is displayed as codons in the correct reading frame for both SEQ ID NO:57 and SEQ ID NO:58 in FIG. **17**.

(176) FIG. **15** shows a view of the current horse genome assembly (EquCab2, GCA_000002305.1) in the UCSC Genome Browser with exon sequences that match the partial DYSF Coding DNA Sequence (SEQ ID NO:57) and the mutant partial DYSF Coding DNA Sequence (SEQ ID NO:58). The DYSF Coding DNA Sequences correspond to Exons G, H, and I as indicated by the regions of sequence similarity of the translated genomic DNA to DYSF protein sequences from human (Homo), orangutan (Pongo), cattle (Bos), mouse (Mus), rat (Rattus) and DYSF protein sequences from the more distantly-related zebrafish (Danio). The sequence of Exon I with 10 bp of flanking intron sequence and its coordinates in the current horse assembly is displayed below the image from the UCSC Genome Browser (SEQ ID NO: 59).

(177) FIG. **16** shows models of part of the normal protein sequence encoded by horse DYSF (XP_023474694.1, presented here as SEQ ID NO:60) corresponding to a translation of SEQ ID NO:57 shown in FIG. **14** and part of the altered protein sequence encoded by horse DYSF with the base substitution at chr15:31,225,630 (based on XP_023474694.1, presented here as SEQ ID NO:61) corresponding to a translation of SEQ ID NO:58 shown in FIG. **14**. The portion of the protein encoded by Exon I as shown in FIG. **15** is indicated in bold, while the amino acid position affected by the base substitution of an adenine (A) for a cytosine (C) at the chr15:31,225,630 position, corresponding to rs1136366555 in dbSNP as shown in FIG. **13**, is underlined. The amino acid positions in XP_023474694.1 are indicated at the beginning and end of the sequence.

(178) FIG. **17** shows horse DYSF Exon I and flanking genomic DNA sequence from which PCR primers to amplify genomic DNA containing the site of the DYSF-P1290T mutation would be most appropriately derived. Genomic coordinates are as in FIG. **13**. Exon I from chr15:31,225,648 to chr15:31,225,619 is shown broken into codons in the correct reading frame for the wild-type allele (SEQ ID NO:62) and the DYSF-P1290T allele (SEQ ID NO:63). Only the reference sequence from the assembly is shown for the flanking sequences. The codon affected by the G to T mutation site at nucleotide position chr15:31,225,630, corresponding to rs1136366555 in dbSNP as shown in FIG. **13** (C to A in the reverse complement as shown), is shown in bold, with the position of the base substitution indicated by underlining. The base substitution changes the bold three base codon from one coding for a proline (CCT) to one coding for a threonine (ACT). Example primers used experimentally to amplify genomic DNA containing the mutation site are shown in lower case [5″-GGTTGCAAACTCCCAACTGT-3′ (SEQ ID NO:64) and 5′-GATTTTTCAAGCTGCCGAAG-3′ (SEQ ID NO:65)].

(179) Genomic DNA obtained from horses can be genotyped by amplifying a region containing a variant in the DYSF gene using Polymerase Chain Reaction (PCR), then sequencing the amplified DNA using Sanger sequencing. The variant allele DYSF-P1290T is abbreviated as P6, while the common or wild-type allele is abbreviated as N. The results can be scored as homozygous for the common or wild-type allele (N/N), heterozygous for the nucleotide substitution (N/P6), or homozygous for the nucleotide substitution (P6/P6).

(180) FIG. **18** shows traces from Sanger DNA sequencing of amplified DYSF genomic DNA using primers shown in FIG. **17** (SEQ ID NO:64 and SEQ ID NO:65). The sequence of the forward strand is shown. The arrows in the figure indicate nucleotide position chr15:31,225,630, the site of a substitution of a thymine (T) for a guanine (G) in this position, corresponding to rs1136366555 in dbSNP, that creates the DYSF-P1290T variant. The traces show, from left to right, results for a horse homozygous for the wild-type or common allele (N/N), results for a horse heterozygous for the substitution (N/P6), and results for a horse homozygous for the substitution (P6/P6).
(181) The sequence of the human DYSF coding sequence is presented in FIG. **7**, while the sequence of the human DYSF protein sequence is presented in FIG. **8**, as discussed above.
(182) Comparison of the equine DYSF-P1290T missense allele to the pathogenic human alleles presented in TABLE 3 is informative. FIG. **19** shows a comparison of that portion of the protein sequence of DYSF encoded by horse Exon I from wild type (XP_023474694.1, shown here as SEQ ID NO:67) and DYSF-P1290T (P1290T, shown here as SEQ ID NO:68) to the protein sequence of DYSF encoded by human (NP_003485.1, shown here as SEQ ID NO:66). Between the sequences of the horse and human proteins in the alignment, an asterisk (*) indicates an identical amino acid in that position, while a space ( ) indicates the nonconservative substitution of an glycine (G) in horse for an arginine (R) in human at human position 1297, and a plus sign (+) indicates the conservative substitution of an alanine (A) in horse for a serine (S) in human at human position 1267, a tyrosine (Y) in horse for a histidine (H) in human position 1285, an aspartic acid (D) in horse for a glutamic acid (E) in human position 1294, and a glutamic acid (E) in horse for an aspartic acid (D) in human position 1300. No other amino acid substitutions are seen in comparison of the human sequence (SEQ ID NO:66) and the wild-type horse sequence (SEQ ID NO:67). The sequence from horse bearing the DYSF-P1290T mutation (SEQ ID NO:68) has a nonconservative substitution of a threonine (T) for a proline (P) that is found in human (SEQ ID NO:66) and the wild-type horse sequence (SEQ ID NO:67). This position, corresponding to position 1290 in horse (SEQ ID NO:67, SEQ ID NO:68) and position 1288 in human (SEQ ID NO:66) is indicated by bold and underlining.
(183) Pathogenic and potentially pathogenic missense alleles are spread across the whole of dysferlin. Besides the pathogenic missense alleles described in the C2 calcium-binding domains C2B, C2A, and C2E described earlier and shown in TABLE 1 and TABLE 2, there are pathogenic and potentially pathogenic missense alleles in the interdomain region between C2D and C2E, as shown in TABLE 3.
(184) TABLE-US-00004 TABLE 3 Pathogenic missense alleles of the C2D - C2E interdomain region of human DYSF. Substitution Domain Reference P1247L C2D-C2E Interdomain ClinVar R1254W C2D-C2E Interdomain ClinVar T1261M C2D-C2E Interdomain ClinVar G1268E C2D-C2E Interdomain ClinVar L1270R C2D-C2E Interdomain ClinVar L1270P C2D-C2E Interdomain ClinVar L1276P C2D-C2E Interdomain Jin 2016 S1302P C2D-C2E Interdomain ClinVar T1305K C2D-C2E Interdomain ClinVar V1321F C2D-C2E Interdomain ClinVar R1331L C2D-C2E Interdomain UMD, Hofhuis 2017 E1335G C2D-C2E Interdomain UMD
(185) Cited sources in this table are: ClinVar (https://www.ncbi.nlm.nih.gov/clinvar/), UMD (http://www.umd.be/DYSF/) Hofhuis 2017 (Hofhuis J et al. 2017 J Cell Sci 130:841-852 DOI: 10.1242/jcs.198861) Jin 2016 (Jib S-Q et al. 2016 Chin Med J 129:2287-2293 DOI: 10.4103/0366-6999.190671)
(186) FIG. **20** shows features of the dysferlin protein encoded by the human DYSF gene. (A) The protein contains seven C2 calcium-binding domains designated C2A through C2G, shaded in gray. The positions of four additional conserved domains (DysfN, DysfC, annexin binding, and transmembrane) are also indicated (C.Therrien et al. 2006 J. Neurological Sciences 250: 71-78), also shaded in gray. The interdomain region between the C2D and C2E domains, affected by the horse DYSF-P1290T mutation, is indicated in light gray. (B) The interdomain region between the C2D and C2E domains is shown expanded, with positions of pathogenic and potentially pathogenic

missense alleles listed in TABLE 3 shown. The horse DYSF-P1290T substitution corresponds to a proline at position 1288 in human, with no pathogenic or potentially pathogenic alleles identified at that position in human.

(187) The extent of sequence conservation between horse and human is high, and it is of value to examine the extent of sequence conservation among a larger number of species, in order to assess the likelihood that the horse DYSF-P1290T allele is pathogenic.

(188) FIG. **21** shows amino acid sequences of proteins encoded by part of the C2D-C2E interdomain region of DYSF, including the position of the equine DYSF-P1290T substitution. Part of the portion of the human DYSF corresponding to the C2D-C2E interdomain region, as identified by UniProt annotation (positions 1261-1320 of NP_003485.1, identical to 075923) was used as a probe in BLASTP searches of the protein sequence database to identify the corresponding region in the orthologous protein in a set of mammals. Extending this approach to more distantly related organisms (birds, reptile, amphibians, and fish) was not successful; either no match was found, or the top match was to a nonorthologus protein (e.g. myoferlin). In the alignment shown in FIG. **11**, sequences that are identical over this range of positions in different species are clustered as a single SEQ ID NO.

(189) The 46 species in the alignment shown in FIG. **21** are presented below with their common names, their scientific names, and the protein database ID from which the amino acid sequence is derived. Amino acid sequences identical over this range in different species have been clustered into single SEQ ID NOs for the alignment. The species associated with the SEQ ID NOs shown in the alignment are: SEQ ID NO:69 [Human (*Homo sapiens*, XP_520786.2), Chimpanzee (*Pan troglodytes*, XP_016804234.1), Bonobo (*Pan paniscus*, XP_003808233.1), Western lowland gorilla (*Gorilla gorilla gorilla*, XP_004029453.2)], SEQ ID NO:70 [Sumatran orangutan (*Pongo abelii*, XP_024097764.1), Panamanian white-throated capuchin (*Cebus capucinus imitator*, XP_017377462.1)], SEQ ID NO:71 [Mouse (*Mus musculus*, XP_006506242.1), Rat (*Rattus norvegicus*, XP_006236834.1)], SEQ ID NO:72 [David's myotis (*Myotis davidii*, XP_006775352.1), Great roundleaf bat (*Hipposideros armiger*, XP_019503204.1)], SEQ ID NO:73 [Dromedary (*Camelus dromedarius*, XP_010992957.1), Bactrian camel (*Camelus bactrianus*, XP_010962003.1), Alpaca (*Vicugna pacos*, XP_015092494.1)], SEQ ID NO:74 [Cat (*Felis catus*, XP_003984159.2), Leopard (*Panthera pardus*, XP_019297033.1)], SEQ ID NO:75 [Cattle (*Bos taurus*, NP_001095960.1), Goat (*Capra hircus*, XP_017910552.1), Mouflon (*Ovis aries musimon*, XP_011978926.1)], SEQ ID NO:76 [Wild boar (*Sus scrofa*, XP_013851496.2), Sunda pangolin (*Manis javanica*, XP_017524151.1)], SEQ ID NO:77 Olive baboon (*Papio anubis*, XP_021780643.1), SEQ ID NO:78 Rhesus macaque (*Macaca mulatta*, XP_014968113.1), SEQ ID NO:79 Black snub-nosed monkey (*Rhinopithecus bieti*, XP_017733087.1), SEQ ID NO:80 Gray mouse lemur (*Microcebus murinus*, XP_012605531.1), SEQ ID NO:81 Philipine tarsier (*Carlito syrichta*, XP_008058745.1), SEQ ID NO:82 Northern greater galago (*Otolemur garnettii*, XP_012662869.1), SEQ ID NO:83 Coquerel's sifika (*Propithecus coquereli*, XP_012501928.1), SEQ ID NO:84 Chinese hamster (*Cricetulus griseus*, XP_027284348.1), SEQ ID NO:85 Prairie vole (*Microtus ochrogaster*, XP_005369881.1), SEQ ID NO:86 Mongolian gerbil (*Meriones unguiculatus*, XP_021497376.1), SEQ ID NO:87 Thirteen-lined ground squirrel (*Ictidomys tridecemlineatus*, XP_005322200.1), SEQ ID NO:88 Long-tailed chinchilla (*Chinchilla lanigera*, XP_005385648.1), SEQ ID NO:89 Alpine marmot (*Marmota marmota marmota*, XP_015335021.1), SEQ ID NO:90 American beaver (*Castor canadensis*, XP_020009964.1), SEQ ID NO:91 Ferret (*Mustela putorius furo*, XP_004742233.1), SEQ ID NO:92 Damora mole-rat (*Fukomys damarensis*, XP_010607925.1), SEQ ID NO:93 Brandt's bat (*Myotis brandtii*, XP_005867829.1), SEQ ID NO:94 Large flying fox (*Pteropus vampyrus*, XP_023389621.1), SEQ ID NO:95 Horse (*Equus caballus*, XP_023474694.1), SEQ ID NO:96 Donkey (*Equus asinus*, XP_014692152.1), SEQ ID NO:97 African bush elephant (*Loxodonta africana*, XP_023408298.1), SEQ ID NO:98 Minke whale (*Balaenoptera acutorostrata scammoni*, XP_007175594.1), SEQ ID

NO:99 Giant panda (*Ailuropoda melanoleuca*, XP_019650381.1), SEQ ID NO:100 Killer whale (*Orcinus orca*, XP_004277100.1), SEQ ID NO:101 Koala (*Phascolarctos cinereus*, XP_020856831.1), SEQ ID NO:102 Platypus (*Ornithorhynchus anatinus*, XP_007665017.1)
(190) The next to the last line (labeled CLUSTAL) shows the consensus sequence, where positions with fully conserved amino acids are represented by an asterisk (*), positions with strongly conserved amino acids are indicated by a colon (:), positions with weakly conserved amino acids are indicated are indicated by period (.), and nonconserved positions are indicated by a blank space ( ). The last line shows the position of the DYSF-P1290T substitution in horse in bold. The position of the DYSF-P1290T substitution is indicated in bold in all of the sequences.
(191) The overall sequence conservation of the C2D-C2E interdomain region of dysferlin is quite high across mammals. The amino acid alteration seen in the equine DYSF-P1290T substitution is not seen in any of the 44 species; all have a proline (P) at this position. The comparison of dysferlin C2D-C2E interdomain sequences presented here refutes the hypothesis that the DYSF-P1290T substitution is selectively neutral, and supports the claim that the DYSF-P1290T mutation found in horses with PSSM2 is pathogenic.
(192) Pyridine Nucleotide-Disulfide Oxidoreductase Domain-Containing Protein 1 (PYROXD1)
(193) A mutation in the equine PYROXD1 coding region (encoding the muscle protein pyridine nucleotide-disulfide oxidoreductase domain-containing protein 1) is present in many populations of horses affected by Polysaccharide Storage Myopathy type 2 (PSSM2), also known as Myofibrillar Myopathy (MFM). The difference in the genomic DNA between horses with PSSM2, also known as MFM, and control horses include a G-to-C substitution in PYROXD1 Exon 12 (as defined in FIG. **15**) at nucleotide position chr6:47,661,977 of the current horse genome assembly (EquCab2, GCA_000002305.1), as shown in FIG. **13**).
(194) FIG. **22** shows a portion of the current horse genome assembly (EquCab2, GCA_000002305.1) with coordinates as displayed in the UCSC Genome Browser centered on the chr6:47,661,977 position, the site of a substitution of a cytosine (C) for a guanine (G) that results in the substitution of a histidine (H) for an aspartate (D) at amino acid position 492 in pyridine nucleotide-disulfide oxidoreductase domain-containing protein 1 (PYROXD1) as shown in FIG. **16** (SEQ ID NO:60). The single nucleotide polymorphism (SNP) defined by this base substitution is identified as rs1136260157 in dbSNP.
(195) There are five predicted transcripts of the PYROXD1 coding region in public databases. These models differ somewhat in ways not relevant to the PYROXD1-D492H mutation. All five isoforms share a common segment that includes the chr6:47,661,977 position. This common segment is shown in FIG. **23**.
(196) FIG. **23** shows a portion of the normal equine PYROXD1 Coding DNA Sequence (SEQ ID NO:104) and a portion of the mutant PYROXD1 Coding DNA Sequence (SEQ ID NO:105) bearing the G to C mutation at nucleotide position 1548 in this figure, corresponding to chr6:47,661,977 as shown in SEQ ID NO:103 (FIG. **22**). This sequence is a region of perfect consensus among five different experimentally predicted mRNA isoforms. The numbering in FIG. **23** is that of isoform X1 (XM_001502130.5), with the segment from 238 to 1583 shown. The numbering of the start and end positions for X2 (XM_023643216.1) is 148-1493. The numbering of the start and end positions for X3 (XM_023643217.1) is 187-1532. The numbering of the start and end positions for X4 (XM_023643218.1) is 151-1496. The numbering of the start and end positions for X5 (XM_023643219.1) is 191-1536. In both sequences the sequence of Exon 12 as shown in FIG. **24** is indicated in bold. The site of a G to C mutation at nucleotide position 1548, corresponding to 47,661,977 in SEQ ID NO:103 (FIG. **22**), and to rs1136260157 in dbSNP, is underlined. The region of sequence comprising Exon 12 as shown in FIG. **24** is displayed as codons in the correct reading frame for both SEQ ID NO:104 and SEQ ID NO:1057 in FIG. **26**.
(197) For the sake of simplicity, the amino acid substitution caused by the substitution of an cytosine (C) for a guanine (G) at the chr6:47,661,977 position, identified as rs1136260157 in

dbSNP, will be referred to as PYROXD1-D492H, based on SEQ ID NO:108. The amino acid substitution caused by this mutation will remain the same regardless of the numerical position of the affected codon in alternative gene models. The wild-type (e.g., also referred to herein as "normal" or "unaffected") allele of this coding region may be referred to as D492 and the mutant allele D492H.

(198) FIG. **24** shows a view of the current horse genome assembly (EquCab2, GCA_000002305.1) in the UCSC Genome Browser with exon sequences that match the PYROXD1 Coding DNA Sequence (SEQ ID NO:104) and the mutant PYROXD1 Coding DNA Sequence (SEQ ID NO:105) in a BLAT search. The PYROXD1 Coding DNA Sequences in horse correspond to PYROXD1 coding sequences in other species as indicated by the regions of sequence similarity of the translated genomic DNA to PYROXD1 protein sequences from human (Homo), orangutan (Pongo), cattle (Bos), mouse (Mus), rat (Rattus), African clawed frog (Xenopus), zebrafish (Danio), chicken (Gallus), and fruit fly (Drosophila). Exon 12, which contains the guanine (G) to cytosine (C) variant at chr6:47,661,977 as shown in SEQ ID NO:55 (FIG. **13**), is indicated below the image of the browser window. The sequence of horse Exon 12 with 10 nucleotides of flanking intron sequence and its coordinates in the current horse assembly is displayed below the image from the UCSC Genome Browser (SEQ ID NO:106).

(199) Conceptual translation of the nucleic acid sequences shown in FIG. **23** (SEQ ID NO:104 and SEQ ID NO:105) yields the amino acid sequences for the proteins shown in FIG. **25**. FIG. **25** shows models of part of the normal protein sequence encoded by horse PYROXD1 (XP_001502180.3, presented here as SEQ ID NO:107) corresponding to a translation of SEQ ID NO:104 shown in FIG. **23** and part of the altered protein sequence encoded by mutant horse PYROXD1 (adapted from XP_001502180.3, presented here as SEQ ID NO:108) with the base substitution at chr6:47,661,977 corresponding to a translation of SEQ ID NO:105 shown in FIG. **23**. The portion of the protein encoded by Exon 12 is indicated in bold, while the amino acid at position 492 affected by the base substitution of a cytosine (C) for a guanine (G) at the chr6:47,661,977 position as shown in FIG. **23**, is underlined. The amino acid positions in XP_001502180.3 are indicated at the beginning and end of the sequence.

(200) FIG. **26** shows horse PYROXD1 Exon 12 and flanking genomic DNA sequence from which PCR primers to amplify genomic DNA containing the site of the PYROXD1-D492H mutation would be most appropriately derived. Genomic coordinates are as in FIG. **22**. Exon 12 from chr6:47,661,764 to chr6:47,662,012 is shown broken into codons in the correct reading frame for the wild-type allele (SEQ ID NO:104) and the PYROXD1-D492H allele (SEQ ID NO:105). Only the reference sequence from the assembly is shown for the flanking sequences. The codon affected by the G to C mutation site at nucleotide position chr6:47,661,977, as shown in FIG. **22** is shown in bold, with the position of the base substitution indicated by underlining. The base substitution changes the bold three base codon from one coding for an aspartate (GAT) to one coding for a histidine (CAT). Example primers used experimentally to amplify genomic DNA containing the mutation site [5′-CAGATTTTCTGCTGGCCATT-3′ (SEQ ID NO:111) and 5-TGGTCATCATTAAATCAGTGCAA-3′ (SEQ ID NO:112)] are shown in lower case in the figure.

(201) Genomic DNA obtained from horses can be genotyped by amplifying a region containing a variant in the PYROXD1 gene using Polymerase Chain Reaction (PCR), then sequencing the amplified DNA using Sanger sequencing. The variant allele PYROXD1-D492H is abbreviated as P8, while the common or wild-type allele is abbreviated as n. The results can be scored as homozygous for the common or wild-type allele (n/n), heterozygous for the nucleotide substitution (n/P8), or homozygous for the nucleotide substitution (P8/P8).

(202) FIG. **27** shows traces from Sanger DNA sequencing of amplified PYROXD1 genomic DNA using primers shown in FIG. **26** (SEQ ID NO:111 and SEQ ID NO:112). The arrows in the figure indicate nucleotide position chr6:47,661,977, the site of a substitution of a cytosine (C) for a guanine (G) in this position that creates the PYROXD1-D492H variant. The traces show, from left

to right, results for a horse homozygous for the wild-type or common allele (n/n), results for a horse heterozygous for the substitution (n/P8), and results for a horse homozygous for the substitution (P8/P8).

(203) The human ortholog of the equine PYROXD1 gene encodes pyridine nucleotide-disulfide oxidoreductase domain-containing protein 1, one member of a family of nuclear-cytoplasmic pyridine nucleotide-disulfide reductases (PNDR). These are flavoproteins, bound to flavin adenine dinucleotide (FAD), that catalyze pyridine-nucleotide-dependent (nicotinamide adenine dinucleotide or NAD) reduction of thiol residues in other proteins. PYROXD1 differs from five other human class I PNDRs, dihydrolipoamide dehydrogenase (DLD), glutathione reductase (GSR), thioredoxin 1, 2, and 3 (TXNRD1, TXNRD2, and TXNRD3), in not having a consensus redox active site in the oxidoreductase domain, and in not having a conserved C-terminal dimerization domain found in all other class I PNDRs; instead, it bears a highly conserved C-terminal nitrile reductase domain. Recessive mutations causing a partial loss of function of PYROXD1 are associated with early-onset myopathy in humans (O'Grady et al. 2016 Am J Hum Genet. 99:1086-1105 DOI: 10.1016/j.ajhg.2016.09.005).

(204) Human patients with various mutations in PYROXD1 show progressive weakness, reduction in muscle bulk, and some experience difficulty walking in the second decade of life. Muscle biopsies show central nuclei and disorganized inclusions with positive staining for the Z disc proteins desmin (encoded by DES) and myotilin (encoded by MYOT). Electron microscopy shows sarcomeric disorganization including disorganized Z discs and nemaline rods (a finding associated with several different types of Nemaline Myopathy). The myopathy associated with mutations in PYROXD1 in humans therefore has features associated with several different types of inherited myopathies in humans (Centronuclear Myopathy, Myofibrillar Myopathy, and Nemaline Myopathy).

(205) In mouse, the Knockout Mouse Project (KOMP), a high-throughput gene knockout project, generated a loss-of-function allele of PYROXD1. Mice homozygous for the knockout allele die as embryos prior to organogenesis. Heterozygotes were smaller than normal and had abnormal vertebrae morphology. The muscle phenotype of heterozygotes has not been studied in detail, but it is evident from these results that a reduction in the level of pyridine nucleotide-disulfide oxidoreductase domain-containing protein 1 has phenotypic consequences, that is, the gene is haploinsufficient, and the activity of the pyridine nucleotide-disulfide oxidoreductase domain-containing protein 1 polypeptide is limiting.

(206) In zebrafish, injection of morpholinos (a synthetic molecule containing DNA bases attached to a backbone of methylenemorpholine rings linked through phosphorodiamidate groups) targeting expression of ryroxd1 (the zebrafish ortholog of horse and human PYROXD1) causes reduction of the Ryroxd1 protein. Treated embryos show fragmentation of muscle fibers, loss of Z disc structure, and the formation of nemaline bodies. Treated embryos exhibit impaired swimming performance. These defects can be partially reversed through the injection of human PYROXD1 mRNA. These experiments show that PYROXD1 function is necessary for both normal muscle structure and function in zebrafish (O'Grady et al. 2016 Am J Hum Genet. 99:1086-1105).

(207) In Drosophila, two different lines ubiquitously expressing RNA interference constructs for CG10721 (the Drosophila homolog of horse and human PYROXD1) were developed (Saha et al. 2018 Physiol Genomics doi:10.1152). Flies bearing these constructs failed to complete development, failing to emerge from their pupal cases, demonstrating the importance of PYROXD1 and its homologs in development.

(208) It is therefore informative to compare the horse PYROXD1 gene and the protein that it encodes with the human ortholog. FIG. **28** shows the partial sequence of the human PYROXD1 coding sequence derived from NM_024854.4 (SEQ ID NO:113). This sequence is a region of perfect consensus among three different experimentally predicted mRNA isoforms. The numbering of the first and last nucleotides corresponds to that of NM_024854.4, with the segment from 619 to

1630 shown. The numbering of the start and end positions for transcript variant 2 (NM_001350912.1) is 1032-2043. The numbering of the start and end positions for transcript variant 3 (NM_001350913.1) is 545-1556. The sequence of Exon 12 is indicated in bold. The 3′ UTR has been removed; the sequence begins with beginning of the consensus among the three isoforms and ends with the TAA stop codon.

(209) FIG. **29** shows the partial sequence of the human PYROXD1 protein sequence, showing a translation of SEQ ID NO:113, equivalent to NP_079130.2 (SEQ ID NO:114). The numbering of the first and last amino acids corresponds to that of NP_079130.2.

(210) FIG. **30** shows a comparison of that portion of the protein sequence of PYROXD1 encoded by horse Exon 12 from wild type (XP_001502180.3, shown here as SEQ ID NO:116) and PYROXD1-D492H (D492H, derived from XP_001502180.3 and shown here as SEQ ID NO:117) to the protein sequence of PYROXD1 encoded by human Exon 12 (NP_079130.2, shown here as SEQ ID NO:115). Between the sequences of the horse and human proteins in the alignment, an asterisk (*) indicates an identical amino acid in that position, while a plus sign (+) indicates the following conservative substitutions: a glutamine (Q) for an arginine (R) at human position 444, a valine (V) for an isoleucine (I) at human position 447, an alanine (A) for a serine (S) at human position 483, and an aspartate (D) for an asparagine (N) at human position 492. The sequence from horse bearing the PYROXD1-D492H mutation (SEQ ID NO:117) has a nonconservative substitution of a histidine (H) for an aspartate (D) at horse position 492, corresponding to human position 490. This position is indicated in bold for all three sequences. The wild-type horse sequence (SEQ ID NO:116) matches the human sequence (SEQ ID NO:115) at this position.

(211) FIG. **31** shows features of the pyridine nucleotide-disulfide oxidoreductase domain-containing protein 1 protein encoded by the human PYROXD1 gene. The human protein has 500 amino acids and two domains: the pyridine nucleotide-disulfide oxidoreductase domain (amino acids 39-361) and the NADH-dependent nitrite reductase domain (447-494) as described in O'Grady et al. 2016 Am J Hum Genet. 99:1086-1105. Positions of two human pathogenic missense alleles (N1555 and Q372H, O'Grady et al. 2016 Am J Hum Genet. 99:1086-1105) are shown, as is the position of the horse D492H mutation described herein.

(212) The sequence comparison between the proteins encoded by horse and human PYROXD1 shows the high degree of sequence conservation between these two species. The extent of sequence conservation is so high that it is difficult to evaluate whether this comparison offers any evidence that the equine PYROXD1-D492H allele would be expected to be pathogenic.

(213) FIG. **32** shows partial amino acid sequences of proteins encoded by PYROXD1, including the position of the equine PYROXD1-D492H substitution. A portion of the human PYROXD1 protein sequence was used as a probe to identify the corresponding region in the orthologous protein across a wide range of species, including mammals, birds, reptiles, amphibians, and fish. In the alignment shown in FIG. **32**, sequences that are identical over this range of positions are clustered as a single SEQ ID NO.

(214) The 92 species in the alignment shown in FIG. **32** are presented below with their common names, their scientific names, and the protein database ID from which the amino acid sequence is derived. Amino acid sequences identical over this range in different species have been clustered into single SEQ ID NOs for the alignment. The species associated with the SEQ ID NOs shown in the alignment are: SEQ ID NO:118 [Chimpanzee (*Pan troglodytes*, XP_520786.2), Bonobo (*Pan paniscus*, XP_003828944.1), Western lowland gorilla (*Gorilla gorilla gorilla*, XP_018894407.1), Olive baboon (*Papio anubis*, XP_009178608.1), Rhesus macaque (*Macaca mulatta*, EHH20571.1), Panamanian white-throated capuchin (*Cebus capucinus imitator*, XP_017378442.1), Philipine tarsier (*Carlito syrichta*, XP_008064020.1), Coquerel's sifika (*Propithecus coquereli*, XP_012496437.1)], SEQ ID NO:119 [Sumatran orangutan (*Pongo abelii*, NP_001127199.1), Gray mouse lemur (*Microcebus murinus*, XP_012640956.1), Large flying fox (*Pteropus vampyrus*, XP_011379886.1), Great roundleaf bat (*Hipposideros armiger*, XP_019485916.1), African bush

elephant (*Loxodonta africana*, XP_003405731.1)], SEQ ID NO:120 [Mouse (*Mus musculus*, XP_017177048.1), Mongolian gerbil (*Meriones unguiculatus*, XP_021514531.1), Sunda pangolin (*Manis javanica*, XP_017517789.1)], SEQ ID NO:121 [Black snub-nosed monkey (*Rhinopithecus bieti*, XP_017744103.1), Thirteen-lined ground squirrel (*Ictidomys tridecemlineatus*, XP_021584516.1), Alpine marmot (*Marmota marmota marmota, XP*_015353707.1)], SEQ ID NO:122 [David's myotis (*Myotis davidii*, ELK23147.1), Brandt's bat (*Myotis brandtii*, XP_014397831.1)], SEQ ID NO:123 [Horse (*Equus caballus*, XP_005611026.1), Donkey (*Equus asinus*, XP_014687887.1), Przewalski's horse (*Equus przewalskii*, XP_008533269.1)], SEQ ID NO:124 [Dromedary (*Camelus dromedarius*, XP_010987132.1), Bactrian camel (*Camelus bactrianus*, XP_010944895.1), Wild boar (*Sus scrofa*, XP_020947912.1), Cattle (*Bos taurus*, XP_005206989.1), Goat (*Capra hircus*, XP_017903901.1), Sheep (*Ovis aries*, XP_012030611.1), Mouflon (*Ovis aries musimon*, XP_012018062.1), Central European red deer (*Cervus elaphus hippelaphus*, OWK03976.1), Minke whale (*Balaenoptera acutorostrata scammoni*, XP_007195125.1), Killer whale (*Orcinus orca*, XP_004270982.1)], SEQ ID NO:125 [Cat (*Felis catus*, XP_006933619.1), Leopard (*Panthera pardus*, XP_019307710.1)], SEQ ID NO:126 Human (*Homo sapiens*, NP_079130.2), SEQ ID NO:127 Northern greater galago (*Otolemur garnettii*, XP_003792495.1), SEQ ID NO:128 Rat (*Rattus norvegicus*, EDM01504.1), SEQ ID NO:129 Chinese hamster (*Cricetulus griseus*, ERE65977.1), SEQ ID NO:130 Prairie vole (*Microtus ochrogaster*, XP_005364695.1), SEQ ID NO:131 Long-tailed chinchilla (*Chinchilla lanigera*, XP_005379114.1), SEQ ID NO:132 American beaver (*Castor canadensis*, XP_020016064.1), SEQ ID NO:133 Ferret (*Mustela putorius furo*, XP_004755775.1), SEQ ID NO:134 Damora mole-rat (*Fukomys damarensis*, XP_010613965.1), SEQ ID NO:135 Alpaca (*Vicugna pacos*, XP_015091099.1), SEQ ID NO:136 Giant panda (*Ailuropoda melanoleuca*, XP_002926616.1), SEQ ID NO:137 Koala (*Phascolarctos cinereus*, XP_020849955.1), SEQ ID NO:138 Tasmanian devil (*Sarcophilus harrisii*, XP_003771207.1), SEQ ID NO:139 Gray short-tailed opossum (*Monodelphis domestica*, XP_001365452.2), SEQ ID NO:140 [Bald eagle (*Haliaeetus leucocephalus*, XP_010564402.1), Golden eagle (*Aquila chrysaetos canadensis*, XP_011574851.1), Chicken (*Gallus gallus*, NP_001264205.1), Downy woodpecker (*Picoides pubescens*, XP_009904475.1), Turkey (*Meleagris gallopavo*, XP_003202569.1), Speckled mousebird (*Colius striatus*, XP_010208275.1), Emperor penguin (*Aptenodytes forsteri*, XP_009277280.1), Rock dove (*Columba livia*, XP_021151605.1), Band-tailed pigeon (*Patagioenas fasciata monilis*, OPJ70652.1), Adélie penguin (*Pygoscelis adeliae*, KFW73339.1), Ruff (*Calidris pugnax*, XP_014815036.1), Japanese quail (*Coturnix japonica*, XP_015723825.1), Crested ibis (*Nipponia nippon*, XP_009463403.1), Anna's hummingbird (*Calypte anna*, XP_008500751.1), Little egret (*Egretta garzetta*, KFP11017.1), Chimney swift (*Chaetura pelagica*, KFU93618.1), Scaled quail (*Callipepla squamata*, OXB63337.1), Common cuckoo (*Cuculus canorusz, XP*_009558366.1), Peregrine falcon (*Falco peregrinus*, XP_013156768.1)], SEQ ID NO:141 [Zebra finch (*Taeniopygia guttata*, XP_002200247.1), Society finch (*Lonchura striata domestica*, XP_021407730.1)], SEQ ID NO:142 Northern carmine bee-eater (*Merops nubicus*, KFQ27091.1), SEQ ID NO:143 North Island brown kiwi (*Apteryx australis mantelli*, XP_013802597.1), SEQ ID NO:144 American crow (*Corvus brachyrhynchos*, KF060564.1), SEQ ID NO:145 Hooded crow (*Corvus cornix cornix*, XP_010402574.2), SEQ ID NO:146 Atlantic canary (*Serinus canaria*, XP_009093423.1), SEQ ID NO:147 Common starling (*Sturnus vulgaris*, XP_014735476.1), SEQ ID NO:148 Blue-crowned manakin (*Lepidothrix coronata*, XP_017660339.1), SEQ ID NO:149 Turquoise-fronted amazon (*Amazona aestiva*, KQK78799.1), SEQ ID NO:150 Schlegel's Japanese gecko (*Gekko japonicus*, XP_015277871.1), SEQ ID NO:151 Green anole (*Anolis carolinensis*, XP_003220806.1), SEQ ID NO:152 Burmese python (*Python bivittatus*, XP_007420993.1), SEQ ID NO:153 Central bearded dragon (*Pogona vitticeps*, XP_020669436.1), SEQ ID NO:154 Brown spotted pit viper (*Protobothrops mucrosquamatus*, XP_015682645.1), SEQ ID NO:155 King cobra (*Ophiophagus hannah*, ETE62148.1), SEQ ID NO:156 Common garter snake (*Thamnophis *sirtalis*, XP_013928623.1), SEQ ID NO:157 West Indian Ocean coelacanth (*Latimeria chalumnae*, XP_005998922.1), SEQ ID NO:158 Whale shark (*Rhincodon typus*, XP_020373590.1), SEQ ID NO:159 Australian ghostshark (*Callorhinchus milii*, XP_007884060.1), SEQ ID NO:160 Atlantic salmon (*Salmo salar*, XP_014008456.1), SEQ ID NO:161 African clawed frog (*Xenopus laevis*, XP_018107310.1), SEQ ID NO:162 Common carp (*Cyprinus carpio*, KTF74061.1)

(215) The next to the last line (labeled CLUSTAL) shows the consensus sequence, where positions with fully conserved amino acids are represented by an asterisk (*), positions with strongly conserved amino acids are indicated by a colon (:), positions with weakly conserved amino acids are indicated are indicated by period (.), and nonconserved positions are indicated by a blank space ( ). The last line shows the position of the PYROXD1-D492H substitution in horse in bold. The position of the PYROXD1-D492H substitution is indicated in bold in all of the aligned sequences.

(216) The overall sequence conservation of this portion of the PYROXD1 protein is quite high. The amino acid alteration seen in the equine PYROXD1-D492H is not seen in any of the 92 species; all have an aspartate (D) or asparagine (N) at this position. The comparison of PYROXD1 sequences presented here refutes the hypothesis that the PYROXD1-D492H substitution is selectively neutral, and supports the claim that the PYROXD1-D492H mutation found in horses with Polysaccharide Storage Myopathy type 2 (PSSM2), also known as Myofibrillar Myopathy (MFM), is pathogenic.

(217) Collagen Type VI Alpha 3 Chain (COL6A3)

(218) A mutation in the equine COL6A3 coding region (encoding the muscle protein collagen type VI alpha 3 chain) is present in many populations of horses affected by Polysaccharide Storage Myopathy type 2 (PSSM2). The difference in the genomic DNA between horses with PSSM2 and control horses include a G-to-C substitution in COL6A3 Exon 26 (as defined in FIG. **33**) at nucleotide position chr6:23,480,621 of the current horse genome assembly (EquCab2, GCA_000002305.1), as shown in FIG. **33**.

(219) FIG. **33** shows portion of the current horse genome assembly (EquCab2, GCA_000002305.1) with coordinates as displayed in the UCSC Genome Browser centered on the chr6:23,480,621 position, the site of a substitution of a guanine (G) for a cytosine (C) on the forward strand that results in the substitution of an alanine (A) for a glycine (G) at amino acid position 2182 in collagen type VI alpha 3 chain (COL6A3) as shown in FIG. **36** (SEQ ID NO:168). The reverse complement sequence is shown, with the site of a substitution of a cytosine (C) for a guanine (G) on the reverse strand as indicated (SEQ ID NO:163). The single nucleotide polymorphism (SNP) defined by this base substitution is identified as rs1139437410 in dbSNP.

(220) The NCBI database contains five models for the mature horse mRNA containing the coding sequence (CDS) for COL6A3: XM_023642645.1, XM_023642646.1, XM_014740385.2, XM_023642647.1, and XM_023642648.1. There is a complete consensus among the five models for the portion of the coding sequence that encodes the five collagen-like repeats in the middle of the COL6A3 protein (see FIG. **34**). For XM_023642645.1 (transcript variant X1), the coordinates are 6356-7363. The other four isoforms have coordinates as follows: XM_023642646.1 (5751-6758), XM_014740385.2 (5733-6740), XM_023642647.1 (5130-6137), and XM_023642648.1 (4530-5537).

(221) FIG. **34** shows a portion of the normal equine COL6A3 Coding DNA Sequence (SEQ ID NO:164) and a portion of the mutant COL6A3 Coding DNA Sequence (SEQ ID NO:165) bearing the G to C mutation at nucleotide position 6792 in this figure, corresponding to chr6:23,480,621 as shown in SEQ ID NO:163 (FIG. **33**). The numbering in FIG. **34** is that of the COL6A3 coding sequence (CDS) model XM_023642645.1; the sequence presented comprises the coding sequence for the five collagen-like domains in the middle of the protein. In both sequences the sequence of the third collagen-like domain, partially encoded by Exon 26, as shown in FIG. **35** is indicated in bold. The site of a G to C mutation at nucleotide position 6792, corresponding to 23,480,621 in SEQ ID NO:163 (FIG. **33**), and to rs1139437410 in dbSNP, is underlined. The region of sequence comprising Exon 26 is displayed as codons in the correct reading frame for both SEQ ID NO:164

and SEQ ID NO:165 in FIG. **37**.
(222) FIG. **35** shows a view of the current horse genome assembly (EquCab2, GCA_000002305.1) in the UCSC Genome Browser with exon sequences that match the COL6A3 Coding DNA Sequence (SEQ ID NO:164) and the mutant COL6A3 Coding DNA Sequence (SEQ ID NO:165) in a BLAT search. The COL6A3 Coding DNA Sequences in horse correspond to COL6A3 coding sequences in other species as indicated by the regions of sequence similarity of the translated genomic DNA to COL6A3 protein sequences from human (Homo), mouse (Mus), and dog (Canus). Exon 26, which contains the guanine (G) to cytosine (C) variant at chr6:23,480,621 on the reverse strand as shown in SEQ ID NO:163 (FIG. **33**), is indicated below the image of the browser window. The sequence of horse Exon 26 with 10 nucleotides of flanking intron sequence and its coordinates in the current horse assembly is displayed below the image from the UCSC Genome Browser (SEQ ID NO:166).
(223) The translation of horse COL6A3 CDS does not always produce a significant match to the protein sequences of COL6A3 from other species, as shown in the UCSC Genome Browser window. This is particularly apparent for Exon 26. There are two reasons for this. First, the protein sequence of the collagen-like repeat portion of the protein, as discussed below, is a repeating sequence of Gly-X-Y. With every third residue a glycine, the sequence is of low complexity. Second, Exon 26 is very short. A short sequence with low complexity will not always produce a significant match to orthologous proteins, even if the proteins are truly orthologous.
(224) Conceptual translation of the nucleic acid sequences shown in FIG. **34** (SEQ ID NO:164 and SEQ ID NO:165) yields the amino acid sequences for the proteins shown in FIG. **36**. FIG. **36** shows models of part of the normal protein sequence encoded by horse COL6A3 (XP_023498413.1, presented here as SEQ ID NO:167) corresponding to a translation of SEQ ID NO:164 shown in FIG. **34** and part of the altered protein sequence encoded by horse COL6A3 (adapted from XP_023498413.1, presented here as SEQ ID NO:168) with the base substitution at chr6:23,480,621 corresponding to a translation of SEQ ID NO:165 shown in FIG. **34**. The portion of the protein corresponding to all five collagen-like domains is shown; the portion corresponding to the third collagen-like domain, partially encoded by Exon 26, is indicated in bold, while the amino acid at position 2182 affected by the base substitution of a cytosine (C) for a guanine (G) at the chr6:23,480,621 position as shown in FIG. **33**, is underlined. The amino acid positions in XP_023498413.1 are indicated at the beginning and end of the sequence.
(225) For the sake of simplicity, the amino acid substitution caused by the substitution of an cytosine (C) for a guanine (G) at the chr6:23,480,621 position on the reverse strand as shown in FIG. **33**, identified as rs1139437410 in dbSNP, will be referred to as COL6A3-G2182A, based on SEQ ID NO:168. The amino acid substitution caused by this mutation will remain the same regardless of the numerical position of the affected codon in alternative gene models. The wild-type (e.g., also referred to herein as “normal” or “unaffected”) allele of this coding region may be referred to as G2182 and the mutant allele G2182A.
(226) FIG. **37** shows horse COL6A3 Exon 26 and flanking genomic DNA sequence from which PCR primers to amplify genomic DNA containing the site of the COL6A3-G2182A mutation would be most appropriately derived. Genomic coordinates are as in FIG. **24**. Exon 26 from chr6:23,480,631 to chr6:23,480,578 is shown broken into codons in the correct reading frame for the wild-type allele (SEQ ID NO:169) and the COL6A3-G2182A allele (SEQ ID NO:170). Only the reference sequence from the assembly is shown for the flanking sequences. The codon affected by the G to C mutation site at nucleotide position chr6:23,480,621 on the reverse strand, as shown in FIG. **33**, is shown in bold, with the position of the base substitution indicated by underlining. The base substitution changes the bold three base codon from one coding for a glycine (GGG) to one coding for an alanine (GCG). Example primers used experimentally to amplify genomic DNA containing the mutation site [5′-AGATGGGGCACAGATCAAAC-3′ (SEQ ID NO:172) and 5′-TTCCCAGACTCTCCTGTGCT-3′ (SEQ ID NO:171)] are shown in lower case in the figure.

(227) Genomic DNA obtained from horses can be genotyped by amplifying a region containing a variant in the COL6A3 gene using Polymerase Chain Reaction (PCR), then sequencing the amplified DNA using Sanger sequencing. The variant allele COL6A3-G2182A is abbreviated as K1, while the common or wild-type allele is abbreviated as n. The results can be scored as homozygous for the common or wild-type allele (n/n), heterozygous for the nucleotide substitution (n/K1), or homozygous for the nucleotide substitution (K1/K1).
(228) FIG. **38** shows traces from Sanger DNA sequencing of amplified COL6A3 genomic DNA using primers shown in FIG. **37** (SEQ ID NO:171 and SEQ ID NO:172). The arrows in the figure indicate nucleotide position chr6:23,480,621, the site of a substitution of a cytosine (C) for a guanine (G) in this position on the reverse strand that creates the COL6A3-G2182A variant. The traces show, from left to right, the sequence of the forward strand for a horse homozygous for the wild-type or common allele (n/n), and results for a horse heterozygous for the substitution (n/K1).
(229) Collagen is the main structural protein of the extracellular matrix. Collagens are synthesized with N- and C-terminal extensions that are cleaved off proteolytically upon export from the cell. The N- and C-terminal extensions assist in localization within the extracellular matrix. The core of the collagen protein is the triple helical domain, consisting of Gly-X-Y repeats. The collagen triple helix consists of a right-handed intertwining of three left-handed helices. The glycine residues are critical, because every third residue occupies a position at the center of the triple helix, where there is no room for any other R group besides the hydrogen found in glycine. The collagen triple helix is stiffened by the post-translation modification of proline and lysine residues to hydroxyproline and hydroxylysine. As collagen fibers assemble, each triple helix is joined to its neighbors. In the fully assembled array, each triple helix overlaps with a neighboring triple helix with an offset of about one quarter of its length. The array is stabilized by covalent links through interchain disulfide bonds and post-translationally modified hydroxylysine.
(230) The human ortholog of the equine COL6A3 gene encodes collagen type VI alpha 3 chain, one of a set of collagen type VI genes. The others are COL6A1, COL6A2, COL6A5, and COL6A6 (COL6A4 is a pseudogene). Collagens play important roles in maintaining extracellular matrix structure and function. Members of the collagen VI family, like COL6A3, form distinct networks of microfibrils in connective tissue and interact with other extracellular matrix components. Mutations in the human genes COL6A1, COL6A2, and COL6A3 are associated with Bethlem Myopathy, Ullrich Congenital Muscular Dystrophy, and dystonia. Mutations may be either dominant or recessive, with the age of onset and the severity of the condition varying with the exact mutation.
(231) In mouse, an in-frame intragenic deletion of six Gly-X-Y repeats of COL6A3 (Col6a.sup.2tm.2.1Chu) is a dominant mutation causing abnormal muscle fiber morphology and myopathy. Targeted mutation of COL6A1 is also a dominant mutation causing myopathy. Electron microscopy of tendons from homozygous mice shows a reduced diameter of collagen fibrils (Pan et al. 2013 J Biol Chem, 288(20):14329-14331).
(232) It is therefore informative to compare the horse COL6A3 gene and the protein that it encodes with the human ortholog. FIG. **39** shows the partial sequence of the human COL6A3 coding sequence derived from NM_004369.3 (SEQ ID NO:173). This sequence is a region of perfect consensus among multiple different experimentally predicted mRNA isoforms. The numbering of the first and last nucleotides corresponds to that of NM_004369.3. The sequence encoding the third collagen-like domain is indicated in bold.
(233) FIG. **40** shows a partial sequence of the human COL6A3 protein, showing a translation of SEQ ID NO:175, equivalent to NP_004360.2 (SEQ ID NO:176). The numbering of the first and last amino acids corresponds to that of NP_004360.2. The sequence of the third collagen-like domain is indicated in bold.
(234) FIG. **41** shows a comparison of the portion of the protein sequence of COL6A3 comprising the five collagen-like repeats from human (NP_079130.2, shown here as SEQ ID NO:175) and

horse (XP_001502180.3, shown here as SEQ ID NO:176). The position of the COL6A3-G2182A substitution is shown below the horse sequence. Between the sequences of the horse and human proteins in the alignment, an asterisk (*) indicates an identical amino acid in that position, while a plus sign (+) indicates a conservative substitution, and a space ( ) indicates a nonconservative substitution. The positions of glycine residues that are part of the Gly-X-Y structure of the collagen triple helix are indicated by an asterisk (*) in reverse text (white text on a black background) between the horse and human sequences. All of these glycine residues are conserved between human and horse. The horse G2182A sequence has a nonconservative substitution of an alanine (A) for a glycine (G) at position 2182; the wild-type horse sequence matches the human sequence at this position.
(235) FIG. **42** shows features of the collagen type VI alpha 3 chain protein encoded by the human COL6A3 gene. A) The human protein has 3,162 amino acids. It contains five collagen-like domains consisting of Gly-X-Y repeats, labeled as Collagen 1-5 in the figure. Toward the N-terminus of the collagen-like repeats are 10 Von Willebrand Factor-like repeats, labeled as VWFA 1-10 in the figure. Toward the C-terminus of the collagen-like repeats are two additional Von Willebrand Factor-like repeats, labeled as VWFA 11 and 12 in the figure. Near the C-terminus is a fibronectin-like repeat, labeled as Fibronectin in the figure, and a Protease Inhibitor domain (Pancreatic trypsin inhibitor Kunitz domain). B) The portion of the protein containing the five collagen-like domains extends from position 2038 to 2373 of the human protein. The five collagen-like domains are numbered. Positions of 19 pathogenic human alleles listed in TABLE 2 below are indicated above the segment shown, as is the position of the horse G2182A variant. The position of the cysteine involved in the interchain disulfide bond (C2087S-S) is indicated below the segment shown.
(236) TABLE 4 shows nineteen missense alleles of human COL6A3 shown in FIG. **42**. Although pathogenic missense alleles of human COL6A3 outside of the collagen-like repeats in the middle of the protein have been described, we focus on mutations in this region in humans because of the relevance to the COL6A3-G2182A allele in horse. All nineteen mutations are substitutions of other amino acid residues for the glycine residues that are part of the Gly-X-Y repeat structure of the collagen-like region. Human missense alleles of COL6A3 are associated with Bethlem Myopathy and with Ullrich Congenital Muscular Dystrophy. Bethlem Myopathy has also been referred to as Limb-Girdle Muscular Dystrophy D5 (LGMDD5). Both dominant and recessive forms of Bethlem Myopathy, caused by mutations in COL6A1, COL6A2, and COL6A3, are known. Symptoms include progressive muscular weakness with muscle atrophy of the trunk and limbs and joint contractures. Missense alleles affecting the glycine residues of the collagen repeats account for almost one-third of all pathogenic alleles. The symptoms of Ullrich Congenital Muscular Dystrophy overlap those of Bethlem Myopathy. Ullrich Congenital Muscular Dystrophy is caused by mutations in COL6A1, COL6A2, and COL6A3.
(237) TABLE-US-00005 TABLE 4 Pathogenic missense alleles of human COL6A3 and associated diseases. Substitution Domain Disease Reference G2047D Collagen 1 Bethlem Myopathy Lampe 2005 G2053V Collagen 1 Not provided ClinVar G2056E Collagen 1 Not provided ClinVar G2056R Collagen 1 Bethlem Myopathy Pepe 1999 G2059C Collagen 1 Bethlem Myopathy ClinVar G2065S Collagen 1 Bethlem Myopathy/ ClinVar Ulrich congenital muscular dystrophy G2055R Collagen 1 Bethlem Myopathy/ ClinVar Ulrich congenital muscular dystrophy G2071D Collagen 1 Bethlem Myopathy/ ClinVar Ulrich congenital muscular dystrophy G2074S Collagen 1 Bethlem Myopathy/ ClinVar Ulrich congenital muscular dystrophy G2077D Collagen 1 Bethlem Myopathy/ ClinVar Ulrich congenital muscular dystrophy G2077V Collagen 1 Bethlem Myopathy ClinVar G2080S Collagen 1 Ullrich congenital ClinVar muscular dystrophy G2080R Collagen 1 Not provided ClinVar G2080D Collagen 1 Bethlem Myopathy/ ClinVar, Ulrich congenital Lampe 2005 muscular dystrophy G2083V Collagen 1 Not provided ClinVar G2098V Interdomain Bethlem Myopathy/ ClinVar Ulrich congenital muscular dystrophy G2174S Collagen 3 Bethlem Myopathy

ClinVar G2267S Collagen 4 Bethlem Myopathy ClinVar G2285R Collagen 4 Bethlem Myopathy ClinVar G2297A Collagen 4 Bethlem Myopathy/ ClinVar Ulrich congenital muscular dystrophy

(238) Cited sources in this table are: ClinVar (https://www.ncbi.nlm.nih.gov/clinvar/) Lampe 2005 (Lampe A K et al. 2005 J Med Genet 42:108-120 DOI: 10.1136/jmg.2004.023754) Pepe 1999 (Pepe G et al. 1999 Neuromuscul Disord 9:264-271)

(239) The glycine substitutions seen in human pathogenic alleles of COL6A3 are clustered in the amino-terminal portion of the collagen-like domain, as shown in FIG. **42** and TABLE 4. This may represent a functional domain within the triple helix (Butterfield et al. Hum Mutat. 2013 34(11):1558-1567). The most severe cases are caused by glycine substitutions in Gly-X-Y triplets 10 to 15. Glycine substitutions in this region are reported as dominant; the only cases of recessive glycine substitutions are located outside of this region in the C-terminal portion of the triple helix. This generalization applies to glycine substitutions in COL6A1, COL6A2, and COL6A3.

(240) The position of the COL6A3-G2182A mutation in horse is shown in FIG. **42**. COL6A3-G2182A affects Gly-X-Y triplet 48, as shown in FIG. **41**. The position of the COL6A3-G2182A mutation in horse is closest to the human COL6A3-G2174S allele. The phenotype produced by human pathogenic glycine substitution alleles depends not only on the position of the mutation, but also on the amino acid substitution. There is a published destabilization scale for glycine substitutions in the triple helix (Ala<Ser<Cys<Arg<Val<Glu<Asp<Trp) (Persikov et al. Hum Mutat. 2004 24(4):330-337). The horse COL6A3-G2182A allele is therefore the least damaging substitution in a moderately damaging position in the protein, consistent with the moderate phenotype of affected horses described below.

(241) The sequence comparison between the proteins encoded by horse and human COL6A3 shows the high degree of sequence conservation between these two species. The extent of sequence conservation is so high that it is difficult to evaluate whether this comparison offers any evidence that the equine COL6A3-G2182A allele would be expected to be pathogenic.

(242) FIG. **43** shows partial amino acid sequences of proteins encoded by COL6A3, including the position of the equine COL6A3-G2182A substitution. A portion of the human COL6A3 protein sequence was used as a probe to identify the corresponding region in the orthologous protein across a wide range of species, including mammals, birds, reptiles, amphibians, and fish. In the alignment shown in FIG. **43**, sequences that are identical over this range of positions are clustered as a single SEQ ID NO.

(243) The 89 species in the alignment shown in FIG. **43** are presented below with their common names, their scientific names, and the protein database ID from which the amino acid sequence is derived. Amino acid sequences identical over this range in different species have been clustered into single SEQ ID NOs for the alignment. The species associated with the SEQ ID NOs shown in the alignment are:

(244) SEQ ID NO:177 [Human (*Homo sapiens*, NP_004360.2), Chimpanzee (*Pan troglodytes*, XP_001153544.3), Bonobo (*Pan paniscus*, XP_003813155.1), Western lowland gorilla (*Gorilla gorilla gorilla*, XP_018878003.1), Sumatran orangutan (*Pongo abelii*, XP_009236555.2)], SEQ ID NO:178 [Thirteen-lined ground squirrel (*Ictidomys tridecemlineatus*, XP_013214606.1), Alpine marmot (*Marmota marmota marmota*, XP_015337108.1)], SEQ ID NO:179 [David's myotis (*Myotis davidii*, XP_015417293.1), Brandt's bat (*Myotis brandtii*, XP_014393090.1)], SEQ ID NO:180 [Horse (*Equus caballus*, XP_023498413.1), Donkey (*Equus asinus*, XP_014698264.1), Przewalski's horse (*Equus przewalskii*, XP_008528449.1)], SEQ ID NO:181 [Dromedary (*Camelus dromedarius*, XP_010994640.1), Bactrian camel (*Camelus bactrianus*, XP_010946449.1), Alpaca (*Vicugna pacos*, XP_015098979.1)], SEQ ID NO:182 Ferret [(*Mustela putorius furo*, XP_012907287.1), Cat (*Felis catus*, XP_019694498.2)], SEQ ID NO:183 [Cattle (*Bos taurus*, XP_024846030.1), Goat (*Capra hircus*, XP_017896230.1), Sheep (*Ovis aries*, XP_014947129.1), Mouflon (*Ovis aries musimon*, XP_011989955.1)], SEQ ID NO:184 [Leopard (*Panthera pardus*, XP_019301005.1), Giant panda (*Ailuropoda melanoleuca*, XP_019662500.1)], SEQ ID NO:185

Olive baboon (*Papio anubis*, XP_021779879.1), SEQ ID NO:186 Rhesus macaque (*Macaca mulatta*, XP_014966890.1), SEQ ID NO:187 Panamanian white-throated capuchin (*Cebus capucinus imitator*, XP_017393233.1), SEQ ID NO:188 Black snub-nosed monkey (*Rhinopithecus bieti*, XP_017745835.1), SEQ ID NO:189 Gray mouse lemur (*Microcebus murinus*, XP_012634739.2), SEQ ID NO:190 Philipine tarsier (*Carlito syrichta*, XP_008057564.1), SEQ ID NO:191 Northern greater galago (*Otolemur garnettii*, XP_003798279.2), SEQ ID NO:192 Coquerel's sifika (*Propithecus coquereli*, XP_012504274.1), SEQ ID NO:193 Mouse (*Mus musculus*, NP_001229937.1), SEQ ID NO:194 Rat (*Rattus norvegicus*, XP_008756297.1), SEQ ID NO:195 Chinese hamster (*Cricetulus griseus*, XP_016824341.1), SEQ ID NO:196 Prairie vole (*Microtus ochrogaster*, XP_005361761.1), SEQ ID NO:197 Mongolian gerbil (*Meriones unguiculatus*, XP_021513097.1), SEQ ID NO:198 Long-tailed chinchilla (*Chinchilla lanigera*, XP_005411290.1), SEQ ID NO:199 American beaver (*Castor canadensis*, XP_020043158.1), SEQ ID NO:200 Damora mole-rat (*Fukomys damarensis*, XP_010624430.1), SEQ ID NO:201 Large flying fox (*Pteropus vampyrus*, XP_023380958.1), SEQ ID NO:202 Sunda pangolin (*Manis javanica*, XP_017531172.1), SEQ ID NO:203 Great roundleaf bat (*Hipposideros armiger*, XP_019485542.1), SEQ ID NO:204 African bush elephant (*Loxodonta africana*, XP_010595897.1), SEQ ID NO:205 Wild boar (*Sus scrofa*, XP_020931400.1), SEQ ID NO:206 Minke whale (*Balaenoptera acutorostrata scammoni*, XP_007184961.1), SEQ ID NO:207 Killer whale (*Orcinus orca*, XP_004262601.1), SEQ ID NO:208 Koala (*Phascolarctos cinereus*, XP_020827927.1), SEQ ID NO:209 Tasmanian devil (*Sarcophilus harrisii*, XP_023359576.1), SEQ ID NO:210 Platypus (*Ornithorhynchus anatinus*, XP_007666682.1), SEQ ID NO:211 [American crow (*Corvus brachyrhynchos*, XP_017602594.1), Hooded crow (*Corvus cornix cornix*, XP_019146502.1)], SEQ ID NO:212 Bald eagle (*Haliaeetus leucocephalus*, XP_010576193.1), SEQ ID NO:213 Golden eagle (*Aquila chrysaetos canadensis*, XP_011584187.1), SEQ ID NO:214 Chicken (*Gallus gallus*, NP_990865.2), SEQ ID NO:215 Downy woodpecker (*Picoides pubescens*, XP_009899132.1), SEQ ID NO:216 Turkey (*Meleagris gallopavo*, XP_010711547.1), SEQ ID NO:217 Northern carmine bee-eater (*Merops nubicus*, XP_008936095.1), SEQ ID NO:218 Speckled mousebird (*Colius striatus*, XP_010208169.1), SEQ ID NO:219 North Island brown kiwi (*Apteryx australis mantelli*, XP_013798018.1), SEQ ID NO:220 Emperor penguin (*Aptenodytes forsteri*, XP_019329168.1), SEQ ID NO:221 Rock dove (*Columba livia*, XP_021142382.1), SEQ ID NO:222 Band-tailed pigeon (*Patagioenas fasciata monilis*, OPJ80253.1), SEQ ID NO:223 Adélie penguin (*Pygoscelis adeliae*, XP_009327902.1), SEQ ID NO:224 Ruff (*Calidris pugnax*, XP_014807202.1), SEQ ID NO:225 Japanese quail (*Coturnix japonica*, XP_015723569.1), SEQ ID NO:226 Crested ibis (*Nipponia nippon*, XP_009460310.1), SEQ ID NO:227 Anna's hummingbird (*Calypte anna*, XP_008493236.1), SEQ ID NO:228 Little egret (*Egretta garzetta*, XP_009633772.1), SEQ ID NO:229 Atlantic canary (*Serinus canaria*, XP_009086444.1), SEQ ID NO:230 Common starling (*Sturnus vulgaris*, XP_014737483.1), SEQ ID NO:231 Zebra finch (*Taeniopygia guttata*, XP_004175063.1), SEQ ID NO:232 Society finch (*Lonchura striata domestica*, XP_021391774.1), SEQ ID NO:233 Chimney swift (*Chaetura pelagica*, XP_010003753.1), SEQ ID NO:234 Scaled quail (*Callipepla squamata*, OXB61690.1), SEQ ID NO:235 Common cuckoo (*Cuculus canorus*, XP_009568416.1), SEQ ID NO:236 Blue-crowned manakin (*Lepidothrix coronata*, XP_017691139.1), SEQ ID NO:237 Peregrine falcon (*Falco peregrinus*, XP_005244480.1), SEQ ID NO:238 Budgerigar (*Melopsittacus undulatus*, XP_012985362.1), SEQ ID NO:239 Schlegel's Japanese gecko (*Gekko japonicus*, XP_015273310.1), SEQ ID NO:240 Green anole (*Anolis carolinensis*, XP_008104462.1), SEQ ID NO:241 Burmese python (*Python bivittatus*, XP_025027494.1), SEQ ID NO:242 Brown spotted pit viper (*Protobothrops mucrosquamatus*, XP_015676224.1), SEQ ID NO:243 King cobra (*Ophiophagus hannah*, ETE68401.1), SEQ ID NO:244 Common garter snake (*Thamnophis sirtalis*, XP_013925229.1), SEQ ID NO:245 West Indian Ocean coelacanth (*Latimeria chalumnae*, XP_014344739.1), SEQ ID NO:246 Whale shark (*Rhincodon typus*, XP_020391663.1), SEQ ID

NO:247 Australian ghostshark (*Callorhinchus milii*, XP_007904071.1), SEQ ID NO:248 Atlantic salmon (*Salmo salar, XP*_014028364.1), SEQ ID NO:249 African clawed frog (*Xenopus laevis*, KTG32498.1), SEQ ID NO:250 Common carp (*Cyprinus carpio*, KTG32498.1).

(245) The next to the last line (labeled CLUSTAL) shows the consensus sequence, where positions with fully conserved amino acids are represented by an asterisk (*), positions with strongly conserved amino acids are indicated by a colon (:), positions with weakly conserved amino acids are indicated are indicated by period (.), and nonconserved positions are indicated by a blank space ( ). The last line shows the position of the COL6A3-G2182A substitution in horse. The position of the COL6A3-G2182A substitution is indicated in bold in all of the aligned sequences.

(246) The strongly conserved positions in the aligned sequences are all glycine (G) residues that are part of the Gly-X-Y repeat structure of the collagen triple helix. The sequences to the N-terminal position of the strongly conserved glycine residues also show a Gly-X-Y repeat structure that is evident from the examination of individual sequences, but a variable number of amino acids between the Gly-X-Y repeats showing sequence conservation and the Gly-X-Y repeats at the N-terminal position prevents this from showing in the multiple alignment. Separation of Gly-X-Y repeats by sequences not conforming to the Gly-X-Y pattern is common among collagens.

(247) The overall sequence conservation of this portion of the COL6A3 protein is quite high. The amino acid alteration seen in the equine COL6A3-G2182A is not seen in any of the 89 species; all have a glycine (G) at this position. The comparison of COL6A3 sequences presented here refutes the hypothesis that the COL6A3-G2182A substitution is selectively neutral, and supports the claim that the COL6A3-G2182A mutation found in horses with PSSM2 is pathogenic.

(248) Phenotypic Effects of the Genetic Variants Associated with PSSM2

(249) This disclosure, together with a prior filing, identifies eight variants associated with PSSM2: MYOT-S232P (P2), FLNC-E753K (P3a), FLNC-A1207T (P3b), MYOZ3-S42L (P4), DYSF-R285W (P5), PYROXD1-D492H (P8), and COL6A3-G2182A (K1). These eight variants correspond to single-base substitutions having the coordinates given as follows on the forward strand in the public horse genome assembly (EquCab 2): chr14:38519183 A/G (P2), chr4:83736244 G/A (P3a), chr4:83738769 G/A (P3b), chr14:27399222 G/A (P4), chr15:31306949 G/A (P5), chr15: 5:31,225,630 T/G (P6), chr6:47,661,977 G/C (P8), and chr6:23,480,621 C/G (K1). The P3a and P3b variants are a haplotype, that is, among hundreds of horses tested, horses with the P3a variant also have the P3b variant and vice versa, whether homozygous or heterozygous. Only two types of chromosomes are seen: those with both P3a and P3b, or those with the wild-type or common alleles of both variants. The haplotype with both variants is therefore abbreviated as P3.

(250) One of the earliest symptoms of PSSM2 is a change in behavior apparently associated with pain. Owners note a difference in temperament, with horses reacting badly to being ridden or even saddled. Common behaviors include biting at the flanks or even at the rider or trainer, and bucking, rearing, and other displays of resistance that trainers often blame on lack of discipline from the owner.

(251) Another early symptom is stifle problems. The stifle is the largest joint in the horse's body, equivalent to the human knee, but in contrast to the human knee, the equine stifle is held at an angle when the horse is standing still. Stifle problems commonly result from injury or arthritis, degenerative joint disease, or injury. In stifle problems resulting from Polysaccharide Storage Myopathy type 2 (PSSM2), there will be no radiographic findings. Stifle problems are one example of shifting lameness. A horse with Polysaccharide Storage Myopathy type 2 (PSSM2) will exhibit lameness that appears first in one limb, then another. There will be no radiographic findings.

(252) Changes in gait are often apparent. These include stiffness in the hindquarters and limited range of motion of the hind legs (“short-gaited”). At canter, disunited canter (“cross-firing”) and “bunny hopping” (bringing both hind legs forward at the same time) are seen. “Rope walking” (placing one foot directly in front of the other along the centerline as if walking a tightrope) is sometimes seen in all four legs or in the rear legs only.

(253) Other gait changes resulting from weakness in the hind limbs are described by horse owners as “heavy on the forehand, not able to come from behind.” This means that the horse's gait is altered in such a way that it appears to be pulling itself forward with its front hooves instead of pushing from the rear. Farriers note this as a pattern of wear in the front hooves for unshod horses.
(254) There is no evidence of cardiomyopathy.
(255) Phenotypic Effects of the DYSF-R285W and DYSF-P1290T Variants
(256) The DYSF-R253W variant (hereafter abbreviated as P5) was discovered by analysis of whole genome sequencing data from a Shire draft horse (E016) diagnosed via muscle biopsy with Polysaccharide Storage Myopathy type 2 (PSSM2). This horse was homozygous for the P5 allele (P5/P5) and homozygous wild type for six other variants with PSSM2: MYOT-S232P (P2), FLNC-E753K (P3a), FLNC-A1207T (P3b), MYOZ3-S42L (P4), PYROXD1-D492H (P8), and COL6A3-G2182A (K1). The P6 allele of DYSF (DYSF-P1290T) was not present.
(257) We analyzed additional horses using PCR amplification and Sanger sequencing to identify P5/P5 homozygotes. A Shire horse (E799) scored as P5/P5 and homozygous for the wild-type alleles of P2, P3, P4, P5, P8, and K1 was identified at 11 years of age. This horse had been retired from regular work the prior year following an episode of lameness, muscle fasciculation, and profuse sweating during work.
(258) Other individual horses of the Shire breed and the closely related Clydesdale breed that were determined to be normal, that is, not affected by PSSM2, were found to be heterozygous for the P5 allele (N/P5) or homozygous for the wild-type allele (N/N). Therefore, the P5 variant appears to be recessive, as is typical for missense alleles of the human DYSF gene shown in TABLE 1, TABLE 2, and TABLE 3.
(259) The DYSF-P1290T variant (hereafter abbreviated as P6) was discovered by analysis of whole genome sequencing data from a number of horses diagnosed via muscle biopsy with Polysaccharide Storage Myopathy type 2 (PSSM2). Each of these horses had other variants, and all were found to be heterozygous of P6 (N/P6). Because mutations in DYSF analyzed in human patients and mice have all been recessive, we analyzed additional horses using PCR amplification and Sanger sequencing to identify P6/P6 homozygotes.
(260) An Icelandic horse (E885) scored as P6/P6 and homozygous for the wild-type alleles of P2, P3, P4, P5, P8, and K1 was identified at 25 years of age. The horse had been retired from work for at least five years. The owner reported that this horse had gait abnormalities, described by the owner as “tripping.” The owner refused an exercise challenge, expected to induce a rise in serum creatine kinase (CK); serum CK measured in this horse in the absence of an exercise challenge was in the normal range, as were the serum CK levels in eight other Icelandics tested at the same time in the same location. All eight of the other Icelandics were also homozygous for the wild-type alleles of P2, P3, P4, P8, and K1; their genotypes with respect to DYSF were N/N (4), N/P6 (3), and N/P5 (1). None of these other eight Icelandics was reported to be symptomatic.
(261) A Thoroughbred (E456) scored as P6/P6, n/P3 and homozygous for the wild-type alleles of P2, P4, P8, and K1 was identified. This horse was also homozygous for the wild-type allele of P5. This Thoroughbred was reported by the owner to have shown frequent tie-ups at the track, was only raced once, and was described by an experienced trainer as “not quite right.” The horse died of colic at four years of age.
(262) Two compound heterozygotes scored as P5/P6 and homozygous for the wild-type alleles of P2, P3, P4, P8, and K1 were identified. Both the P5/P6 Percheron (E797) and the P5/P6 Clydesdale (E823) were described by the owners as symptomatic, as were two other P5/P5 Clydesdales (E818 and E820) with the owner of the P5/P6 Clydesdale (E823).
(263) Recombination between the P5 and P6 base substitution can produce a chromosome with both of the mutations in cis. Because both whole-genome sequencing on the Illumina platform and Sanger sequencing of DNA amplified by PCR do not yield phase information, such a chromosome could only be definitively identified in a horse scored as homozygous for P5 and heterozygous for

P6, or in a horse scored as homozygous for P6 and heterozygous for P5. No such result has been seen in any sample.

(264) The phenotype of human patients with the various alleles listed in TABLE 1, TABLE 2, and TABLE 3 suggests that horses homozygous for the DYSF-R253W allele (P5) would be expected to show increased serum creatine kinase (CK) and aspartate aminotransferase (AST), especially upon an exercise challenge. Affected horses are also expected to show abnormalities on muscle biopsy, including necrosis and regeneration, which may lead to adipose and fibrotic infiltration of the endomysial space. We do not expect heterozygous horses to be affected.

(265) The novel DYSF-P1290T (P6) allele does not precisely replicate any known human mutation. Nevertheless, the phenotype observed in P6/P6 homozygotes and in P5/P6 compound heterozygotes suggests that these horses would also be expected to show increased serum creatine kinase (CK) and aspartate aminotransferase (AST), especially upon an exercise challenge, as well as defects observable by muscle biopsy, as for horses homozygous for the DYSF-R253W (P5) allele.

(266) Phenotypic Effects of the PYROXD1-D492H Variant

(267) The PYROXD1-D492H variant (hereafter abbreviated as P8) was discovered by analysis of whole genome sequencing data from Icelandic, Arabian, Thoroughbred, and Quarter Horses. Some of these horses were diagnosed via muscle biopsy with Polysaccharide Storage Myopathy type 2 (PSSM2) or Myofibrillar Myopathy (MFM). These horses were heterozygous for the P8 allele (n/P8); some were homozygous wild-type for seven other variants associated with PSSM2: MYOT-S232P (P2), FLNC-E753K (P3a), FLNC-A1207T (P3b), MYOZ3-S42L (P4), DYSF-R285W (P5), DYSF-P1290T (P6) and COL6A3-G2182A (K1).

(268) An Icelandic horse (E013) free of the P2, P3, P4, P5, P6, and K1 variants was diagnosed as having PSSM2 by muscle biopsy. The horse was heterozygous for P8 (n/P8). The symptoms observed in this horse correspond to the general description of the symptoms of PSSM2, and are similar to symptoms produced by homozygosity or heterozygosity for the P2, P3, P4, P5, and K1 variants in various combinations.

(269) An Arabian horse (E361) scored as n/P8 and homozygous for the wild-type alleles of P2, P3, P4, P5, P6, and K1 was diagnosed as having Recurrent Exertional Rhabdomyolysis (RER) from a severe episode of exercise intolerance during which serum creatine kinase (CK) and aspartate aminotransferase (AST) rose to levels above 20,000 units; the baseline is below 500. Myoglobinuria (dark brown urine resulting from the presence of myoglobin) was also observed.

(270) An Arabian horse (E049) scored as n/P8 and homozygous for the wild-type alleles of P2, P3, P4, P5, P6, and K1 was highly symptomatic and was euthanized at 12 years of age. A necropsy showed gluteal muscles affected but normal muscles in the lower limbs, neck, and inner thighs. In the most affected regions, muscle cells were replaced by fat cells. Up to 50% of the muscle cells were replaced by fat in some regions.

(271) A horse owner who had submitted samples for a number of Arabians received results for P8 several months after submission. One of her Arabian mares (E682) was n/P2 and P8/P8 and homozygous for the wild-type alleles of P3, P4 and not yet scored for P5, P6, and K1. The owner reported that this mare had never been able to keep weight on her, and “looked like a living skeleton no matter what diet was fed.” This horse is shown in FIG. **44**. Muscle wasting in the pelvic girdle (hindquarters), shoulder girdle (topline), and proximal limbs is evident. The horse is pregnant in the photo. The horse was reported to be symptomatic with gait abnormalities (“rope walking”), and died of a choking incident between the time of submission of the sample and the test results. Difficulty in swallowing is observed in human patients with Myofibrillar Myopathy 8 (MFM8) due to mutations in PYROXD1 (O'Grady et al. 2016 Am J Hum Genet. 99:1086-1105 DOI: 10.1016/j.ajhg.2016.09.005).

(272) The phenotype of human patients with the missense alleles described here suggests that horses heterozygous or homozygous for PYROXD1-D492H (P8) would be expected to show an

array of myopathic changes resembling Centronuclear Myopathy, Myofibrillar Myopathy, and Nemaline Myopathy in muscle biopsies. Specifically, we expect to see ectopic aggregates of desmin-positive material outside of the Z disc, Z disc streaming, and other changes seen in Myofibrillar Myopathy.

(273) There is a high incidence of the P8 variant among Arabians (TABLE 7). Arabian horses diagnosed with Myofibrillar Myopathy (MFM) by muscle biopsy and symptoms of exercise intolerance were analyzed in a recent published study (Valberg et al. 2018 Physiol Genomics doi: 10.1152). Samples were taken pre- and post-exercise and analyzed by RNA-Seq and iTRAQ (proteomics). Differential expression was seen for genes involved in pathways for structure morphogenesis, fiber organization, tissue development, and cell differentiation. Proteomic analysis showed lower levels of the antioxidant protein peroxiredoxin 6 in resting muscle; the authors proposed that altered cysteine metabolism and a deficiency of cysteine-containing antioxidants produced oxidative stress during aerobic exercise. Irreversible oxidation of cysteine residues in proteins that are structural components of the contractile apparatus, for example desmin, might induce proteolytic degradation of these proteins, causing Z disc fragmentation and streaming seen in muscle biopsies. While the genotype of the affected horses in this study is unknown, the defect in PYROXD1 described in this disclosure, and the phenotype of PYROXD1 mutants in humans, suggests that the affected horses in this study are n/P8 or P8/P8.

(274) Three Appaloosas kept at the same farm under the same conditions were coincidentally tested for vitamin E levels around the same time that they were genotyped with respect to P8. All three horses were clear for P2, P3, P4, P5, and K1. Two are n/n for P8; one is n/P8.

(275) Vitamin E levels (reference range is 200-1000 ug/dL)

(276) TABLE-US-00006 Horse 1 (n/n) 255 ug/dL Horse 2 (n/n) 383 ug/dL Horse 3 (n/P8) 164 ug/dL

Vitamin E is an antioxidant that is consumed when it scavenges hyperoxyl radicals in lipid membranes. Increased levels of oxidative stress would be expected to deplete vitamin E.

(277) Prospects for treatment: If the myopathy caused by the defect in PYROXD1 described in this disclosure results from oxidative stress, it is possible that treatments with antioxidants would be effective at preventing symptoms or the progression of the disease. For example, dietary supplementation with vitamin E might be effective. Also, the combination of methylsulfonylmethane (MSM) and vitamin C has been shown to be effective at combatting oxidative stress causes by sustained exercise in horses, leading to an increase in glutathione synthesis and in levels of glutathione transferase (Marñón G et al. 2008 Acta Vet Scand 50:45 DOI: 10.1186/1751-0147-50-45; Williams CA 2016 J Anim Sci 94:4067-4075 DOI: 10.2527/jas.2015-9988). N-acetyl cysteine is a safe, low-cost compound that increases the level of glutathione, and has been shown to protect against muscle damage in mice subjected to oxidative stress through a mutation in CASQ1, the gene encoding calsequestrin (Paolini C et al. Skelet Muscle 5:10 DOI: 10.1186/s13395-015-0035-9).

(278) Phenotypic Effects of the COL6A3-G2182A Variant

(279) The COL6A3-G2182A variant (hereafter abbreviated as K1) was discovered by analysis of whole genome sequencing data from Icelandic, Arabian, Thoroughbred, and Quarter Horses. Some of these horses were diagnosed via muscle biopsy with Polysaccharide Storage Myopathy type 2 (PSSM2). These horses were heterozygous for the K1 allele; some were homozygous wild-type for six other variants previously associated with PSSM2: MYOT-S232P (P2), FLNC-E753K (P3a), FLNC-A1207T (P3b), MYOZ3-S42L (P4), DYSF-R285W (P5), DYSF-P1290T (P6), and PYROXD1-D492H (P8).

(280) A Paint horse (E008) scored as n/K1 and homozygous for the wild-type alleles of P2, P3, P4, P5, P6, and P8 was diagnosed as having PSSM2 by muscle biopsy. The symptoms observed in this horse correspond to the general description of the symptoms of PSSM2, and are similar to symptoms produced by homozygosity or heterozygosity for the P2, P3, P4, P5, and P8 variants in

various combinations.
(281) The phenotype of human patients with missense alleles of COL6A3 and of mice with the targeted allele suggests that muscle biopsies from horses bearing the COL6A3-G2182A mutation (K1) will show changes in the connective tissue layer that ensheaths the muscle fiber. The size of the connective tissue layer is expected to increase. In late stages of the disease, replacement of some of this layer with adipose or scar tissue is expected, without evidence of necrosis. This condition has been called skeletal muscle endomysial fibrosis. It has not yet been directly observed in muscle biopsies from n/K1 or K1/K1 horses.
(282) Incidence of Genetic Variants by Breed
(283) The incidence of the previously described genetic variants (P2, P3, and P4) and the genetic variants described in this disclosure (P5, P6, P8, and K1) varies among breeds. It is useful to have a measurement of the allele frequency of different variants among breeds, as this can facilitate a further examination of the phenotype produced by a specific variant by identifying a breed in which it occurs at high incidence, perhaps with other genetic variants at a low incidence.
(284) TABLE 5 shows the observed incidence of P2, P3, P4, P5, P6, P8, and K1 in Quarter Horse-related breeds. For purposes of this tabulation, a “Quarter Horse-related” breed is a Quarter Horse, Appendix Quarter Horse (the result of a cross of a Quarter Horse to a Thoroughbred), Paint Horse, Appaloosa, or a horse resulting from a cross of any two of these types to each other. This sample cannot be considered unselected, as the majority of horse owners volunteering their horses for study did so because their horses had symptoms of some kind. When we attempted to recruit breeding herds, the majority of horse owners did not volunteer; it is possible that those volunteering did so because they observed symptoms of exercise intolerance in their herd.
(285) Quarter Horse-related breeds are of relatively recent origin, with gene flow from other breeds. For example, the American Quarter Horse Association allows the registration of Appendix Quarter Horses that have a Thoroughbred parent. In contrast, Thoroughbreds have a closed breeding book, meaning that a horse can only be registered as a Thoroughbred if both parents are Thoroughbreds. All seven genetic variants are found in Quarter Horse-related breeds. The incidence ranges from an allele frequency of 0.204 for P2 to an allele frequency of 0.007 for K1. Quarter Horse-related breeds are therefore a good choice for the further study of the interaction of all seven genetic variants. It should be noted that P2 (chr14:38519183 A/G) and P4 (chr14:27399222 G/A) are located approximately 11.2 Megabases (Mb) apart on chromosome 14 and are expected to display genetic linkage. Two different genotypes confirm that the chromosome having both P2 and P4 in cis exists in horse populations: P2/P2 n/P4 and n/P2 P4/P4 individuals have been observed in Quarter Horse-related breeds. No horse that is P2/P2 P4/P4, or more properly written, P2 P4/P2 P4, has been observed to date in Quarter Horse-related breeds or in any other breed in a sample of nearly 1,000 horses tested for both variants.
(286) TABLE 6 shows the observed incidence of P2, P3, P4, P5, P6, P8, and K1 in Thoroughbreds. Because as noted above, Thoroughbreds have a closed breeding book, it is possible that Thoroughbreds may be entirely free of one or more of these genetic variants. In the sample of modest size tested to date, P5 and K1 have not yet been observed in Thoroughbreds, while the observed frequency of other variants ranges from 0.028 for P8 to 0.193 for P6. The Thoroughbreds included in this sample are less highly selected for symptomatic horses than is the Quarter Horse-related sample because the method of recruitment of horse owners was different. However, some horse owners were aware of the nature of the research and likely volunteered symptomatic horses. The results show that Thoroughbreds are well suited for the further study of the P6 genetic variant, as it occurs at a reasonably high frequency in the absence of P5, the other allele of DYSF.
(287) TABLE 7 shows the observed incidence of P2, P3, P4, P5, P6, P8, and K1 in Arabians. So far, P3 has not yet been observed in Arabians, nor has P5 or P6. P2, P4, and K1 are observed. P8 is observed at an allele frequency of 0.218. As noted previously in this disclosure, it is possible that the PYROXD1 variant P8 is responsible for the defect in the redox state of cysteine-containing

antioxidants seen in Arabians diagnosed with Myofibrillar Myopathy (MFM) in a recent published study (Valberg et al. 2018 Physiol Genomics doi: 10.1152). This shows that Arabians are ideal for the further study of the effects of the PYROXD1 variant P8.
(288) TABLE 8 shows the observed incidence of P2, P3, P4, P5, P6, P8, and K1 in draft breeds, defined for this sample as Shires, Clydesdales, and Percherons, or horses derived from crosses among these breeds. There are many other draft breeds for which we have obtained samples of very modest size, but we initially concentrated on these three breeds because the observed incidence of P5 was very high. The observed frequency of P5 in this modest sample is 0.543. The observed frequency of P6 in this modest sample is 0.136. These frequencies are high enough that compound heterozygotes (P5/P6) can be expected; indeed, two symptomatic compound heterozygotes have been identified. The observed frequency of P4, P8, and K1 in this small sample is zero; P2 and P3 are present. This shows that Shires, Clydesdales, and Percherons are ideal for the further study of the effects of the DYSF variants P5 and P6.
(289) TABLE-US-00007 TABLE 5 Incidence of genetic variants in Quarter Horse-related breeds P2 n/n n/P2 P2/P2 Total P2 allele frequency 390 167 38 595 0.204 P3 n/n n/P3 P3/P3 Total P3 allele frequency 503 92 4 599 0.083 P4 n/n n/P4 P4/P4 Total P4 allele frequency 587 81 4 587 0.076 P5 n/n n/P5 P5/P5 Total P5 allele frequency 35 3 0 37 0.041 P6 n/n n/P6 P6/P6 Total P6 allele frequency 64 8 0 72 0.056 P8 n/n n/P8 P8/P8 Total P8 allele frequency 79 7 0 86 0.041 K1 n/n n/K1 K1/K1 Total K1 allele frequency 145 2 0 147 0.007
(290) TABLE-US-00008 TABLE 6 Incidence of genetic variants in Thoroughbreds P2 n/n n/P2 P2/P2 Total P2 allele frequency 149 51 8 208 0.161 P3 n/n n/P3 P3/P3 Total P3 allele frequency 170 42 2 214 0.107 P4 n/n n/P4 P4/P4 Total P4 allele frequency 177 34 1 212 0.085 P5 n/n n/P5 P5/P5 Total P5 allele frequency 102 0 0 102 0.0 P6 n/n n/P6 P6/P6 Total P6 allele frequency 69 33 4 106 0.193 P8 n/n n/P8 P8/P8 Total P8 allele frequency 116 7 0 123 0.028 K1 n/n n/K1 K1/K1 Total K1 allele frequency 121 0 0 121 0.0
(291) TABLE-US-00009 TABLE 7 Incidence of genetic variants in Arabians P2 n/n n/P2 P2/P2 Total P2 allele frequency 43 17 4 64 0.195 P3 n/n n/P3 P3/P3 Total P3 allele frequency 64 0 0 64 0.0 P4 n/n n/P4 P4/P4 Total P4 allele frequency 58 8 1 65 0.077 P5 n/n n/P5 P5/P5 Total P5 allele frequency 22 0 0 22 0.0 P6 n/n n/P6 P6/P6 Total P6 allele frequency 16 0 0 16 0.0 P8 n/n n/P8 P8/P8 Total P8 allele frequency 38 21 3 62 0.218 K1 n/n n/K1 K1/K1 Total K1 allele frequency 18 3 0 21 0.071
(292) TABLE-US-00010 TABLE 8 Incidence of genetic variants in Draft breeds P2 n/n n/P2 P2/P2 Total P2 allele frequency 15 5 5 25 0.300 P3 n/n n/P3 P3/P3 Total P3 allele frequency 24 1 0 25 0.020 P4 n/n n/P4 P4/P4 Total P4 allele frequency 23 0 0 23 0.0 P5 n/n n/P5 P5/P5 Total P5 allele frequency 6 9 8 23 0.543 P6 n/n n/P6 P6/P6 Total P6 allele frequency 16 6 0 22 0.136 P8 n/n n/P8 P8/P8 Total P8 allele frequency 22 0 0 22 0.0 K1 n/n n/K1 K1/K1 Total K1 allele frequency 23 0 0 23 0.0
(293) In the preceding description and following claims, the term “and/or” means one or all of the listed elements or a combination of any two or more of the listed elements; the terms “comprises,” “comprising,” and variations thereof are to be construed as open ended—i.e., additional elements or steps are optional and may or may not be present; unless otherwise specified, “a,” “an,” “the,” and “at least one” are used interchangeably and mean one or more than one; and the recitations of numerical ranges by endpoints include all numbers subsumed within that range (e.g., 1 to 5 includes 1, 1.5, 2, 2.75, 3, 3.80, 4, 5, etc.).
(294) In the preceding description, particular embodiments may be described in isolation for clarity. Unless otherwise expressly specified that the features of a particular embodiment are incompatible with the features of another embodiment, certain embodiments can include a combination of compatible features described herein in connection with one or more embodiments.
(295) For any method disclosed herein that includes discrete steps, the steps may be conducted in any feasible order. And, as appropriate, any combination of two or more steps may be conducted

simultaneously.
(296) This is illustrated by the following examples. It is to be understood that the particular examples, materials, amounts, and procedures are to be interpreted broadly in accordance with the scope and spirit of the invention as set forth herein.

EXAMPLES

Example 1

Method of Detecting DNA Mutations Associated with Equine Polysaccharide Storage Myopathy Type 2 (PSSM2), Also Known as Myofibrillar Myopathy (MFM)

(297) The complete DNA sequences of the horse DYSF, PYROXD1, and COL6A3 coding regions were obtained from the current version of the public horse genome assembly (EquCab2).
(298) Using the DYSF, PYROXD1, and COL6A3 sequences, PCR primers are developed that can amplify the site of genomic DNA containing the DYSF-R253W, DYSF-P1290T, PYROXD1-D492H, and COL6A3-G2182A mutations. For example, a PCR primer pair that has been successfully and reliably used to amplify the region including DYSF-R253W from isolated horse DNA samples lies in the region around Exon B (FIG. **3**). These sequences are
(299) TABLE-US-00011 (SEQ ID NO: 14) 5′-CCCGAGATTTCTGGCTTTCT-3′ and (SEQ ID NO: 15) 5′-CTCGACAAGTTCTGGGGTGT-3′.
A PCR primer pair that has been successfully and reliably used to amplify the region including DYSF-P1290T from isolated horse DNA samples lies in the region around Exon I (FIG. **3**). These sequences are
(300) TABLE-US-00012 (SEQ ID NO: 64) 5′-GGTTGCAAACTCCCAACTGT-3′ and (SEQ ID NO: 65) 5′-GATTTTTCAAGCTGCCGAAG-3′.
A PCR primer pair that has been successfully and reliably used to amplify the region including PYROXD1-D492H from isolated horse DNA samples lies in the region around Exon 12 (FIG. **15**). These sequences are
(301) TABLE-US-00013 (SEQ ID NO: 111) 5′-CAGATTTTCTGCTGGCCATT-3′ and (SEQ ID NO: 112) 5′-TGGTCATCATTAAATCAGTGCAA-3′]
(302) A PCR primer pair that has been successfully and reliably used to amplify the region including COL6A3-G2182A from isolated horse DNA samples lies in the region around Exon 12 (FIG. **26**). These sequences are 5′-AGATGGGGCACAGATCAAAC-3′ (SEQ ID NO:172) and 5′-TTCCCAGACTCTCCTGTGCT-3′ (SEQ ID NO:171). Many other primer pairs are also possible.
(303) Using the above PCR primers to amplify the region, the genotype of any horse (G/G, G/A, or A/A) for the DNA sequence of the forward strand at chr15:31,306,949, and R/R, R/W, or W/W for the amino acid sequence of the DYSF-R253W variant can be obtained. In this method, the amplified DNA may be cloned and then sequenced or sequenced directly without cloning. Alternatively, the appearance of amplified product in the presence of primers specific to the wild type or mutant allele may be monitored in real time using a qPCR instrument designed for this purpose. Many other methods of detecting the nucleotides at the positions of the horse DYSF sequence are possible.
(304) Using the above PCR primers to amplify the region, the genotype of any horse (G/G, G/T, or T/T) for the DNA sequence of the forward strand at chr15:31,306,949, and P/P, P/T, or T/T for the amino acid sequence of the DYSF-P1290T variant can be obtained. In this method, the amplified DNA may be cloned and then sequenced or sequenced directly without cloning. Alternatively, the appearance of amplified product in the presence of primers specific to the wild type or mutant allele may be monitored in real time using a qPCR instrument designed for this purpose. Many other methods of detecting the nucleotides at the positions of the horse DYSF sequence are possible.
(305) Using the above PCR primers to amplify the region, the genotype of any horse (G/G, G/C, or C/C) for the DNA sequence of the forward strand at chr6:47,661,977, and D/D, D/H, or H/H for the amino acid sequence of the PYROXD1-D492H variant can be obtained. In this method, the

amplified DNA may be cloned and then sequenced or sequenced directly without cloning. Alternatively, the appearance of amplified product in the presence of primers specific to the wild type or mutant allele may be monitored in real time using a qPCR instrument designed for this purpose. Many other methods of detecting the nucleotides at the positions of the horse PYROXD1 sequence are possible.

(306) Using the above PCR primers to amplify the region, the genotype of any horse (G/G, G/C, or C/C) for the DNA sequence of the forward strand at chr6:23,480,621, and G/G, G/A, or A/A for the amino acid sequence of the COL6A3-G2182A variant can be obtained. In this method, the amplified DNA may be cloned and then sequenced or sequenced directly without cloning. Alternatively, the appearance of amplified product in the presence of primers specific to the wild type or mutant allele may be monitored in real time using a qPCR instrument designed for this purpose. Many other methods of detecting the nucleotides at the positions of the horse COL6A3 sequence are possible.

(307) DNA testing now provides veterinarians and veterinary pathologists with a means to more accurately determine if a horse with the clinical signs of Polysaccharide Storage Myopathy type 2 (PSSM2) has the heritable and common form of the disease that can be specifically attributed to the DYSF-R253W, DYSF-P1290T, PYROXD1-D492H, or COL6A3-G2182A coding region mutation. All that is needed is a tissue sample containing the individual's DNA (typically hair root or blood) and appropriate PCR and sequence analysis technology to detect the distinct nucleotide changes. Such PCR primers are based in (1) DYSF Exon B (as shown in FIG. **3**) and the flanking intron sequences, as shown in FIG. **3**, or in other DNA sequences of this gene, (2) DYSF Exon I (as shown in FIG. **15**) and the flanking intron sequences, as shown in FIG. **15**, or in other DNA sequences of this gene, (3) PYROXD1 Exon 12 (as shown in FIG. **24**) and the flanking intron sequences, as shown in FIG. **24**, or in other DNA sequences of this gene, or (4) COL6A3 Exon 26 (as shown in FIG. **35**) and the flanking intron sequences, as shown in FIG. **35**, or in other DNA sequences of this gene.

(308) Also, DNA testing provides owners and breeders with a means to determine if any horse can be expected to produce offspring with these forms of Polysaccharide Storage Myopathy type 2 (PSSM2). Abbreviating the DYSF-R253W allele as P5, a P5/P5 horse would produce a carrier foal 100% of the time, while an N/P5 horse would produce a carrier foal 50% of the time when mated to an N/N horse. Mating of an N/P5 horse to an N/P5 horse would produce an affected foal 25% of the time and a carrier foal 50% of the time. Abbreviating the DYSF-P1290T allele as P6, a P6/P6 horse would produce a carrier foal 100% of the time, while an N/P6 horse would produce a carrier foal 50% of the time when mated to an N/N horse. Mating of an N/P6 horse to an N/P6 horse would produce an affected foal 25% of the time and a carrier foal 50% of the time. Abbreviating the PYROXD1-D492H allele as P8, a P8/P8 horse would produce a heterozygous foal 100% of the time when mated to an n/n horse, while an n/P8 horse would produce a heterozygous foal 50% of the time when mated to an n/n horse. Mating of an n/P8 horse to an n/P8 horse would produce an affected P8/P8 foal 25% of the time, and an affected n/P8 foal 50% of the time. Abbreviating the COL6A3-G2182A allele as K1, a K1/K1 horse would produce a heterozygous foal 100% of the time when mated to an n/n horse, while an n/K1 horse would produce a heterozygous foal 50% of the time when mated to an n/n horse. Mating of an n/K1 horse to an n/K1 horse would produce an affected K1/K1 foal 25% of the time, and an affected n/K1 foal 50% of the time. Breeding programs could incorporate this information in the selection of parents that could eventually reduce and even eliminate these forms of Polysaccharide Storage Myopathy type 2 (PSSM2) in their herds.

(309) The complete disclosure of all patents, patent applications, and publications, and electronically available material (including, for instance, nucleotide sequence submissions in, e.g., GenBank and RefSeq, and amino acid sequence submissions in, e.g., SwissProt, PIR, PRF, PDB, and translations from annotated coding regions in GenBank and RefSeq) cited herein are

incorporated by reference in their entirety. In the event that any inconsistency exists between the disclosure of the present application and the disclosure(s) of any document incorporated herein by reference, the disclosure of the present application shall govern. The foregoing detailed description and examples have been given for clarity of understanding only. No unnecessary limitations are to be understood therefrom. The invention is not limited to the exact details shown and described, for variations obvious to one skilled in the art will be included within the invention defined by the claims.
(310) Unless otherwise indicated, all numbers expressing quantities of components, molecular weights, and so forth used in the specification and claims are to be understood as being modified in all instances by the term "about." Accordingly, unless otherwise indicated to the contrary, the numerical parameters set forth in the specification and claims are approximations that may vary depending upon the desired properties sought to be obtained. At the very least, and not as an attempt to limit the doctrine of equivalents to the scope of the claims, each numerical parameter should at least be construed in light of the number of reported significant digits and by applying ordinary rounding techniques.
(311) Notwithstanding that the numerical ranges and parameters setting forth the broad scope of the invention are approximations, the numerical values set forth in the specific examples are reported as precisely as possible. All numerical values, however, inherently contain a range necessarily resulting from the standard deviation found in their respective testing measurements.
(312) All headings are for the convenience of the reader and should not be used to limit the meaning of the text that follows the heading, unless so specified.

Claims

1. A method of mating a horse comprising: (a) detecting in a biological sample obtained from said horse, said sample comprising nucleic acids that include the coding regions for dysferlin (DYSF), polynucleotide-disulfide oxidoreductase domain-containing protein 1 (PYROXD1) and collagen type VI alpha 3 chain (COL6A3): (1) a thymine (T) or a cytosine (C) at nucleotide 1001 of SEQ ID NO: 1, wherein detection of the thymine indicates the presence of variant P5; (2) an adenine (A) or cytosine (C) at nucleotide 1001 of SEQ ID NO:56, wherein detection of the adenine indicates the presence of variant P6; (3) a cytosine (C) or a guanine (G) at nucleotide 1001 of SEQ ID NO:103, wherein detection of the cytosine indicates the presence of variant P8; and (4) a cytosine (C) or a guanine (G) at nucleotide 1001 of SEQ ID NO: 163, wherein detection of the cytosine indicates the presence of variant K1; wherein in each case the presence of the nucleotide can be inferred from detecting the nucleotide present at the complement thereof; wherein the presence of any one or more of said variants P5, P6, P8 and K1 in said nucleic acid is indicative of inherited equine myopathy in said horse; wherein said horse is not homozygous for the presence of P5, P6, P8 or K1; wherein said horse is heterozygous for at least one of P5, P6, P8 or K1; and (b) mating said horse with a mare or sire.
2. The method according to claim 1 wherein said mare or sire is free of P5, P6, P8 and K1.
3. The method according to claim 1 wherein said mare or sire is heterozygous for variant P5, P6, P8 or K1.
4. The method according to claim 1, further comprising: contacting the nucleic acids of the sample with at least one oligonucleotide probe to form a hybridized nucleic acid; and amplifying the hybridized nucleic acid.
5. The method according to claim 4, wherein: Exon B of the equine dysferlin coding region (DYSF), as defined in SEQ ID NO: 5, or a portion thereof is amplified; Exon I of the equine dysferlin coding region (DYSF), as defined in SEQ ID NO: 59, or a portion thereof is amplified; Exon 12 of the equine pyridine nucleotide-disulfide oxidoreductase domain-containing protein 1 (PYROXD 1), as defined in SEQ ID NO: 106, or a portion thereof is amplified; or Exon 26 of the

equine collagen type VI alpha 3 chain coding region (COL6A3), as defined in SEQ ID NO: 166, or a portion thereof is amplified.
6. The method according to claim 4, wherein the hybridized nucleic acid is amplified using polymerase chain reaction, strand displacement amplification, ligase chain reaction, or nucleic acid sequence-based amplification.
7. The method according to claim 4, wherein at least one oligonucleotide probe is immobilized on a solid surface or a semisolid surface.
8. The method according to claim 1, wherein the nucleic acid is genomic DNA.
9. The method according to claim 4, wherein the nucleic acid is genomic DNA.
10. The method according to claim 5, wherein the nucleic acid is genomic DNA.
11. The method according to claim 6, wherein the nucleic acid is genomic DNA.
12. The method according to claim 7, where in the nucleic acid is genomic DNA.
13. The method according to claim 1 wherein the nucleic acid is RNA.
14. The method according to claim 4 wherein the nucleic acid is RNA.
15. The method according to claim 5 wherein the nucleic acid is RNA.
16. The method according to claim 6 wherein the nucleic acid is RNA.
17. The method according to claim 7 wherein the nucleic acid is RNA.
18. The method according to claim 13, wherein the method comprises converting the RNA into DNA by reverse transcriptase.
19. The method according to claim 14, wherein the method comprises converting the RNA into DNA by reverse transcriptase.
20. The method according to claim 15, wherein the method comprises converting the RNA into DNA by reverse transcriptase.
21. The method according to claim 16, wherein the method comprises converting the RNA into DNA by reverse transcriptase.
22. The method according to claim 17, wherein the method comprises converting the RNA into DNA by reverse transcriptase.
23. A method for detecting a biomarker in a horse, the method comprising: (a) obtaining a biological sample from said horse, the biological sample comprising nucleic acid that includes the coding regions for dysferlin (DYSF) corresponding to SEQ ID NOs: 2 and 57, polynucleotide-disulfide oxidoreductase domain-containing protein 1 (PYROXD1) corresponding to SEQ ID NO: 104 and collagen type VI alpha 3 chain (COL6A3) corresponding to SEQ ID NO: 164; and detecting: (1) a thymine (T) or a cytosine (C) at nucleotide 264 of SEQ ID NO:2, wherein detection of the thymine indicates the presence of variant P5; (2) an adenine (A) or cytosine (C) at nucleotide 448 of SEQ ID NO:57, wherein detection of the adenine indicates the presence of variant P6; (3) a cytosine (C) or a guanine (G) at nucleotide 1311 of SEQ ID NO:104, wherein detection of the cytosine indicates the presence of variant P8; and (4) a cytosine (C) or a guanine (G) at nucleotide 437 of SEQ ID NO: 164, wherein detection of the cytosine indicates the presence of variant K1 wherein in each case the presence of the nucleotide can be inferred from detecting the nucleotide present at the complement thereof; wherein the presence of any one or more of said variants P5, P6, P8 and K1 in said nucleic acid is indicative of inherited equine myopathy in said horse; wherein said horse is not homozygous for the presence of P5, P6, P8 or K1; wherein said horse is heterozygous for at least one of P5, P6, P8 or K1; and (b) mating said horse with a mare or sire.
24. The method according to claim 23 wherein said mare or sire is free from variants P5, P6, P8 and K1.
25. The method according to claim 24 wherein said mare or sire is heterozygous for variant P5, P6, P8 or K1.
26. The method of claim 23, further comprising: contacting the nucleic acid with at least one oligonucleotide probe to form a hybridized nucleic acid; and amplifying the hybridized nucleic acid.

27. The method of claim 26, wherein the hybridized nucleic acid is amplified using polymerase chain reaction, strand displacement amplification, ligase chain reaction, or nucleic acid sequence-based amplification.
28. The method of claim 26, wherein at least one oligonucleotide probe is immobilized on a solid surface or semisolid surface.